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SOLUTIONS

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PROBLEMS FOR JUNIORS

JP.541 If a, b, c sides in acute $\triangle ABC$ with s – semiperimeter; r – inradii and

$x, y, z \in \left(0, \frac{\pi}{2}\right)$ are such that:

$$\cos x = \frac{a}{b+c}; \cos y = \frac{b}{c+a}; \cos z = \frac{c}{a+b}$$

then:

$$\left(\tan^2 \frac{x}{2} + \tan^2 \frac{y}{2} + \tan^2 \frac{z}{2}\right) \tan^2 \frac{x}{2} \tan^2 \frac{y}{2} \tan^2 \frac{z}{2} = \frac{r^2}{s^2}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x} = \frac{1 - \frac{a}{b+c}}{1 + \frac{a}{b+c}} = \frac{b+c-a}{a+b+c} = \frac{a+b+c-2a}{a+b+c} = \frac{2s-2a}{2s} = \frac{s-a}{s}$$

Analogous:

$$\tan^2 \frac{y}{2} = \frac{s-b}{s}; \tan^2 \frac{z}{2} = \frac{s-c}{s}$$

$$\tan^2 \frac{x}{2} + \tan^2 \frac{y}{2} + \tan^2 \frac{z}{2} = \frac{s-a}{s} + \frac{s-b}{s} + \frac{s-c}{s} =$$

$$= \frac{3s - (a+b+c)}{s} = \frac{3s - 2s}{s} = \frac{s}{s} = 1$$

$$\tan^2 \frac{x}{2} \tan^2 \frac{y}{2} \tan^2 \frac{z}{2} = \frac{s-a}{s} \cdot \frac{s-b}{s} \cdot \frac{s-c}{s} =$$

$$= \frac{(s-a)(s-b)(s-c)}{s^3} = \frac{s(s-a)(s-b)(s-c)}{s^4} = \frac{F^2}{s^4} = \frac{r^2 s^2}{s^4} = \frac{r^2}{s^2}$$

$$\left(\tan^2 \frac{x}{2} + \tan^2 \frac{y}{2} + \tan^2 \frac{z}{2}\right) \tan^2 \frac{x}{2} \tan^2 \frac{y}{2} \tan^2 \frac{z}{2} = 1 \cdot \frac{r^2}{s^2} = \frac{r^2}{s^2}$$

Equality holds for $a = b = c$.

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Solution 2 by Marin Chirciu-Romania

Using $\tan^2 \frac{x}{2} = \frac{1-\cos x}{1+\cos x} = \frac{1-\frac{a}{b+c}}{1+\frac{a}{b+c}} = \frac{b+c-a}{b+c+a} = \frac{2s-2a}{2s} = \frac{s-a}{s}$ we obtain:

$$LHS = \sum \tan^2 \frac{x}{2} \prod \tan^2 \frac{x}{2} = \sum \frac{s-a}{s} \prod \frac{s-a}{s} = \frac{s}{s} \cdot \frac{r^2 s}{s^3} = \frac{r^2}{s^2} = RHS$$

Remark: In the same way: If a, b, c sides in acute $\triangle ABC$ with s – semiperimeter; r –

inradii, R – circumradii and $x, y, z \in \left(0, \frac{\pi}{2}\right)$ are such that $\cos x = \frac{a}{b+c}$, $\cos y = \frac{b}{c+a}$, $\cos z =$

$\frac{c}{a+b}$ then: $\left(\cot^2 \frac{x}{2} + \cot^2 \frac{y}{2} + \cot^2 \frac{z}{2}\right) \cot^2 \frac{x}{2} \cot^2 \frac{y}{2} \cot^2 \frac{z}{2} = \frac{s^2(4R+r)}{r^3}$

Marin Chirciu

Solution

Using $\cot^2 \frac{x}{2} = \frac{1+\cos x}{1-\cos x} = \frac{1+\frac{a}{b+c}}{1-\frac{a}{b+c}} = \frac{b+c+a}{b+c-a} = \frac{2s}{2s-2a} = \frac{s}{s-a}$ we obtain:

$$\begin{aligned} LHS &= \sum \cot^2 \frac{x}{2} \prod \cot^2 \frac{x}{2} = \sum \frac{s}{s-a} \prod \frac{s}{s-a} = \\ &= \frac{s(4R+r)}{sr} \cdot \frac{s^3}{r^2 s} = \frac{s^2(4R+r)}{r^3} = RHS \end{aligned}$$

JP.542 Solve for real numbers:

$$\begin{cases} \frac{6x+6y}{9+4xy} = z \\ \frac{6y+6z}{9+4yz} = x \\ \frac{6z+6x}{9+4zx} = y \end{cases}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{cases} 6x+6y=9z+4xyz \\ 6y+6z=9x+4xyz \\ 6z+6x=9y+4xyz \end{cases} \Rightarrow \begin{cases} x+y=\frac{3z}{2}+\frac{2xyz}{3} \\ y+z=\frac{3x}{2}+\frac{2xyz}{3} \\ z+x=\frac{3y}{2}+\frac{2xyz}{3} \end{cases}$$

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$$\Rightarrow \begin{cases} y + z - (x + y) = \frac{3x}{2} + \frac{2xyz}{3} - \left(\frac{3z}{2} + \frac{2xyz}{3}\right) \\ z + x - (y + z) = \frac{3y}{2} + \frac{2xyz}{3} - \left(\frac{3x}{2} + \frac{2xyz}{3}\right) \end{cases}$$

$$\Rightarrow \begin{cases} z - x = +\frac{3}{2}(x - z) \\ x - y = \frac{3}{2}(y - x) \end{cases} \Rightarrow \begin{cases} \frac{5}{2}(x - z) = 0 \\ \frac{5}{2}(y - x) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x - z = 0 \\ y - x = 0 \end{cases} \Rightarrow x = y = z$$

$$6x + 6x = 9x + 4x^3$$

$$4x^3 - 3x = 0 \Rightarrow x(4x^2 - 3) = 0$$

$$\text{Solutions: } \begin{cases} x = 0 \\ y = 0 \\ z = 0 \end{cases} \quad \begin{cases} x = \frac{\sqrt{3}}{2} \\ y = \frac{\sqrt{3}}{2} \\ z = \frac{\sqrt{3}}{2} \end{cases} \quad \begin{cases} x = -\frac{\sqrt{3}}{2} \\ y = -\frac{\sqrt{3}}{2} \\ z = -\frac{\sqrt{3}}{2} \end{cases}$$

Solution 2 by Marin Chirciu-Romania

$$\begin{cases} \frac{6x + 6y}{9 + 4xy} = z \\ \frac{6y + 6z}{9 + 4yz} = x \\ \frac{6z + 6x}{9 + 4zx} = y \end{cases} \Leftrightarrow \begin{cases} 6x + 6y = 9z + 4xyz \\ 6y + 6z = 9x + 4xyz \\ 6z + 6x = 9y + 4xyz \end{cases}$$

Subtract the equations (1) and (2) $\Rightarrow 6(x - z) = 9(z - x) \Rightarrow x = z$.

Subtract the equations (2) and (3) $\Rightarrow 6(y - x) = 9(x - y) \Rightarrow x = y$.

Using the equation (1) and $x = y = z \Rightarrow 4x^3 - 3x = 0 \Rightarrow x \in \left\{-\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\right\}$

The set of solutions of the equation is $S = \left\{\left(-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right), (0, 0, 0), \left(\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)\right\}$

Remark.

The inequality can be developed.

Let $0 < \lambda < 2n$ fixed. Solve for real numbers:

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$$\begin{cases} \frac{\lambda nx + \lambda ny}{\lambda^2 + n^2 xy} = z \\ \frac{\lambda ny + \lambda nz}{\lambda^2 + n^2 yz} = x \\ \frac{\lambda nz + \lambda nx}{\lambda^2 + n^2 zx} = y \end{cases}$$

Marin Chirciu

Solution:

$$\begin{cases} \frac{\lambda nx + \lambda ny}{\lambda^2 + n^2 xy} = z \\ \frac{\lambda ny + \lambda nz}{\lambda^2 + n^2 yz} = x \\ \frac{\lambda nz + \lambda nx}{\lambda^2 + n^2 zx} = y \end{cases} \Leftrightarrow \begin{cases} \lambda nx + \lambda ny = \lambda^2 z + n^2 xyz \\ \lambda ny + \lambda nz = \lambda^2 x + n^2 xyz \\ \lambda nz + \lambda nx = \lambda^2 y + n^2 xyz \end{cases}$$

Subtract the equations (1) and (2) $\Rightarrow \lambda n(x - z) = \lambda^2(z - x) \Rightarrow x = z$.

Subtract the equations (2) and (3) $\Rightarrow \lambda n(y - x) = \lambda^2(x - y) \Rightarrow x = y$.

Using equation (1)

$$\text{and } x = y = z \Rightarrow n^2 x^3 - (2\lambda n - \lambda^2)x = 0 \Rightarrow x \in \left\{ -\frac{\sqrt{2\lambda n - \lambda^2}}{n}, 0, \frac{\sqrt{2\lambda n - \lambda^2}}{n} \right\}$$

The set of solutions of the equation is:

$$S = \left\{ \left(-\frac{\sqrt{2\lambda n - \lambda^2}}{n}, -\frac{\sqrt{2\lambda n - \lambda^2}}{n}, -\frac{\sqrt{2\lambda n - \lambda^2}}{n} \right), (0, 0, 0), \left(\frac{\sqrt{2\lambda n - \lambda^2}}{n}, \frac{\sqrt{2\lambda n - \lambda^2}}{n}, \frac{\sqrt{2\lambda n - \lambda^2}}{n} \right) \right\}$$

Note.

For $\lambda = 3, n = 2$ we obtain Problem JP.542 from RMM Nr. 37 – Summer 2025

JP.543 Find $x, y, z > 0$ such that $x + y + z = 1$ and

$$\left(\frac{x^5}{yz + 1} + \frac{y^5}{zx + 1} + \frac{z^5}{xy + 1} \right) \left(\frac{x^7}{yz + 1} + \frac{y^7}{zx + 1} + \frac{z^7}{xy + 1} \right) = \frac{1}{72900}$$

Proposed by Daniel Sitaru – Romania

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Solution 1 by proposer

$$\begin{aligned}
 \sum_{cyc} \frac{x^5}{yz+1} &= \sum_{cyc} \frac{x^6}{xyz+x} = \sum_{cyc} \frac{(x^3)^2}{xyz+x} \geq \\
 &\stackrel{BERGSTROM}{\geq} \frac{(x^3+y^3+z^3)^2}{3xyz+x+y+z} \stackrel{AM-GM}{\geq} \frac{(x^3+y^3+z^3)^2}{3 \cdot \left(\frac{x+y+z}{3}\right)^3 + 1} = \\
 &= \frac{(x^3+y^3+z^3)^2}{3 \cdot \frac{1}{27} + 1} \stackrel{JENSEN}{\geq} \frac{1}{\frac{1}{9} + 1} \cdot \left(3 \cdot \left(\frac{x+y+z}{3}\right)^3\right)^2 = \\
 &= \frac{9}{10} \cdot 9 \cdot \frac{1}{3^6} = \frac{3^4}{3^6 \cdot 10} = \frac{1}{90}
 \end{aligned}$$

Analogous:

$$\sum_{cyc} \frac{x^7}{yz+1} \geq \frac{1}{810}$$

By multiplying:

$$\left(\sum_{cyc} \frac{x^5}{yz+1}\right) \cdot \left(\sum_{cyc} \frac{x^7}{yz+1}\right) \geq \frac{1}{72900}$$

Equality holds for $x = y = z = \frac{1}{3}$.

Solution 2 by Marin Chirciu-Romania

$$\sum \frac{x^5}{yz+1} \stackrel{Holder}{\geq} \frac{(\sum x)^5}{3^3 \sum (yz+1)} = \frac{1^5}{3^3 (\sum yz + 3)} \stackrel{sos}{\geq} \frac{1}{3^3 \left(\frac{1}{3} + 3\right)} = \frac{1}{90},$$

with equality for $x = y = z = \frac{1}{3}$.

We have used $\sum yz \leq \frac{1}{3}$, see: $1 = (x+y+z)^2 \geq 3(xy+yz+zx) \Rightarrow xy+yz+zx \leq \frac{1}{3}$

Analogous:

$$\sum \frac{x^7}{yz+1} \stackrel{Holder}{\geq} \frac{(\sum x)^7}{3^5 \sum (yz+1)} = \frac{1^7}{3^5 (\sum yz + 3)} \stackrel{sos}{\geq} \frac{1}{3^5 \left(\frac{1}{3} + 3\right)} = \frac{1}{810}$$

with equality for $x = y = z = \frac{1}{3}$

It follows:

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$$\sum \frac{x^5}{yz+1} \cdot \sum \frac{x^7}{yz+1} \geq \frac{1}{90} \cdot \frac{1}{810} = \frac{1}{72900}, \text{ with equality for } x = y = z = \frac{1}{3}.$$

We deduce that $x = y = z = \frac{1}{3}$ is the solution of the problem.

Remark.

The inequality can be developed.

Let $n \in \mathbb{N}$. Find $x, y, z > 0$ such that $x + y + z = 1$ and

$$\sum \frac{x^{n+2}}{yz+1} \cdot \sum \frac{x^{n+4}}{yz+1} = \frac{1}{9^n \cdot 100}$$

Marin Chirciu

Solution

$$\sum \frac{x^{n+2}}{yz+1} \stackrel{\text{Holder}}{\geq} \frac{(\sum x)^{n+2}}{3^n \sum (yz+1)} = \frac{1^{n+2}}{3^n (\sum yz + 3)} \stackrel{\text{sos}}{\geq} \frac{1}{3^n \left(\frac{1}{3} + 3\right)} = \frac{1}{3^{n-1} \cdot 10},$$

with equality for $x = y = z = \frac{1}{3}$

We have used $\sum yz \leq \frac{1}{3}$, see: $1 = (x + y + z)^2 \geq 3(xy + yz + zx) \Rightarrow xy + yz + zx \leq \frac{1}{3}$.

Analogous:

$$\sum \frac{x^{n+4}}{yz+1} \stackrel{\text{Holder}}{\geq} \frac{(\sum x)^{n+4}}{3^{n+2} \sum (yz+1)} = \frac{1^{n+4}}{3^{n+2} (\sum yz + 3)} \stackrel{\text{sos}}{\geq} \frac{1}{3^{n+2} \left(\frac{1}{3} + 3\right)} = \frac{1}{3^{n+1} \cdot 10},$$

with equality for $x = y = z = \frac{1}{3}$.

It follows:

$$\sum \frac{x^{n+2}}{yz+1} \cdot \sum \frac{x^{n+4}}{yz+1} \geq \frac{1}{3^{n-1} \cdot 10} \cdot \frac{1}{3^{n+1} \cdot 10} = \frac{1}{3^{2n} \cdot 100}, \text{ with equality for } x = y = z = \frac{1}{3}.$$

We deduce that $x = y = z = \frac{1}{3}$ is the solution of the problem.

Remark: For $n = 3$ we obtain Problem JP.543 from RMM Nr. 37 – Summer 2025

JP.544 If $x, y \in \mathbb{R}$ then:

$$\log(1 + 3 \sin^2 x) \cdot \log(1 + 3 \cos^2 x \sin^2 y) \cdot \log(1 + 3 \cos^2 x \cos^2 y) \leq \log^3 2$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned}
 & \log(1 + 3 \sin^2 x) \cdot \log(1 + 3 \cos^2 x \sin^2 y) \cdot \log(1 + 3 \cos^2 x \cos^2 y) \leq \\
 & \stackrel{AM-GM}{\leq} \left(\frac{\log(1 + 3 \sin^2 x) + \log(1 + 3 \cos^2 x \sin^2 y) + \log(1 + 3 \cos^2 x \cos^2 y)}{3} \right)^3 = \\
 & = \frac{1}{27} \cdot \left(\log \left((1 + 3 \sin^2 x) \cdot (1 + 3 \cos^2 x \sin^2 y) \cdot (1 + 3 \cos^2 x \cos^2 y) \right) \right)^3 \leq \\
 & \stackrel{AM-GM}{\leq} \frac{1}{27} \cdot \left(\log \left(\frac{1 + 3 \sin^2 x + 1 + 3 \cos^2 x \sin^2 y + 1 + 3 \cos^2 x \cos^2 y}{3} \right)^3 \right)^3 = \\
 & = \frac{1}{27} \cdot \left(3 \log \frac{3 + 3 \sin^2 x + 3 \cos^2 x (\sin^2 y + \cos^2 y)}{3} \right)^3 = \\
 & = \frac{1}{27} \cdot 27 \cdot \left(\log \frac{3 + 3(\sin^2 x + \cos^2 x)}{3} \right)^3 = \left(\log \frac{3 + 3}{3} \right)^3 = \log^3 2
 \end{aligned}$$

Equality holds for $x = \pm \arctan \frac{1}{2}; y = \frac{\pi}{4}$

Solution 2 by Marin Chirciu-Romania

$$\begin{aligned}
 & \text{Using } \log a \cdot \log b \cdot \log c \stackrel{AM-GM}{\leq} \left(\frac{\log a + \log b + \log c}{3} \right)^3 \stackrel{(1)}{\leq} \log^3 2, \\
 & \text{for } (a, b, c) = (1 + 3 \sin^2 x, 1 + 3 \cos^2 x \sin^2 y, 1 + 3 \cos^2 x \cos^2 y) \text{ and} \\
 & (1) \Leftrightarrow \frac{\log a + \log b + \log c}{3} \leq \log 2 \Leftrightarrow \log a + \log b + \log c \leq \log 8 \Leftrightarrow \\
 & \Leftrightarrow \log(abc) \leq \log 8 \Leftrightarrow abc \leq 8, \text{ see } abc \stackrel{AM-GM}{\leq} \left(\frac{a+b+c}{3} \right)^3 \stackrel{(2)}{\leq} 8 \Leftrightarrow \\
 & \Leftrightarrow \frac{a+b+c}{3} \leq 2 \Leftrightarrow a+b+c \leq 16 \Leftrightarrow 1 + 3 \sin^2 x + 1 + 3 \cos^2 x \sin^2 y + 1 + \\
 & \quad + 3 \cos^2 x \cos^2 y \leq 6 \Leftrightarrow \\
 & \Leftrightarrow 3 \sin^2 x + 3 \cos^2 x \sin^2 y + 3 \cos^2 x \cos^2 y \leq 3 \Leftrightarrow \\
 & \Leftrightarrow \sin^2 x \cos^2 x \sin^2 y + \cos^2 x \cos^2 y \leq 1 \Leftrightarrow \\
 & \Leftrightarrow \sin^2 x + \cos^2 x (\sin^2 y + \cos^2 y) \leq 1 \Leftrightarrow \sin^2 x + \cos^2 x \cdot 1 \leq 1 \Leftrightarrow 1 \leq 1, \text{ obviously.}
 \end{aligned}$$

Equality holds if and only if:

$$\begin{aligned}
 a = b = c & \Leftrightarrow 1 + 3 \sin^2 x = 1 + 3 \cos^2 x \sin^2 y = 1 + 3 \cos^2 x \cos^2 y \Leftrightarrow \\
 & \Leftrightarrow 3 \sin^2 x = 3 \cos^2 x \sin^2 y = 3 \cos^2 x \cos^2 y \Leftrightarrow
 \end{aligned}$$

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$$\sin^2 x = \cos^2 x \sin^2 y = \cos^2 x \cos^2 y$$

From $\cos^2 x \sin^2 y = \cos^2 x \cos^2 y$ we obtain $\cos^2 x = 0$ or $\sin^2 y = \cos^2 y \Leftrightarrow$

$$\tan^2 y = 1$$

JP.545 If $x, y, z > 0$ then:

$$\frac{x}{7x+5y+5z} + \frac{y}{5x+7y+5z} + \frac{z}{5x+5y+7z} \leq \frac{3}{17}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\text{Denote: } \begin{cases} 7x+5y+5z=u \\ 5x+7y+5z=v \\ 5x+5y+7z=w \end{cases} \Rightarrow \begin{cases} x=\frac{1}{34}(12u-5v-5w) \\ y=\frac{1}{34}(-5u+12v-5w) \\ z=\frac{1}{34}(-5u-5v+12w) \end{cases}$$

$$\begin{aligned} \sum_{cyc} \frac{x}{7x+5y+5z} &= \sum_{cyc} \frac{12u-5v-5w}{34u} = \frac{1}{34} \sum_{cyc} \left(12 - 5 \left(\frac{v}{u} + \frac{w}{u} \right) \right) = \\ &= \frac{12 \cdot 3}{34} - \frac{5}{34} \sum_{cyc} \left(\frac{v}{u} + \frac{w}{v} \right) \leq \frac{36}{34} - \frac{5 \cdot 6}{34} = \frac{6}{34} = \frac{3}{17} \end{aligned}$$

Equality holds for $x = y = z = 1$.

Solution 2 by Marin Chirciu-Romania

Return the inequality: $\sum \frac{x}{7x+5y+5z} \leq \frac{3}{17} \Leftrightarrow \sum \frac{y+z}{7x+5y+5z} \geq \frac{6}{17}$, which follows from:

$$\begin{aligned} \sum \frac{y+z}{7x+5y+5z} &= \sum \frac{(y+z)^2}{(y+z)(7x+5y+5z)} \stackrel{CS}{\geq} \frac{(\sum (y+z))^2}{\sum (y+z)(7x+5y+5z)} = \\ &= \frac{4(\sum x^2 + 2\sum yz)}{10\sum x^2 + 24\sum yz} \stackrel{(1)}{\geq} \frac{6}{17}, \end{aligned}$$

$$\text{where (1)} \Leftrightarrow \frac{4(\sum x^2 + 2\sum yz)}{10\sum x^2 + 24\sum yz} \geq \frac{6}{17} \Leftrightarrow \sum x^2 \geq \sum yz \Leftrightarrow \sum (x-y)^2 \geq 0.$$

Equality holds if and only if $x = y = z$. Remark: The inequality can be developed: If

$$x, y, z > 0 \text{ and } \lambda \geq n > 0 \text{ then: } \sum \frac{x}{\lambda x + ny + nz} \leq \frac{3}{\lambda + 2n}$$

Marin Chirciu

Solution:

Return the inequality: $\sum \frac{x}{\lambda + ny + nz} \leq \frac{3}{\lambda + 2n} \Leftrightarrow \sum \frac{y+z}{\lambda x + ny + nz} \geq \frac{6}{\lambda + 2n}$, which follows from:

$$\begin{aligned} \sum \frac{y+z}{\lambda x + ny + nz} &= \sum \frac{(y+z)^2}{(y+z)(\lambda x + ny + nz)} \stackrel{cs}{\geq} \frac{(\sum (y+z))^2}{\sum (y+z)(\lambda x + ny + nz)} = \\ &= \frac{4(\sum x^2 + 2\sum yz)}{2n\sum x^2 + 2(\lambda + n)\sum yz} \stackrel{(1)}{\geq} \frac{6}{2\lambda + n} \end{aligned}$$

where (1) $\Leftrightarrow \frac{4(\sum x^2 + 2\sum yz)}{2n\sum x^2 + 2(\lambda + n)\sum yz} \geq \frac{6}{2\lambda + n} \Leftrightarrow (\lambda - n)\sum x^2 \geq (\lambda - n)\sum yz$, which follows from

the condition for hypothesis $\lambda \geq n \geq 0$ and $\sum x^2 \geq \sum yz \Leftrightarrow \sum (x - y)^2 \geq 0$

Equality holds if and only if $x = y = z$.

Note.

For $\lambda = 7, n = 5$ we obtain Problem JP.545 from RMM Nr. 37 – Summer 2025

Solution 3 by Tapas Das-India

let $x + y + z = p$, then the given problem can be written as

$$\frac{x}{2x + 5p} + \frac{y}{2y + 5p} + \frac{z}{2z + 5p} \leq \frac{3}{17}, \text{ let } f(a) = \frac{a}{2a + 5p}$$

$$\text{where } a > 0, f'(a) = \frac{5p}{(2a + 5p)^2}, f''(a) = \frac{-10p}{(2a + 5p)^3}$$

so f is concave in $a \in (0, \infty)$.

Now using Jensen $f(x) + f(y) + f(z) \leq 3f\left(\frac{x+y+z}{3}\right) = 3f\left(\frac{p}{3}\right)$ or,

$$\sum \frac{x}{2x + 5p} = \sum \frac{x}{7x + 5y + 5z} \leq 3 \frac{\frac{p}{3}}{\frac{2p}{3} + 5p} = \frac{3}{17}$$

Solution 4 by Angel Plaza-Spain

Since the expression is homogeneous, we may assume that $x + y + z = 1$, so the

inequality may be written as

$$\frac{x}{5 + 2x} + \frac{y}{5 + 2y} + \frac{z}{5 + 2z} \leq \frac{3}{17}$$

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with $0 < x, y, z < 1$. $f(x) = \frac{x}{5+2x}$ is concave for $x \in (0, 1)$ because $f''(x) = \frac{-20}{(5+2x)^3} < 0$ for

$x \in (0, 1)$. Therefore by Jensen's inequality,

$$\frac{x}{5+2x} + \frac{y}{5+2y} + \frac{z}{5+2z} \leq 3f\left(\frac{1}{3}\right) = 3 \cdot \frac{\frac{1}{3}}{5+\frac{2}{3}} = \frac{3}{17}$$

JP.546 In $\triangle ABC$ the following relationship holds:

$$\sum \frac{\sin^4 \frac{A}{2} + \sin^4 \frac{B}{2}}{\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}} \geq \frac{3}{4}$$

Proposed by Marin Chircu – Romania

Solution 1 by proposer

Lemma: If $x, y > 0$ then:

$$\sum \frac{x^4 + y^4}{x^2 + y^2} \geq \sum x^2$$

Proof.

$$\sum \frac{x^4 + y^4}{x^2 + y^2} \stackrel{CS}{\geq} \sum \frac{(x^2 + y^2)^2}{x^2 + y^2} = \frac{1}{2} \sum (x^2 + y^2) = \frac{1}{2} \cdot 2 \sum x^2 = \sum x^2$$

Let's get back to the main problem

Using Lemma for $(x, y) = \left(\sin \frac{A}{2}, \sin \frac{B}{2}\right)$ we obtain:

$$\sum \frac{\sin^4 \frac{A}{2} + \sin^4 \frac{B}{2}}{\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}} \stackrel{\text{Lemma}}{\geq} \sum \sin^2 \frac{A}{2} \geq \frac{3}{4}, \text{ see}$$

$$\sum \sin^2 \frac{A}{2} = 1 - \frac{r}{2R} \stackrel{\text{Euler}}{\geq} \frac{3}{4}$$

Equality holds if and only if the triangle is equilateral.

Solution 2 by Tapas Das-India

$$\sum \frac{\sin^4 \frac{A}{2} + \sin^4 \frac{B}{2}}{\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}} \geq \sum \frac{\left(\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}\right)^2}{2\left(\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}\right)} =$$

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$$\sum (\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2}) \frac{1}{2} = \sum \sin^2 \frac{A}{2} = 1 - \frac{r}{2R} \stackrel{\text{Euler}}{\geq} 1 - \frac{1}{4} \geq \frac{3}{4}$$

Equality holds if and only if the triangle is equilateral.

JP.547 In $\triangle ABC$:

$$\sum \frac{\cos^4 \frac{A}{2} + \cos^4 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} \geq \frac{9r}{2R}$$

Proposed by Marin Chirciu – Romania

Solution 1 by proposer

Lemma: If $x, y > 0$ then:

$$\sum \frac{x^4 + y^4}{x^2 + y^2} \geq \sum x^2$$

Proof.

$$\sum \frac{x^4 + y^4}{x^2 + y^2} \stackrel{CS}{\geq} \sum \frac{(x^2 + y^2)^2}{x^2 + y^2} = \frac{1}{2} \sum (x^2 + y^2) = \frac{1}{2} \cdot 2 \sum x^2 = \sum x^2$$

Let's get back to the main problem.

Using Lemma for $(x, y) = (\cos \frac{A}{2}, \cos \frac{B}{2})$ we obtain:

$$\sum \frac{\cos^4 \frac{A}{2} + \cos^4 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} \stackrel{\text{Lemma}}{\geq} \sum \cos^2 \frac{A}{2} \geq \frac{3}{4}, \text{ see}$$

$$\sum \cos^2 \frac{A}{2} = 2 + \frac{r}{2R} = \frac{4R + r}{2R} \stackrel{\text{Euler}}{\geq} \frac{9r}{2R}$$

Equality holds if and only if the triangle is equilateral.

Solution 2 by Tapas Das-India

$$\begin{aligned} \sum \frac{\cos^4 \frac{A}{2} + \cos^4 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} &\stackrel{CBS}{\geq} \sum \frac{(\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2})^2}{2(\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2})} = \\ &= \sum \cos^2 \frac{A}{2} = \left(2 + \frac{r}{2R}\right) = \frac{4R + r}{2R} \stackrel{\text{Euler}}{\geq} \frac{9r}{2R} \end{aligned}$$

Equality holds if and only if the triangle is equilateral.

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Solution 3 by Daniel Sitaru-Romania

$$\begin{aligned}
 \sum \frac{\cos^4 \frac{A}{2} + \cos^4 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} &= \sum \frac{\cos^4 \frac{A}{2} + \cos^4 \frac{B}{2}}{\cos^3 \frac{A}{2} + \cos^3 \frac{B}{2}} \cdot \frac{\cos^3 \frac{A}{2} + \cos^3 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} \stackrel{\text{LEHMER}}{\geq} \\
 &\geq \sum_{\text{cyc}} \left(\frac{\cos^3 \frac{A}{2} + \cos^3 \frac{B}{2}}{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}} \right)^2 \stackrel{\text{LEHMER}}{\geq} \sum_{\text{cyc}} \left(\frac{\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2}}{\cos \frac{A}{2} + \cos \frac{B}{2}} \right)^2 \stackrel{\text{LEHMER}}{\geq} \\
 &\geq \sum_{\text{cyc}} \left(\frac{\cos \frac{A}{2} + \cos \frac{B}{2}}{1 + 1} \right)^2 \stackrel{\text{AM-GM}}{\geq} \sum_{\text{cyc}} \cos \frac{A}{2} \cos \frac{B}{2} \stackrel{\text{AM-GM}}{\geq} \\
 &\geq 3 \cdot \sqrt[3]{\left(\prod_{\text{cyc}} \cos \frac{A}{2} \right)^2} = 3 \sqrt[3]{\left(\frac{s}{4R} \right)^2} = \frac{3}{2R} \sqrt[3]{\frac{1}{2} \cdot Rs^2} \stackrel{\text{EULER}}{\geq} \\
 &\geq \frac{3}{2R} \sqrt[3]{rs^2} \stackrel{\text{MITRINOVIC}}{\geq} \frac{3}{2R} \sqrt[3]{r \cdot 27r^2} = \frac{9r}{2R}
 \end{aligned}$$

Equality holds for $a = b = c$.

JP.548 If $x, y, z \in [0, \infty)$ solve the system:

$$\begin{cases} x^2 = y(y + z) \\ y^2 = z(z + x) \end{cases}$$

Proposed by Cristian Miu – Romania

Solution by proposer

If $x = 0$ or $y = 0$, or $z = 0$ it is easy to see that $x = y = z = 0$. How we can assume that

$$x > 0, y > 0 \text{ and } z > 0.$$

From $x^2 = y(y + z)$ we obtain that $x > y$ and from $y^2 = z(z + x)$ we obtain that $y > z$.

$$\text{So } x > y > z.$$

But $x < y + z$ because $x < y + z \Leftrightarrow x^2 < y^2 + z^2 + 2yz \Leftrightarrow y^2 + yz < y^2 + z^2 + 2yz \Leftrightarrow 0 < z^2 + yz$. So x, y, z are the sides of a triangle XYZ .

Let $YZ = x, XZ = y$ and $XY = z$.

$$x^2 = y^2 + yz \Leftrightarrow \sin^2 X = \sin^2 Y + \sin Y \sin Z \Leftrightarrow$$

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$$\frac{1 - \cos 2X}{2} = \frac{1 - \cos 2Y}{2} + \sin Y \sin Z \Leftrightarrow \cos 2Y - \cos 2X = 2 \sin Y \sin Z \Leftrightarrow$$

$$\Leftrightarrow 2 \sin(y + X) \sin(X - Y) = 2 \sin Y \sin Z \Leftrightarrow \sin(X - Y) \sin Y \Leftrightarrow$$

$$\Leftrightarrow 2 \sin \frac{X - 2Y}{2} \cos \frac{X}{2} = 0 \Leftrightarrow \sin \left(\frac{X - 2Y}{2} \right) = 0 \Leftrightarrow X = 2Y$$

$$\text{Because } -\frac{\pi}{2} < \frac{X - 2Y}{2} < \frac{\pi}{2}$$

In the same way from

$$\sin^2 Y = \sin^2 Z + \sin Z \sin X \text{ we obtain } Y = 2Z$$

So $X = 2Y, Y = 2Z$ and $X + Y + Z = \pi$. We obtain

$$X = \frac{4\pi}{7}, Y = \frac{2\pi}{7}, Z = \frac{\pi}{7}$$

$$\text{So } \frac{x}{\sin \frac{4\pi}{7}} = \frac{y}{\sin \frac{2\pi}{7}} = \frac{z}{\sin \frac{\pi}{7}} \text{ and from here}$$

$$x = k \sin \frac{4\pi}{7}, y = k \sin \frac{2\pi}{7}, z = k \sin \frac{\pi}{7}$$

JP.549 Let ABC be a triangle with inradius r and circumradius R . Prove that:

$$\sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} \leq \left(\frac{R}{r} \right)^2$$

The sum is over all cyclic permutations of (A, B, C) .

Proposed by George Apostolopoulos –Greece

Solution 1 by proposer

We will prove that $\frac{a^3 + b^3}{a^5 + b^5} \leq \frac{1}{ab}$. We have

$$a^5 + b^5 - ab(a^3 + b^3) = a^5 + b^5 - a^4b - ab^4 = (a - b)^2(a + b)(a^2 + b^2) \geq 0$$

Similarly $\frac{b^3 + c^3}{b^5 + c^5} \leq \frac{1}{bc}$, and $\frac{c^3 + a^3}{c^5 + a^5} \leq \frac{1}{ca}$, where a, b, c be the lengths of the sides of the ΔABC .

Adding up the (3) inequalities, we get:

$$\sum_{cyc} \frac{a^3 + b^3}{a^5 + b^5} \leq \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}$$

It is well-known the inequality $x^2 + y^2 + z^2 \geq xy + yz + zx$ for $x, y, z \in \mathbb{R}$. So,

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$$\sum_{cyc} \frac{a^3+b^3}{a^5+b^5} \leq \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad (*)$$

Now, will prove that $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \leq \frac{1}{4r^2}$. We have

$$(b-c)^2 \geq 0 \Leftrightarrow a^2 - (b-c)^2 \leq a^2 \Leftrightarrow \frac{1}{a^2} \leq \frac{1}{a^2 - (b-c)^2} = \frac{1}{(a+b-c)(a-b+c)}$$

Let $2s = a + b + c$, then $\frac{1}{a^2} \leq \frac{1}{4(s-c)(s-b)}$. Similarly

$$\frac{1}{b^2} \leq \frac{1}{4(s-a)(s-c)}, \text{ and } \frac{1}{c^2} \leq \frac{1}{4(s-b)(s-a)}.$$

$$\text{So } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \leq \frac{1}{4} \left(\frac{1}{(s-c)(s-b)} + \frac{1}{(s-a)(s-c)} + \frac{1}{(s-b)(s-a)} \right) =$$

$$= \frac{1}{4} \cdot \frac{s-a+s-b+s-c}{(s-a)(s-b)(s-c)} = \frac{1}{4} \cdot \frac{3s-2s}{(s-a)(s-b)(s-c)} = \frac{1}{4} \cdot \frac{s^2}{s(s-a)(s-b)(s-c)}$$

We know (Heron) that $F = r \cdot s = \sqrt{s(s-a)(s-b)(s-c)}$ where F is the area of $\triangle ABC$.

$$\text{So, } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \leq \frac{s^2}{4F^2} = \frac{s^2}{4r^2s^2} = \frac{1}{4r^2}. \text{ Namely } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \leq \frac{1}{4r^2}.$$

Using the law of Sines, we get from (*)

$$\sum_{cyc} \frac{(2R \sin A)^3 + (2R \sin B)^3}{(2R \sin A)^5 + (2R \sin B)^5} \leq \frac{1}{4r^2}$$

or

$$\sum_{cyc} \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} \leq \left(\frac{R}{r} \right)^2$$

Equality holds if and only if the triangle ABC is equilateral.

Solution 2 by Marin Chirciu-Romania

Using sine theorem we obtain:

$$\begin{aligned} LHS &= \sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} = \sum \frac{\left(\frac{a}{2R}\right)^3 + \left(\frac{b}{2R}\right)^3}{\left(\frac{a}{2R}\right)^5 + \left(\frac{b}{2R}\right)^5} = \\ &= 4R^2 \sum \frac{a^3 + b^3}{a^5 + b^5} \stackrel{(1)}{\leq} 4R^2 \sum \frac{1}{ab} = 4R^2 \cdot \frac{1}{2Rr} = \frac{2R}{r} \stackrel{\text{Euler}}{\leq} \left(\frac{R}{r} \right)^2 = RHS, \end{aligned}$$

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where (1) $\Leftrightarrow \frac{a^3+b^3}{a^5+b^5} \leq \frac{1}{ab} \Leftrightarrow a^5 + b^5 \geq ab(a^3 + b^3) \Leftrightarrow (a-b)(a^4 - b^4) \geq 0$, because

the factors have the same sign.

Equality holds if and only if the triangle is equilateral.

Remark: The inequality can be strengthened.

In $\triangle ABC$:

$$\sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} \leq \frac{2R}{r}$$

Using the sine theorem we obtain:

$$\begin{aligned} LHS &= \sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} = \sum \frac{\left(\frac{a}{2R}\right)^3 + \left(\frac{b}{2R}\right)^3}{\left(\frac{a}{2R}\right)^5 + \left(\frac{b}{2R}\right)^5} = 4R^2 \sum \frac{a^3 + b^3}{a^5 + b^5} \stackrel{(1)}{\leq} 4R^2 \sum \frac{1}{ab} = \\ &= 4R^2 \cdot \frac{1}{2Rr} = \frac{2R}{r} \stackrel{Euler}{\leq} \left(\frac{R}{r}\right)^2 = RHS, \end{aligned}$$

where (1) $\Leftrightarrow \frac{a^3+b^3}{a^5+b^5} \leq \frac{1}{ab} \Leftrightarrow a^5 + b^5 \geq ab(a^3 + b^3) \Leftrightarrow (a-b)(a^4 - b^4) \geq 0$, because

the factors have the same sign.

Equality holds if and only if the triangle is equilateral.

Remark: We can write the inequalities:

In $\triangle ABC$:

$$\sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} \leq \frac{2R}{r} \leq \left(\frac{R}{r}\right)^2.$$

Using the sine theorem we obtain:

$$\begin{aligned} LHS &= \sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} = \sum \frac{\left(\frac{a}{2R}\right)^3 + \left(\frac{b}{2R}\right)^3}{\left(\frac{a}{2R}\right)^5 + \left(\frac{b}{2R}\right)^5} = 4R^2 \sum \frac{a^3 + b^3}{a^5 + b^5} \stackrel{(1)}{\leq} 4R^2 \sum \frac{1}{ab} = \\ &= 4R^2 \cdot \frac{1}{2Rr} = \frac{2R}{r} \stackrel{Euler}{\leq} \left(\frac{R}{r}\right)^2 = RHS, \end{aligned}$$

where (1) $\Leftrightarrow \frac{a^3+b^3}{a^5+b^5} \leq \frac{1}{ab} \Leftrightarrow a^5 + b^5 \geq ab(a^3 + b^3) \Leftrightarrow (a-b)(a^4 - b^4) \geq 0$, because

the factors have the same sign. Equality hold if and only if the triangle is equilateral.

Remark: The problem can be developed and strengthened.

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If $n \in \mathbb{N}$ then in ΔABC holds:

$$\sum \frac{\sin^n A + \sin^n B}{\sin^{n+2} A + \sin^{n+2} B} \leq \frac{2R}{r}$$

Marin Chirciu

Using the sine theorem:

$$\begin{aligned} LHS &= \sum \frac{\sin^n A + \sin^n B}{\sin^{n+2} A + \sin^{n+2} B} = \sum \frac{\left(\frac{a}{2R}\right)^n + \left(\frac{b}{2R}\right)^n}{\left(\frac{a}{2R}\right)^{n+2} + \left(\frac{b}{2R}\right)^{n+2}} = 4R^2 \sum \frac{a^n + b^n}{a^{n+2} + b^{n+2}} \stackrel{(1)}{\leq} \\ &\leq 4R^2 \sum \frac{1}{ab} = 4R^2 \cdot \frac{1}{2Rr} = \frac{2R}{r} = RHS, \end{aligned}$$

where (1)

$$\Leftrightarrow \frac{a^n + b^n}{a^{n+2} + b^{n+2}} \leq \frac{1}{ab} \Leftrightarrow a^{n+2} + b^{n+2} \geq ab(a^n + b^n) \Leftrightarrow (a-b)(a^{n+1} - b^{n+1}) \geq 0$$

because the factors have the same sign.

Equality holds if and only if the triangle is equilateral.

Solution 3 by Tapas Das-India

$$\begin{aligned} \sum \frac{\sin^3 A + \sin^3 B}{\sin^5 A + \sin^5 B} &\stackrel{CEBYSHEV}{\leq} \sum \frac{\sin^3 A + \sin^3 B}{\frac{1}{2}(\sin^3 A + \sin^3 B)(\sin^2 A + \sin^2 B)} = \\ &= \sum \frac{2}{\sin^2 A + \sin^2 B} = 8R^2 \sum \frac{1}{a^2 + b^2} \stackrel{AM-HM}{\leq} 2R^2 \sum \left(\frac{1}{a^2} + \frac{1}{b^2}\right) = \\ &= 4R^2 \sum \frac{1}{a^2} \stackrel{STEINING}{\leq} \frac{4R^2}{4r^2} = \left(\frac{R}{r}\right)^2 \end{aligned}$$

Equality holds if and only if the triangle is equilateral.

JP.550 The non-coplanar points are given: A, B, C and D . If K is the middle of

the segment $[BD]$, ($KM = \text{bisector } \widehat{AKB}, M \in (AB)$),

($KP = \text{bisector } \widehat{AKD}, P \in (AD)$), and $N \in (AC)$, such that $\frac{AC}{AN} - \frac{BD}{2AK} = 1$ then

prove that: $AN \cdot NC + AM \cdot MB \geq 2PD \cdot (AN + AM - AP)$

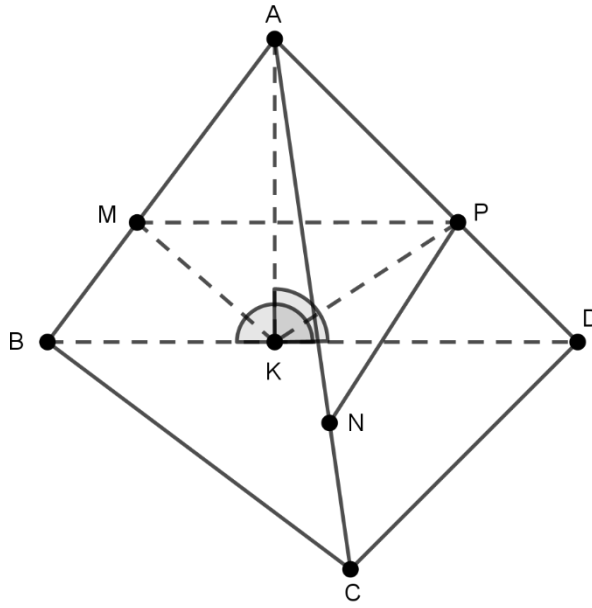
Proposed by Gheorghe Molea – Romania

Solution 1 by proposer

$$(KM = \text{bisector } \widehat{AKB} \Rightarrow \frac{BK}{AK} = \frac{BM}{AM}$$

$$(KP = \text{bisector } \widehat{AKD} \Rightarrow \frac{KD}{KA} = \frac{PD}{PA}$$

$$\text{But } [BK] \equiv [KD] \Rightarrow \frac{BM}{AM} = \frac{PD}{PA} \stackrel{RTT}{\Rightarrow} MP \parallel BD$$



$$\frac{AC}{AN} - \frac{BD}{2AK} = 1 \Leftrightarrow \frac{AC}{AN} - \frac{KD}{AK} = 1 \Leftrightarrow$$

$$\frac{AC}{AN} = 1 + \frac{KD}{AK} \Leftrightarrow \frac{AC}{AN} = 1 + \frac{PD}{PA} \Leftrightarrow \frac{AC}{AN} = \frac{AD}{PA} \stackrel{RTT}{\Rightarrow} PN \parallel CD$$

$$\text{From } MP \parallel BD \Rightarrow \frac{AP}{PD} = \frac{AM}{MB} \Leftrightarrow AP \cdot MB = AM \cdot PD$$

$$\text{Using means inequality } m_a \geq m_g \Rightarrow \frac{AP}{AM} + \frac{MB}{PD} \geq 2\sqrt{\frac{AP \cdot MB}{AM \cdot PD}} = 2 \Rightarrow$$

$$\frac{AP}{AM} + \frac{MB}{PD} \geq 2 \Rightarrow AP \cdot PD + AM \cdot MB \geq 2AM \cdot PD \quad (*)$$

$$\text{From } PN \parallel DC \Rightarrow \frac{AP}{PD} = \frac{AN}{NC} \Leftrightarrow AP \cdot NC = PD \cdot AN$$

$$\text{From means inequality } m_a \geq m_g \text{ it follows: } \frac{AP}{AN} + \frac{NC}{PD} \geq 2\sqrt{\frac{AP \cdot NC}{AN \cdot PD}} = 2 \Rightarrow$$

$$\frac{AP}{AN} + \frac{NC}{PD} \geq 2 \Rightarrow AP \cdot PD + AN \cdot NC \geq 2AN \cdot PD \quad (**)$$

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Adding (*) and (**) we obtain:

$$2AP \cdot PD + AN \cdot NC + AM \cdot MB \geq 2PD(AN + AM)$$

$$\Leftrightarrow AN \cdot NC + AM \cdot MB \geq 2PD(AN + AM + AP)$$

We have equality $\Leftrightarrow [AB] \equiv [AC] \equiv [AD]$

Solution 2 by Marin Chirciu-Romania

Because K is the middle of $[BD] \Rightarrow \frac{BD}{2} = KB = KD \Rightarrow \frac{AC}{AN} - \frac{KB}{AK} = 1$ and $\frac{AC}{AN} - \frac{KD}{AK} = 1$.

Using bisector's theorem in $\triangle AKB$ and $\triangle AKD \Rightarrow \frac{AM}{MB} = \frac{AK}{KB}$ and $\frac{AP}{PD} = \frac{AK}{KD}$.

$$\begin{aligned} \text{From } \frac{AC}{AN} - \frac{KB}{AK} = 1 &\Leftrightarrow \frac{AN+NC}{AN} - \frac{KB}{AK} = 1 \Leftrightarrow 1 + \frac{NC}{AN} - \frac{KB}{AK} = 1 \Leftrightarrow \frac{NC}{AN} = \frac{KB}{AK} \Rightarrow \\ &\Rightarrow \frac{NC}{AN} = \frac{KB}{AK} \xrightarrow{T-bis} \frac{MB}{AM} \Rightarrow \frac{NC}{AN} = \frac{MB}{AM} \xrightarrow{R-Thales} MN \parallel BC \quad (1) \end{aligned}$$

Analogous:

$$\begin{aligned} \text{From } \frac{AC}{AN} - \frac{KD}{AK} = 1 &\Leftrightarrow \frac{AN+NC}{AN} - \frac{KD}{AK} = 1 \Leftrightarrow 1 + \frac{NC}{AN} - \frac{KD}{AK} = 1 \Leftrightarrow \frac{NC}{AN} = \frac{KD}{AK} \Rightarrow \\ &\Rightarrow \frac{NC}{AN} = \frac{KD}{AK} \xrightarrow{T-bis} \frac{PD}{AP} \Rightarrow \frac{NC}{AN} = \frac{PD}{AP} \xrightarrow{R-Thales} PN \parallel DC \quad (2). \end{aligned}$$

$$\text{From } \frac{NC}{AN} = \frac{MB}{AM} \text{ and } \frac{NC}{AN} = \frac{PD}{AP} \Rightarrow \frac{MB}{AM} = \frac{PD}{AP} \xrightarrow{R-Thals} PM \parallel BD \quad (3).$$

From (1), (2) and (3) $\Rightarrow \triangle MNP \simeq \triangle BCD$.

Let's get to the main problem.

Dividing both members by $AM \cdot MB$ we obtain:

$$\begin{aligned} \frac{AN \cdot NC + AM \cdot MB}{AM \cdot MB} &\geq \frac{2PD \cdot (AN + AM - AP)}{AM \cdot MB} \Leftrightarrow \\ \frac{AN}{AM} \cdot \frac{NC}{MB} + 1 &\geq \frac{2PD}{MB} \left(\frac{AN}{AM} + 1 - \frac{AP}{AM} \right) \end{aligned}$$

$$\text{Using } \triangle MNP \simeq \triangle BCD \Rightarrow \frac{AN}{AM} = \frac{AC}{AB}, \frac{NC}{MB} = \frac{AC}{AB}, \frac{PD}{MB} = \frac{AD}{AB} \text{ and } \frac{AP}{AM} = \frac{AD}{AB}$$

The above equalities substituted into the required inequality lead to:

$$\begin{aligned} \frac{AC}{AB} \cdot \frac{AC}{AB} + 1 &\geq \frac{2AD}{AB} \left(\frac{AC}{AB} + 1 - \frac{AD}{AB} \right) \Leftrightarrow AB^2 + AC^2 \geq 2AD(AB + AC - AD) \Leftrightarrow \\ &\Leftrightarrow AB^2 + AC^2 + 2AD^2 \geq 2AD(AB + AC) \Leftrightarrow \end{aligned}$$

$$\Leftrightarrow (AB^2 + AD^2) + (AC^2 + AD^2) \geq 2AD(AB + AC) \Leftrightarrow (AB - AD)^2 + (AC - AD)^2 \geq 0,$$

with equality for $AB = AC = AD$.

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JP.551 Find the real numbers x, y, z knowing that they meet the conditions:

$$x + y + z = 1; xy + (x + y)(z + 1) = \frac{4}{3}$$

Proposed by Gheorghe Molea – Romania

Solution 1 by proposer

From $x + y + z = 1$ it result successively:

$$(x + y + z)^2 = 1 \Leftrightarrow x^2 + y^2 + z^2 + 2(xy + yz + zx) = 1 \Leftrightarrow$$

$$x^2 + y^2 + z^2 + 2\left(\frac{4}{3} - x - y\right) = 1 \text{ (we have used also the second condition for the}$$

$$\text{hypothesis)} \Leftrightarrow x^2 + y^2 + z^2 - 2x - 2y = -\frac{5}{3}$$

$$\Leftrightarrow (x - 1)^2 + (y - 1)^2 + z^2 = \frac{1}{3} \Leftrightarrow \frac{(x-1)^2 + (y-1)^2 + z^2}{3} = \frac{1}{9} \quad (*)$$

$$\text{From } x + y + z = 1 \Rightarrow (x - 1) + (y - 1) + z = -1 \quad (**)$$

$$\text{It is obviously that: } (\alpha - \beta)^2 + (\alpha - \gamma)^2 + (\beta - \gamma)^2 \geq 0$$

$(\forall)\alpha, \beta, \gamma$, with equality only if $\alpha = \beta = \gamma$, which becomes:

$$3(\alpha^2 + \beta^2 + \gamma^2) \geq (\alpha + \beta + \gamma)^2$$

If in this last inequality, we make the substitutions:

$\alpha = x - 1, \beta = y - 1, \gamma = z$ and we consider the relationships (*) and (**) we obtain:

$$1 = 9 \cdot \frac{1}{9} = 3[(x - 1)^2 + (y - 1)^2 + z^2] \geq [(x - 1) + (y - 1) + z]^2 = (-1)^2 = 1, \text{ from}$$

$$x - 1 = y - 1 = z \stackrel{\text{notation}}{=} t \Rightarrow$$

$x = t + 1, y = t + 1, z = t$, and from the first relationship of the hypothesis it follows:

$$t = -\frac{1}{3}, \text{ so } x = \frac{2}{3}, y = \frac{2}{3}, z = -\frac{1}{3}$$

Solution 2 by Marin Chirciu-Romania

$$\text{Denoting } x + y = S \text{ and } xy = P \text{ we have } \begin{cases} S = 1 - z \\ P + S(z + 1) = \frac{4}{3} \end{cases} \Leftrightarrow \begin{cases} S = 1 - z \\ P = \frac{1}{3} + z^2 \end{cases}$$

We form the equation of the second degree knowing the sum $S = 1 - z$ and the product

$$P = \frac{1}{3} + z^2$$

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We obtain $t^2 - St + P = 0 \Leftrightarrow t^2 - (1 - z)t + \frac{1}{3} + z^2 = 0 \Leftrightarrow$

$$\Leftrightarrow 3t^2 - 3(1 - z)t + 3z^2 + 1 = 0, \text{ with}$$

$$\Delta = 9(z - 1)^2 - 12(3z^2 + 1) = -27t^2 - 18z - 3 = -3(3z + 1)^2.$$

Because the solutions x, y of the second-degree equation must be real we must meet the

$$\text{condition } \Delta = -3(3z + 1)^2 \geq 0 \Leftrightarrow z = \frac{-1}{3}$$

Putting $z = \frac{-1}{3}$ in the equation $3t^2 - 3(1 - z)t + 3z^2 + 1 = 0$ we obtain

$$9t^2 - 12t + 4 = 0 \Leftrightarrow (3t - 2)^2 = 0 \Leftrightarrow t = \frac{2}{3}, \text{ wherefrom } x = y = \frac{2}{3}.$$

We deduce that $(x, y, z) = \left(\frac{2}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ is the unique real solution of the system.

Remark: The problem can be developed.

Let $\lambda > 0$ fixed. Solve in real numbers x, y, z the system:

$$\begin{cases} x + y + z = \lambda \\ xy + (x + y)(z + \lambda) = \frac{4\lambda^2}{3} \end{cases}$$

Marin Chirciu

Solution

Denoting $x + y = S$ and $xy = P$ we have $\begin{cases} S = \lambda - z \\ P + S(z + \lambda) = \frac{4\lambda^2}{3} \end{cases} \Leftrightarrow \begin{cases} S = \lambda - z \\ P = \frac{\lambda^2}{3} + z^2 \end{cases}$

Forming the equation of second degree knowing the sum $S = \lambda - z$ and the product

$$P = \frac{\lambda^2}{3} + z^2.$$

We obtain $t^2 - St + P = 0 \Leftrightarrow t^2 - (\lambda - z)t + \frac{\lambda^2}{3} + z^2 = 0 \Leftrightarrow$

$$\Leftrightarrow 3t^2 - 3(\lambda - z)t + 3z^2 + \lambda^2 = 0 \text{ with}$$

$$\Delta = 9(z - \lambda)^2 - 12(3z^2 + \lambda^2) = -27t^2 - 18\lambda z - 3\lambda^2 = -3(3z + \lambda)^2$$

Because the solutions x, y of the second-degree equation must be real we put the condition

$$\Delta = -3(3z + \lambda)^2 \geq 0 \Leftrightarrow z = \frac{-\lambda}{3}$$

Putting $z = \frac{-\lambda}{3}$ in equation $3t^2 - 3(\lambda - z)t + 3z^2 + \lambda^2 = 0$ we obtain

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$$9t^2 - 12\lambda t + 4\lambda^2 = 0 \Leftrightarrow (3t - 2\lambda)^2 = 0 \Leftrightarrow t = \frac{2\lambda}{3}, \text{ wherefrom } x = y = \frac{2\lambda}{3}$$

We deduce that $(x, y, z) = \left(\frac{2\lambda}{3}, \frac{2\lambda}{3}, -\frac{\lambda}{3}\right)$ is the unique real solution of the system.

Note: For $\lambda = 1$ we obtain Problem JP.551 from RMM Nr.37 – Summer 2025.

JP.552 Justify if there are non-zero natural numbers a, b, c, d , different two by two, so that we have:

$$a(b + c - a) = b(a + c - b) = c(a + b - c) = \frac{a + b + c}{d}$$

Proposed by Gheorghe Molea – Romania

Solution 1 by proposer

$$\begin{aligned} \text{We have } \frac{a+b+c}{d} &= a(b + c - a) = b(a + c - b) = c(a + b - c) = \\ &= \frac{b + c - a}{\frac{1}{a}} = \frac{a + c - b}{\frac{1}{b}} = \frac{a + b - c}{\frac{1}{c}} = \frac{b + c - a + a + c - b + a + b - c}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \\ &= \frac{a + b + c}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \Rightarrow \frac{a + b + c}{d} = \frac{a + b + c}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = d \end{aligned}$$

where $a, b, c, d \in \mathbb{N}^*$, different two by two.

We have: $d = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} < \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{11}{6} < 2$ but as $d > 0$, it follows $d = 1$. Let be

$a < b < c$ but

$$1 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} < \frac{1}{a} + \frac{1}{a} + \frac{1}{a} = \frac{3}{a} \text{ wherefrom } a < 3, \text{ but as } a > 1 \Rightarrow a = 2$$

We have: $1 = \frac{1}{2} + \frac{1}{b} + \frac{1}{c}$, where $b, c \in \mathbb{N}^*$, $b < c$, wherefrom

$$\frac{1}{b} + \frac{1}{c} = \frac{1}{2} \Leftrightarrow bc - 2b - 2c = 0 \Leftrightarrow$$

$$\Leftrightarrow bc - 2b - 2c + 4 = 4 \Leftrightarrow b(c - 2) - 2(c - 2) = 4 \Leftrightarrow (b - 2)(c - 2) = 4$$

As $b < c \Rightarrow b - 2 = 1$ and $c - 2 = 4$, wherefrom $b = 3, c = 6$

We have obtained the values: $a = 2, b = 3, c = 6, d = 1$ which substituted in the relationship in the statement lead to: $14 = 15 = -6 = 11$ (F), do $(\nexists) a, b, c, d$.

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Solution 2 by Marin Chirciu-Romania

We have $a(b + c - a) = b(a + c - b) = c(a + b - c) = \frac{b+c-a}{\frac{1}{a}} = \frac{a+c-b}{\frac{1}{b}} = \frac{a+b-c}{\frac{1}{c}} =$

$$= \frac{(a+b-c) + (a+c-b) + (a+b-c)}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \frac{a+b+c}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \frac{a+b+c}{d}, \text{ if } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = d,$$

see $(a, b, c, d) = (2, 3, 6, 1)$, because $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$.

Remark: The problem can be developed.

Justify if exists non-zero natural numbers a, b, c, d, e different in pairs, such that we have:

$$a(b + c + d - a) = b(a + c + d - b) = c(a + b + d - c) =$$

$$= d(a + b + c - d) = \frac{2(a + b + c + d)}{e}$$

Marin Chirciu

Solution: We have

$$a(b + c + d - a) = b(a + c + d - b) = c(a + b + d - c) = d(a + b + c - d) =$$

$$= \frac{b + c + d - a}{\frac{1}{a}} = \frac{a + c + d - b}{\frac{1}{b}} = \frac{a + b + d - c}{\frac{1}{c}} = \frac{a + b + c - d}{\frac{1}{d}} =$$

$$= \frac{(b + c + d - a) + (a + c + d - b) + (a + b + d - c) + (a + b + c - d)}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} =$$

$$= \frac{2(a+b+c+d)}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} = \frac{2(a+b+c+d)}{e}, \text{ if } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} = e,$$

See $(a, b, c, d, e) = (2, 4, 6, 12, 1)$, because $\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{12} = 1$

JP.553 Prove that for any $a, b, c \in \mathbb{R}$, we have the inequality:

$$\frac{a^3 + a}{a^4 + a^2 + 1} + \frac{b^3 + b}{b^4 + b^2 + 1} + \frac{c^3 + c}{c^4 + c^2 + 1} \leq 2$$

Proposed by Laura Molea and Gheorghe Molea – Romania

Solution 1 by proposers

If $a = b = c = 0 \Rightarrow 0 + 0 + 0 \leq 2$ (True)

We suppose $a, b, c \neq 0$ and we will prove that $\frac{x^3+x}{x^4+x^2+1} \leq \frac{2}{3}, (\forall)x \in \mathbb{R}$

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We obtain: $2x^4 + 2x^2 + 2 - 3x^3 - 3x \geq 0 \Leftrightarrow$

$$2x^4 + x^3 + 2x^2 - 4x^3 - 2x^2 - 4x + 2x^2 + x + 2 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow x^2(2x^2 + x + 2) - 2x(2x^2 + x + 2) + (2x^2 + x + 2) \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (2x^2 + x + 2)(x - 1)^2 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (x - 1)^2 \left(x^2 + \frac{1}{2}x + 1 \right) \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (x - 1)^2 \cdot \left(x^2 + \frac{1}{2}x + \frac{1}{16} + \frac{15}{16} \right) \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (x - 1)^2 \cdot \left[\left(x + \frac{1}{4} \right)^2 + \frac{15}{16} \right] \geq 0, \text{ true, } (\forall)x \in \mathbb{R} \text{ with equality if } x = 1.$$

We have $\frac{a^3+a}{a^4+a^2+1} + \frac{b^3+b}{b^4+b^2+1} + \frac{c^3+c}{c^4+c^2+1} \leq \frac{2}{3} \cdot 3 = 2$, with equality $\Leftrightarrow a = b = c = 1$.

Solution 2 by Marin Chirciu-Romania

$$LHS = \sum \frac{a^3 + a}{a^4 + a^2 + 1} \stackrel{(1)}{\leq} \sum \frac{2}{3} = 2 = RHS,$$

$$\text{where (1)} \Leftrightarrow \frac{a^3+a}{a^4+a^2+1} \leq \frac{2}{3} \Leftrightarrow 2a^4 - 3a^3 + 2a^2 - 3a + 2 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (a - 1)^2(2a^2 + a + 2) \geq 0 \text{ with equality for } a = 1.$$

Equality holds if and only if $a = b = c = 1$.

Remark: The inequality can be developed.

If $a, b, c \in \mathbb{R}$ and $-2 < \lambda \leq 2$ then

$$\sum \frac{a^3 + a}{a^4 + \lambda a^2 + 1} \leq \frac{6}{\lambda + 2}$$

Marin Chirciu

Solution:

$$LHS = \sum \frac{a^3 + a}{a^4 + \lambda a^2 + 1} \stackrel{(1)}{\leq} \sum \frac{2}{\lambda + 2} = \frac{6}{\lambda + 2} = RHS,$$

$$\text{Where (1)} \Leftrightarrow \frac{a^3+a}{a^4+\lambda a^2+1} \leq \frac{2}{\lambda+2} \Leftrightarrow 2a^4 - (\lambda+2)a^3 + 2\lambda a^2 - (\lambda+2)a + 2 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (a - 1)^2(2a^2 + (2 - \lambda)a + 2) \geq 0, \text{ with equality for } a = 1, \text{ see } -2 \leq \lambda \leq 2.$$

Equality holds if and only if $a = b = c = 1$.

Note: For $\lambda = 1$ we obtain Problem JP.553 from RMM Nr.37 – Summer 2025

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Solution 3 by Angel Plaza-Spain

The result follows because $\frac{a^3+a}{a^4+a^2+1} \leq \frac{2}{3}$, which it is equivalent to

$$2(a^4 + a^2 + 1) - 3(a^3 + a) \geq 0. \quad 2(a^4 + a^2 + 1) - 3(a^3 + a) = \\ = (a - 1)^2(2a^2 + a + 2) \geq 0$$

JP.554 Solve the equation for integers:

$$x(x-1)^2 + y(y-1)^2 = x(3x+7y)$$

Proposed by Laura Molea and Gheorghe Molea – Romania

Solutions 1 by proposers

The equation becomes:

$$x^3 + y^3 - 5x^2 - 2y^2 + x + y - 7xy = 0 \Leftrightarrow \\ (x+y)(x^2 + y^2 - xy) + (x+y) - 5x(x+y) - 2y(x+y) = 0 \Leftrightarrow \\ \Leftrightarrow (x+y)(x^2 + y^2 - xy + 1 - 5x - 2y) = 0$$

We have the situations:

$$1. x + y = 0 \Rightarrow x = k, y = -k, \text{ where } k \in \mathbb{Z} \text{ and}$$

$$2. x^2 + y^2 - xy + 1 - 5x - 2y = 0 \Leftrightarrow x^2 - 6x - xy + x + y^2 - 2y + 1 = 0 \Leftrightarrow \\ \Leftrightarrow x(x-6) - x(y-1) + (y-1)^2 = 0$$

$$\text{We denote } z = y - 1 \Rightarrow z^2 - xz + x(x-6) = 0$$

$$\Delta = x^2 - 4x(x-6) = 3x(8-x)$$

$\Delta \geq 0 \Leftrightarrow x \in [0; 8]$. As $x \in \mathbb{Z} \Rightarrow x \in \{0, 1, 2, \dots, 8\}$ and we will check, which of these

values leads to $z \in \mathbb{Z}$, where $z = \frac{x \pm \sqrt{3x(8-x)}}{2}$. We obtain: $x = 0 \Rightarrow$

$$\Rightarrow z = 0 \Rightarrow y = 1; x = 2 \Rightarrow z = 4 \text{ or } z = -2$$

$$\Rightarrow y = 5 \text{ or } y = -1; x = 6 \Rightarrow z = 6 \text{ or } z = 0$$

$$\Rightarrow y = 7 \text{ or } y = 1; x = 8 \Rightarrow z = 4 \Rightarrow y = 5 \Rightarrow$$

x	0	2	2	6	6	8
y	1	5	-1	7	1	5

and from 1. $x = k, y = -k$ where $k \in \mathbb{Z}$

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Solution 2 by Marin Chirciu-Romania

$$\text{If } x = 0 \Rightarrow y(y-1)^2 = 0 \Rightarrow y \in \{0, 1\} \Rightarrow (x, y) \in \{(0, 0), (0, 1)\}.$$

$$\text{If } x \neq 0 \Rightarrow x(x-1)^2 + y(y-1)^2 = x(3x^2 + 7y) \Leftrightarrow$$

$$\Leftrightarrow x^3 - 5x^2 + x(1-7y) + y(y-1)^2 = 0$$

As $x \in \mathbb{Z}$, I am looking for solutions among the dividers of the free term $y(y-1)^2$

Solution $x = -y$.

It follows that $(x, y) = (k, -k), k \in \mathbb{Z}$ are integer solutions for the equation from the enunciation.

Remark: In the same way.

Prove that the equation:

$$x(x-1)^2 + y(y-1)^2 + 4x^2 = 0,$$

has an infinity of real solutions.

Marin Chirciu

Solution:

We prove that $(x, y) = (\lambda, -\lambda), \lambda \in \mathbb{R}$ are solutions for the equation from enunciation.

The conclusion follows.

JP.555 In $\triangle ABC$ we know: $m(\hat{A}) = 90^\circ, AD \perp BC, D \in (BC), DE \perp AB,$

$E \in (AB), DF \perp AC, F \in (AC), BE = a, CF = b, BC = c, a, b, c > 0$

Prove that $c \leq 2\sqrt{a^2 + b^2}$.

Proposed by Gheorghe Molea – Romania

Solution 1 by proposer

We denote: $AD = h, AE = y, AF = x, BD = z, CD = t$.

We have:

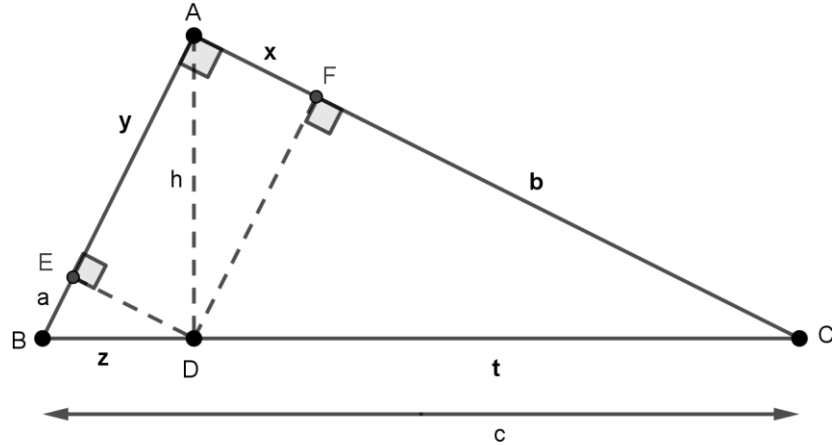
$$h^2 = x^2 + y^2$$

$$x^2 = z^2 - a^2$$

$$y^2 = t^2 - b^2$$

$$c = z + t$$

$$\Rightarrow h^2 = x^2 + y^2 = z^2 - a^2 + t^2 - b^2 = z^2 + t^2 - (a^2 + b^2) \quad (*)$$



$$\text{But } c^2 = (z + t)^2 = z^2 + t^2 + 2zt \Rightarrow c^2 = z^2 + t^2 + 2h^2 \Rightarrow z^2 + t^2 = c^2 - 2h^2$$

The relationship (*) becomes:

$$h^2 = c^2 - 2h^2 - (a^2 + b^2) \Rightarrow h^2 = \frac{c^2 - (a^2 + b^2)}{3}$$

$$\text{But } h \leq \text{the median for } A \Rightarrow h^2 \leq (\text{the median from } A)^2 \Rightarrow \frac{c^2 - (a^2 + b^2)}{3} \leq \left(\frac{c}{2}\right)^2$$

$$\Rightarrow 4c^2 - 4(a^2 + b^2) \leq 3c^2 \Rightarrow c^2 \leq 4(a^2 + b^2)$$

$$\Rightarrow c \leq 2\sqrt{a^2 + b^2}$$

We have equality $\Leftrightarrow a = b \Leftrightarrow \triangle ABC$ is an right isoscel triangle.

Solution 2 by Marin Chirciu-Romania

We denote $AF = x, AE = y \Rightarrow DE = x, DF = y$.

Using the height theorem in right triangles ABD and ADC we obtain:

$$x^2 = ay, y^2 = bx \Rightarrow x^2 + y^2 = ay + bx$$

Using Pythagoras's theorem in $\triangle ABC \Rightarrow c^2(a + y)^2 + (b + x)^2$.

The inequality we must prove $c \leq 2\sqrt{a^2 + b^2} \Leftrightarrow c^2 \leq 4(a^2 + b^2) \Leftrightarrow$

$$\Leftrightarrow (a + y)^2 + (b + x)^2 \leq 4(a^2 + b^2) \Leftrightarrow a^2 + y^2 + 2ay + b^2 + x^2 + 2bx \leq 4(a^2 + b^2)$$

$$\Leftrightarrow (x^2 + y^2) + 2ay + 2bx \leq 3(a^2 + b^2) \stackrel{x^2 + y^2 = ay + bx}{\Leftrightarrow}$$

$$\Leftrightarrow (ay + bx) + 2ay + 2bx \leq 3(a^2 + b^2) \Leftrightarrow$$

$$\begin{aligned} \Leftrightarrow 3(ay + bx) &\leq 3(a^2 + b^2) \Leftrightarrow (ay + bx) \leq (a^2 + b^2) \stackrel{x^2+y^2=ay+bx}{\Leftrightarrow} \\ &\Leftrightarrow (ay + bx)^2 \leq (a^2 + b^2)(y^2 + x^2) \end{aligned}$$

see CBS inequality, with equality for $ax = by$.

PROBLEMS FOR SENIORS

SP.541 Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f\left(\frac{x}{3}\right) - 3f(x) = 15x; (\forall)x \in \mathbb{R}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Replace $x = 0$ in statement:

$$f(0) - 3f(0) = 0 \Rightarrow -2f(0) = 0 \Rightarrow f(0) = 0$$

$$\frac{1}{3}f\left(\frac{x}{3}\right) - f(x) = 5x \quad (1)$$

Replace $x \rightarrow \frac{x}{3}$ in (1):

$$\begin{aligned} \frac{1}{3}f\left(\frac{x}{3^2}\right) - f\left(\frac{x}{3}\right) &= 5 \cdot \frac{x}{3} \\ \frac{1}{3^2}f\left(\frac{x}{3^2}\right) - \frac{1}{3}f\left(\frac{x}{3}\right) &= 5 \cdot \frac{x}{3^2} \quad (2) \end{aligned}$$

Replace $x \rightarrow \frac{x}{3}$ in (2):

$$\begin{aligned} \frac{1}{3^2}f\left(\frac{x}{3^3}\right) - \frac{1}{3}f\left(\frac{x}{3^2}\right) &= 5 \cdot \frac{x}{3^3} \\ \frac{1}{3^3}f\left(\frac{x}{3^3}\right) - \frac{1}{3^2}f\left(\frac{x}{3^2}\right) &= 5 \cdot \frac{x}{3^4} \quad (3) \end{aligned}$$

Replace $x \rightarrow \frac{x}{3}$ in (3):

$$\begin{aligned} \frac{1}{3^3}f\left(\frac{x}{3^4}\right) - \frac{1}{3^2}f\left(\frac{x}{3^3}\right) &= 5 \cdot \frac{x}{3^5} \\ \frac{1}{3^4}f\left(\frac{x}{3^4}\right) - \frac{1}{3^3}f\left(\frac{x}{3^3}\right) &= 5 \cdot \frac{x}{3^6} \quad (4) \end{aligned}$$

Analogous:

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$$\frac{1}{3^{n+1}}f\left(\frac{x}{3^{n+1}}\right) - \frac{1}{3^n}f\left(\frac{x}{3^n}\right) = 5 \cdot \frac{x}{3^{2n}} \quad (n)$$

By adding: (1); (2); (3); (4);...;(n):

$$\frac{1}{3^{n+1}}f\left(\frac{x}{3^{n+1}}\right) - f(x) = 5x\left(1 + \frac{1}{3^2} + \frac{1}{3^4} + \cdots + \frac{1}{3^{2n}}\right)$$

$$\frac{1}{3^{n+1}}f\left(\frac{x}{3^{n+1}}\right) - f(x) = 5x \cdot \frac{\frac{1}{3^{2n+2}} - 1}{\frac{1}{3^2} - 1}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{3^{n+1}}f\left(\frac{x}{3^{n+1}}\right) - f(x) \right) = 15x \lim_{n \rightarrow \infty} \frac{\frac{1}{3^{2n+2}} - 1}{\frac{1}{3^2} - 1}$$

$$\frac{1}{\infty} \cdot f(0) - f(x) = 5x \cdot \frac{\frac{1}{\infty} - 1}{\frac{1}{9} - 1}$$

$$0 \cdot 0 - f(x) = 5x \cdot \frac{0 - 1}{-\frac{8}{9}}$$

$$-f(x) = \frac{45x}{8}$$

$$f(x) = -\frac{45x}{8}$$

Solution 2 by Marin Chirciu-Romania

$$f\left(\frac{x}{3}\right) - 3f(x) = 15x \Leftrightarrow f(x) = \frac{1}{3}f\left(\frac{x}{3}\right) - 5x$$

Given to x the values: $\frac{x}{3}, \frac{x}{3^2}, \dots, \frac{x}{3^n}$ we obtain successively:

$$\begin{aligned} f(x) &= \frac{1}{3}f\left(\frac{x}{3}\right) - 5x = \frac{1}{3}\left[\frac{1}{3}f\left(\frac{x}{3^2}\right) - \frac{5x}{3}\right] - 5x = \frac{1}{3^2}f\left(\frac{x}{3^2}\right) - 5x\left(1 + \frac{1}{3^2}\right) = \\ &= \frac{1}{3^2}\left[\frac{1}{3}f\left(\frac{x}{3^3}\right) - 5\frac{x}{3^2}\right] - 5x\left(1 + \frac{1}{3^2}\right) = \frac{1}{3^3}f\left(\frac{x}{3^3}\right) - 5\frac{x}{3^4} - 5x\left(1 + \frac{1}{3^2}\right) = \\ &= \frac{1}{3^3}f\left(\frac{x}{3^3}\right) - 5x - \left(1 + \frac{1}{3^2} + \frac{1}{3^4}\right) = \cdots = \frac{1}{3^n}f\left(\frac{x}{3^n}\right) - 5x\left(1 + \frac{1}{3^2} + \frac{1}{3^4} + \cdots + \frac{1}{3^{2n}}\right) \end{aligned}$$

Passing the limit for $n \rightarrow \infty$ and taking into account that the function is continuous and

$f(0) = 0$ we obtain:

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$$f(x) = -5x \lim_{n \rightarrow \infty} \left(1 + \frac{1}{3^2} + \frac{1}{3^4} + \cdots + \frac{1}{3^{2n}} \right) = -5x \cdot \frac{9}{8} = \frac{-45}{8}x.$$

We deduce that the searched function is $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{-45}{8}x$

Remark: The problem can be developed.

Let $a > 1, b \in \mathbb{R}$ fixed. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f\left(\frac{x}{a}\right) - af(x) = abx, (\forall)x \in \mathbb{R}$$

Marin Chirciu

Solution

$$f\left(\frac{x}{a}\right) - af(x) = abx \Leftrightarrow f(x) = \frac{1}{a}f\left(\frac{x}{a}\right) - bx$$

Given to x the values: $\frac{x}{a}, \frac{x}{a^2}, \dots, \frac{x}{a^n}$ we obtain successively:

$$\begin{aligned} f(x) &= \frac{1}{a}f\left(\frac{x}{a}\right) - bx = \frac{1}{a}\left[\frac{1}{a}f\left(\frac{x}{a^2}\right) - b\frac{x}{a}\right] - bx = \frac{1}{a^2}f\left(\frac{x}{a^2}\right) - ax\left(1 + \frac{1}{a^2}\right) = \\ &= \frac{1}{a^2}\left[\frac{1}{a}f\left(\frac{x}{a^3}\right) - b\frac{x}{a^2}\right] - bx\left(1 + \frac{1}{a^2}\right) = \frac{1}{a^3}f\left(\frac{x}{a^3}\right) - b\frac{x}{a^4} - bx\left(1 + \frac{1}{a^2}\right) = \\ &= \frac{1}{a^3}f\left(\frac{x}{a^3}\right) - bx\left(1 + \frac{1}{a^2} + \frac{1}{a^4}\right) = \cdots = \frac{1}{a^n}f\left(\frac{x}{a^n}\right) - bx\left(1 + \frac{1}{a^2} + \frac{1}{a^4} + \cdots + \frac{1}{a^{2n}}\right) \end{aligned}$$

Passing to the limit for $n \rightarrow \infty$ and taking into account that the function is continuous and $f(0) = 0$ we obtain:

$$f(x) = -bx \lim_{n \rightarrow \infty} \left(1 + \frac{1}{a^2} + \frac{1}{a^4} + \cdots + \frac{1}{a^{2n}} \right) = -bx \cdot \frac{a^2}{a^2 - 1} = \frac{a^2 b}{1 - a^2}x$$

We deduce that the searched function is $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{a^2 b}{1 - a^2}x$

Note:

For $a = 3, b = 5$ we obtain Problem SP.541. from RMM Nr. 37 – Summer 2025

SP.542 If $z_1, z_2, z_3 \in \mathbb{C}; |z_1| = |z_2| = |z_3| = 1; z_1 + z_2 + z_3 = 1$ then find:

$$\Omega = \left(\sum_{i=1}^n z_i^3 \right) \left(\sum_{i=1}^n z_i^5 \right) \left(\sum_{i=1}^n z_i^7 \right)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let be z_1, z_2, z_3 roots for the equation:

$$z^3 - S_1 z^2 + S_2 z - S_3 = 0$$

$$S_1 = z_1 + z_2 + z_3 = 1$$

$$S_2 = z_1 z_2 + z_2 z_3 + z_3 z_1$$

$$S_3 = z_1 z_2 z_3$$

$$|z_1| = 1 \Rightarrow z_1 \cdot \overline{z_1} = 1 \Rightarrow \overline{z_1} = \frac{1}{z_1}$$

$$|z_1| = |z_2| = |z_3| = 1 \Rightarrow z_1 \neq 0; z_2 \neq 0; z_3 \neq 0$$

$$1 = \overline{1} = \overline{z_1 + z_2 + z_3} = \overline{z_1} + \overline{z_2} + \overline{z_3}$$

$$1 = \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \Rightarrow z_1 z_2 z_3 = z_1 z_2 + z_2 z_3 + z_3 z_1 \Rightarrow S_2 = S_3$$

$$z^3 - z^2 + S_2 z - S_2 = 0$$

$$z^2(z - 1) + S_2(z - 1) = 0$$

$$(z - 1)(z^2 + S_2) = 0 \Rightarrow z_1 = 1$$

$$z^2 = -S_2 \Rightarrow z_2 + z_3 = 0 \Rightarrow z_3 = -z_2$$

$$z_1^3 + z_2^3 + z_3^3 = 1^3 + z_2^3 + (-z_2)^3 = 1 + z_2^3 - z_2^3 = 1$$

$$z_1^5 + z_2^5 + z_3^5 = 1^5 + z_2^5 + (-z_2)^5 = 1 + z_2^5 - z_2^5 = 1$$

$$z_1^7 + z_2^7 + z_3^7 = 1^7 + z_2^7 + (-z_2)^7 = 1 + z_2^7 - z_2^7 = 1$$

$$\Omega = 1 \cdot 1 \cdot 1 = 1$$

Solution 2 by Marin Chirciu-Romania

Using $|z|^2 = z \cdot \overline{z}, z \in \mathbb{C}$ we have: $1 = \overline{1} = \overline{z_1 + z_2 + z_3} = \overline{z_1} + \overline{z_2} + \overline{z_3} = \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} =$

$$\frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 z_2 z_3} = \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{1} \Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 1$$

We form the equation of the third degree knowing the sums of Viète:

$$z_1 + z_2 + z_3 = 1, z_1 z_2 + z_2 z_3 + z_3 z_1 = 1, z_1 z_2 z_3 = 1:$$

We obtain the equation $z^3 - z^2 + z - 1 = 0 \Leftrightarrow (z - 1)(z^2 + 1) = 0$, with the solution

$$\{z_1, z_2, z_3\} = \{1, i, -i\}$$

$$\text{It follows } \sum_{k=1}^3 z_i^3 = 1, \sum_{k=1}^3 z_i^5 = 1, \sum_{k=1}^3 z_i^7 = 1.$$

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We deduce that $\Omega = \sum_{k=1}^3 z_i^3 \sum_{k=1}^3 z_i^5 \sum_{k=1}^3 z_i^7 = 1$

Remark: The problem can be developed.

If $z_1, z_2, z_3 \in \mathbb{C}, |z_1| = |z_2| = |z_3| = 1, z_1 + z_2 + z_3 = 1, z_1 z_2 z_3 = 1$ and $k, n, p \in \mathbb{N}$ then find:

$$\Omega = \sum_{k=1}^3 z_i^{2k+1} \sum_{k=1}^3 z_i^{2n+1} \sum_{k=1}^3 z_i^{2p+1}$$

Marin Chiricu

Solution

We use $|z|^2 = z \cdot \bar{z}, z \in \mathbb{C}$:

$$\begin{aligned} 1 = \bar{1} &= \overline{z_1 + z_2 + z_3} = \bar{z}_1 + \bar{z}_2 + \bar{z}_3 = \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 z_2 z_3} = \\ &= \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{1} \Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 1 \end{aligned}$$

We form the equation of the third degree knowing the sums of Viète

$$z_1 + z_2 + z_3 = 1, z_1 z_2 + z_2 z_3 + z_3 z_1 = 1, z_1 z_2 z_3 = 1:$$

We obtain the equation $z^3 - z^2 + z - 1 = 0 \Leftrightarrow (z - 1)(z^2 + 1) = 0$,

with the solutions : $\{z_1, z_2, z_3\} = \{1, i, -i\}$.

It follows $\sum_{k=1}^3 z_i^{2k+1} = 1, \sum_{k=1}^3 z_i^{2n+1} = 1, \sum_{k=1}^3 z_i^{2p+1} = 1$.

We deduce that $\Omega = \sum_{k=1}^3 z_i^{2k+1} \sum_{k=1}^3 z_i^{2n+1} \sum_{k=1}^3 z_i^{2p+1} = 1$.

Note:

For $k = 1, n = 2, p = 3$ we obtain the problem SP.542 from RMM – 37 – Summer 2025

SP.543 Let be $f: [0, 1] \rightarrow [0, 20]; f(x) = 2 \cdot 3^x + 4 \cdot 5^x - 6$. Find:

$$\Omega = \int_0^{20} f^{-1}(x) dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$f'(x) = 2 \cdot 3^x \log 3 + 4 \cdot 5^x \log 5 > 0$$

f strictly increasing

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$$f(0) = 2 + 4 - 6 = 0; f(1) = 6 + 20 - 6 = 20$$

$$f \text{ continuous} \Rightarrow f([0, 1]) = [0, 20]$$

By Young's theorem:

$$\int_0^1 f(x) dx + \int_0^{20} f^{-1}(x) dx = 1 \cdot 20$$

$$\int_0^1 (2 \cdot 3^x + 4 \cdot 5^x - 6) dx + \int_0^{20} f^{-1}(x) dx = 20$$

$$\begin{aligned} \Omega &= \int_0^{20} f^{-1}(x) dx = 20 - 2 \int_0^1 3^x dx - 4 \int_0^1 5^x dx + 6(1 - 0) = \\ &= 20 - 2 \left(\frac{3^1}{\log 3} - \frac{3^0}{\log 3} \right) - 4 \left(\frac{5^1}{\log 5} - \frac{5^0}{\log 5} \right) + 6 = 26 - \frac{4}{\log 3} - \frac{16}{\log 5} \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

Using Young equality:

If $f: [0, a] \rightarrow [0, b]$ is a derivable function strictly increasing with $f(0) = 0$ and $f(a) = b$,

then the following equality holds $\int_0^a f(x) dx + \int_0^b f^{-1}(x) dx = ab$.

In our case $f: [0, 1] \rightarrow [0, 20]$, $f(x) = 2 \cdot 3^x + 4 \cdot 5^x - 6$ is a derivable function, strictly increasing (exponentials with a superunit base) with $f(0) = 0$ and $f(1) = 20$.

Applying Young equality we obtain:

$$\int_0^1 f(x) dx + \int_0^{20} f^{-1}(x) dx = 20$$

$$\begin{aligned} \text{We have } \int_0^1 f(x) dx &= \int_0^1 (2 \cdot 3^x + 4 \cdot 5^x - 6) dx = \left(2 \cdot \frac{3^x}{\ln 3} + 4 \cdot \frac{5^x}{\ln 5} - 6x \right) \Big|_0^1 = \\ &= \left(2 \cdot \frac{3}{\ln 3} + 4 \cdot \frac{5}{\ln 5} - 6 \right) - \left(2 \cdot \frac{1}{\ln 3} + 4 \cdot \frac{1}{\ln 5} \right) = \frac{4}{\ln 3} + \frac{16}{\ln 5} - 6 \end{aligned}$$

We obtain:

$$\Omega = \int_0^{20} f^{-1}(x) dx = 20 - \left(\frac{4}{\ln 3} + \frac{16}{\ln 5} - 6 \right) = 26 - \frac{4}{\ln 3} - \frac{16}{\ln 5}$$

Remark: In the same way.

Let be $f: [0, 1] \rightarrow [0, 25]$, $f(x) = 3 \cdot 4^x + 4 \cdot 5^x - 7$. Find:

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$$\Omega = \int_0^{25} f^{-1}(x) dx$$

Marin Chirciu

Solution: Using Young equality:

If $f: [0, a] \rightarrow [0, b]$ is a derivable function, strictly increasing with $f(0) = 0$ and $f(a) = b$,

then the following equality $\int_0^a f(x) dx + \int_0^b f^{-1}(x) dx = ab$.

In our case $f: [0, 1] \rightarrow [0, 25]$, $f(x) = 3 \cdot 4^x + 4 \cdot 5^x - 7$ is a derivable function, strictly increasing (exponentials with a superunit base), with $f(0) = 0$ and $f(1) = 25$.

Applying Young's equality we obtain:

$$\int_0^1 f(x) dx + \int_0^{25} f^{-1}(x) dx = 25.$$

$$\begin{aligned} \text{We have } \int_0^1 f(x) dx &= \int_0^1 (3 \cdot 4^x + 4 \cdot 5^x - 7) dx = \left(3 \cdot \frac{4^x}{\ln 4} + 4 \cdot \frac{5^x}{\ln 5} - 6x \right) \Big|_0^1 = \\ &= \left(3 \cdot \frac{4}{\ln 4} + 4 \cdot \frac{5}{\ln 5} - 7 \right) - \left(3 \cdot \frac{1}{\ln 4} + 4 \cdot \frac{1}{\ln 5} \right) = \frac{9}{\ln 3} + \frac{16}{\ln 5} - 7. \end{aligned}$$

We obtain:

$$\Omega = \int_0^{25} f^{-1}(x) dx = 25 - \left(\frac{9}{\ln 4} + \frac{16}{\ln 5} - 7 \right) = 32 - \frac{9}{\ln 4} - \frac{16}{\ln 5}.$$

SP.544 Find the maximum value of $n \in \mathbb{N}^*$ such that:

$$\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k+1} + k\sqrt{k}} < \frac{31}{32}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned} 1 > 0 &\Rightarrow n+1 > n \Rightarrow \sqrt{n+1} > \sqrt{n} \\ \Rightarrow \sqrt{n+1} - \sqrt{n} > 0 &\Rightarrow (\sqrt{n+1} - \sqrt{n})(n+1 - n) > 0 \\ (n+1)(\sqrt{n+1} - \sqrt{n}) - (\sqrt{n+1} - \sqrt{n}) \cdot n &> 0 \\ (n+1)\sqrt{n+1} - (n+1)\sqrt{n} - n\sqrt{n+1} + n\sqrt{n} &> 0 \\ (n+1)\sqrt{n+1} + n\sqrt{n} &> (n+1)\sqrt{n} + n\sqrt{n+1} \end{aligned}$$

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$$\begin{aligned}
 & \frac{1}{(n+1)\sqrt{n+1} + n\sqrt{n}} < \frac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}} = \\
 & = \frac{(n+1)\sqrt{n} - n\sqrt{n+1}}{(n+1)^2 \cdot n - n^2(n+1)} = \frac{(n+1)\sqrt{n} - n\sqrt{n+1}}{n(n+1)(n+1-n)} = \\
 & = \frac{(n+1)\sqrt{n}}{n(n+1)} - \frac{n\sqrt{n+1}}{n(n+1)} = \frac{\sqrt{n}}{n} - \frac{\sqrt{n+1}}{n+1} \\
 & \sum_{k=1}^n \frac{1}{(k+1)\sqrt{k+1} + k\sqrt{k}} < \sum_{k=1}^n \left(\frac{\sqrt{k}}{k} - \frac{\sqrt{k+1}}{k+1} \right) = \\
 & = 1 - \frac{\sqrt{n+1}}{n+1} = 1 - \frac{1}{\sqrt{n+1}} < \frac{31}{32} \Leftrightarrow \frac{1}{\sqrt{n+1}} > \frac{1}{32} \\
 & \sqrt{n+1} < 32 \Rightarrow n+1 < 1024 \Rightarrow n < 1023
 \end{aligned}$$

Solution: $n = 1022$

Solution 2 by Marin Chirciu-Romania

$$\begin{aligned}
 & \sum_{k=1}^n \frac{1}{(k+1)\sqrt{k+1} + k\sqrt{k}} < \sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} = \sum_{k=1}^n \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right) = \\
 & = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{n+1}} \leq 1 - \frac{1}{32} \\
 & \text{From } 1 - \frac{1}{\sqrt{n+1}} < 1 - \frac{1}{32} \Leftrightarrow \frac{1}{\sqrt{n+1}} > \frac{1}{32} \Leftrightarrow \sqrt{n+1} < 32 \Leftrightarrow n+1 < 32^2 \Leftrightarrow \\
 & \Leftrightarrow n < 1023 \Rightarrow n_{\max} = 1022
 \end{aligned}$$

Remark: The problem can be developed.

Let $m \geq 2$ fixed. Find the maximum value of $n \in \mathbb{N}^*$ such that:

$$\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k+1} + k\sqrt{k}} < \frac{m-1}{m}$$

Marin Chirciu

Solution

$$\begin{aligned}
 & \sum_{k=1}^n \frac{1}{(k+1)\sqrt{k+1} + k\sqrt{k}} < \sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} = \sum_{k=1}^n \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right) = \\
 & = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{n+1}} < 1 - \frac{1}{m}
 \end{aligned}$$

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$$\text{From } 1 - \frac{1}{\sqrt{n+1}} < 1 - \frac{1}{m} \Leftrightarrow \frac{1}{\sqrt{n+1}} > \frac{1}{m} \Leftrightarrow \sqrt{n+1} < m \Leftrightarrow n+1 < m^2 \Leftrightarrow \\ \Leftrightarrow n < m^2 - 1 \Rightarrow n_{\max} = m^2 - 2.$$

Note: For $m = 32$ we obtain Problem SP.544 from RMM Nr. 37 – Summer 2025

SP.545 Let a, b, c be positive real numbers such that $a = \max\{a, b, c\}$ and $a^2 b^5 c^5 \geq 1$, and let:

$$F(a, b, c) = \sqrt{\frac{ab + bc + ca}{3}} - \sqrt[3]{abc}$$

Prove that:

$$F(a, b, c) \geq F\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$$

Proposed by Vasile Cîrtoaje and Vasile Mircea Popa – Romania

Solution 1 by proposers

Since $F(a, b, c) \geq 0$ and $F\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right) \geq 0$, it suffices to prove the homogeneous inequality

$$F(a, b, c) \geq (a^2 b^5 c^5)^{\frac{1}{6}} \cdot F\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$$

for $a = \max\{a, b, c\}$. Due to homogeneity, we may set $a = 1$, hence $b, c \leq 1$. Thus, we need to show that

$$F(1, b, c) \geq (bc)^{\frac{5}{6}} \cdot F\left(1, \frac{1}{b}, \frac{1}{c}\right),$$

which is equivalent to

$$\sqrt{\frac{a + b + ab}{3}} - \sqrt[3]{bc} \geq (bc)^{\frac{5}{6}} \left[\sqrt{\frac{b + c + 1}{bc}} - \frac{1}{\sqrt[3]{bc}} \right]$$

$$\text{Denote } s = \frac{b+c}{2}, \quad p = \sqrt{bc},$$

with $p \leq s \leq 1$. For fixed p , the desired inequality is equivalent to $F(s) \geq 0$, where

$$F(s) = \sqrt{2s + p^2} - p^{\frac{2}{3}}\sqrt{3} - p^{\frac{2}{3}}\left(\sqrt{2s + 1} - p^{\frac{1}{3}}\sqrt{3}\right).$$

Since

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$$F'(s) = \frac{1}{\sqrt{2s+p^2}} - \frac{p^{\frac{2}{3}}}{\sqrt{2s+1}} = \frac{\sqrt{2s+1} - p^{\frac{2}{3}}\sqrt{2s+p^2}}{\sqrt{(2s+p^2)(2s+1)}} \geq 0,$$

$F(s)$ is increasing, hence $F(s) \geq F(p)$. So, we need to prove that $F(p) \geq 0$, i.e.

$$\sqrt{2p+p^2} - p^{\frac{2}{3}}\sqrt{3} \geq p^{\frac{2}{3}}\left(\sqrt{2p+1} - p^{\frac{1}{3}}\sqrt{3}\right)$$

Putting $p = x^3$, where $x \leq 1$, the inequality becomes

$$\sqrt{x^3+2} - \sqrt{3x} \geq \sqrt{x}\left(\sqrt{2x^3+1} - x\sqrt{3}\right), \quad \frac{(x+2)(x-1)^2}{\sqrt{x^3+2} + \sqrt{3x}} \geq \frac{\sqrt{x}(2x+1)(x-1)^2}{\sqrt{2x^3+1} + x\sqrt{3}}$$

It is true if

$$\frac{x+2}{\sqrt{x^3+2} + \sqrt{3x}} \geq \frac{\sqrt{x}(2x+1)}{\sqrt{2x^3+1} + x\sqrt{3}},$$

which is equivalent to: $x\sqrt{3}(1-x) + (x+2)\sqrt{2x^3+1} - (2x+1)\sqrt{x^4+2x} \geq 0$.

It suffices to show that: $(x+2)\sqrt{2x^3+1} \geq (2x+1)\sqrt{x^4+2x}$

It is true since: $x+2 - (2x+1) = 1-x \geq 0$

and:

$$2x^3+1 - (x^4+2x) = (1-x)^3(1+x) \geq 0$$

The proof is completed. The equality occurs for $a = b = c \geq 1$.

Remark. The inequality is true in the particular case $a, b, c \geq 1$

(which involves $a^2b^5c^5 \geq 1$).

Solution 2 by Mohamed Amine Ben Ajiba-Tanger-Morocco

Let $x := \sqrt{bc}$. Since $a = \max\{a, b, c\}$ and $a^2b^5c^5 \geq 1$, then $a \geq x$ and

$a^2x^{10} \geq 1$ or $ax^5 \geq 1$. We will write the desired inequality as

$$E(a, b, c) = \sqrt{\frac{ab+bc+ca}{3}} - \sqrt[3]{abc} - \sqrt{\frac{a+b+c}{3abc}} + \frac{1}{\sqrt[3]{abc}} \geq 0.$$

We will first prove that $E(a, b, c) \geq E(a, x, x)$. The inequality is successively equivalent to

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$$\begin{aligned} & \sqrt{\frac{ab+bc+ca}{3}} - \sqrt{\frac{a+b+c}{3abc}} \geq \sqrt{\frac{2a\sqrt{bc}+bc}{3}} - \sqrt{\frac{a+2\sqrt{bc}}{3abc}} \\ & \left(\sqrt{ab+bc+ca} - \sqrt{2a\sqrt{bc}+bc} \right) \sqrt{abc} \geq \sqrt{a+b+c} - \sqrt{a+2\sqrt{bc}} \\ & \frac{a(\sqrt{b}-\sqrt{c})^2 \cdot \sqrt{abc}}{\sqrt{ab+bc+ca} + \sqrt{2a\sqrt{bc}+bc}} \geq \frac{(\sqrt{b}-\sqrt{c})^2}{\sqrt{a+b+c} + \sqrt{a+2\sqrt{bc}}} \\ & \sqrt{a^3bc(a+b+c)} + \sqrt{a^3bc(a+2\sqrt{bc})} \geq \sqrt{ab+bc+ca} + \sqrt{2a\sqrt{bc}+bc} \quad (1) \end{aligned}$$

Since $a = \max\{a, b, c\}$, we have $(a^2bc)^3 \geq a^2b^5c^5 \geq 1$ or $a^2bc \geq 1$, then

$$\begin{aligned} a^3bc(a+b+c) & \geq a(a+b+c) \geq ab+bc+ca \text{ \& } a^3bc(a+2\sqrt{bc}) \geq a(a+2\sqrt{bc}) \\ & \geq 2a\sqrt{bc}+bc, \end{aligned}$$

then (1) is true. So it suffices to prove that $E(a, x, x) \geq 0$, which is equivalent to

$$\sqrt{\frac{2ax+x^2}{3}} - \sqrt[3]{ax^2} \geq \sqrt{\frac{a+2x}{3ax^2}} - \frac{1}{\sqrt[3]{ax^2}}.$$

and since $ax^5 \geq 1$, it suffices to prove that

$$\sqrt{\frac{2ax+x^2}{3}} - \sqrt[3]{ax^2} \geq \sqrt[3]{ax^5} \cdot \left(\sqrt{\frac{a+2x}{3ax^2}} - \frac{1}{\sqrt[3]{ax^2}} \right)$$

Setting $t^6 = \frac{a}{x} \geq 1$. The inequality is equivalent to

$$\begin{aligned} & \sqrt{\frac{2t^6+1}{3}} - t^2 \geq t^2 \left(\sqrt{\frac{t^6+2}{3t^6}} - \frac{1}{t^2} \right) \Leftrightarrow t\sqrt{2t^6+1} - \sqrt{t^6+2} \geq \sqrt{3}t(t^2-1) \\ & \Leftrightarrow \frac{(t^4-1)(2t^4-t^2+2)}{t\sqrt{2t^6+1} + \sqrt{t^6+2}} \geq \sqrt{3}t(t^2-1) \\ & \stackrel{t \geq 1}{\Leftrightarrow} (t^2+1)(2t^4-t^2+2) \geq t^2\sqrt{3(2t^6+1)} + t\sqrt{3(t^6+2)} \quad (2) \end{aligned}$$

By AM – GM inequality, we have

$$\begin{aligned} RHS_{(2)} &\leq \frac{3t^4 + (2t^6 + 1)}{2} + \frac{3t^2 + (t^6 + 2)}{2} = \frac{3}{2}(t^2 + 1)(t^4 + 1) \stackrel{?}{\geq} LHS_{(2)} \\ &\Leftrightarrow \frac{3}{2}(t^4 + 1) \leq 2t^4 - t^2 + 2 \Leftrightarrow \frac{1}{2}(t^2 - 1)^2 \geq 0, \end{aligned}$$

which is true and the proof is complete. Equality holds iff $a = b = c = 1$.

SP.546 If $a, b, c \in (0, 1)$ then:

$$\sqrt{3a - a^2} + \sqrt{5b - b^2} + \sqrt{7c - c^2} \leq \sqrt{15(a + b + c) - (a + b + c)^2}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let be $x, y, z > 0$ such that:

$$\frac{a}{3} = x; \frac{b}{5} = y; \frac{c}{7} = z$$

$$\begin{aligned} &\sqrt{3a - a^2} + \sqrt{5b - b^2} + \sqrt{7c - c^2} = \\ &= \sqrt{9\left(\frac{a}{3} - \frac{a^2}{9}\right)} + \sqrt{25\left(\frac{b}{5} - \frac{b^2}{25}\right)} + \sqrt{49\left(\frac{c}{7} - \frac{c^2}{49}\right)} \\ &= 3\sqrt{\frac{a}{3} - \left(\frac{a}{3}\right)^2} + 5\sqrt{\frac{b}{5} - \left(\frac{b}{5}\right)^2} + 7\sqrt{\frac{c}{7} - \left(\frac{c}{7}\right)^2} \end{aligned}$$

$$\text{Let be } f: (0, 1) \rightarrow \mathbb{R}; f(x) = \sqrt{x - x^2}$$

$$f'(x) = \frac{-1}{4\sqrt{x-x^2}(x-x^2)} < 0 \Rightarrow f \text{ concave}$$

By Jensen's inequality:

$$f\left(\frac{3}{15}x + \frac{5}{15}y + \frac{7}{15}z\right) \geq \frac{3}{15}f(x) + \frac{5}{15}f(y) + \frac{7}{15}f(z)$$

$$15f\left(\frac{3x + 5y + 7z}{15}\right) \geq 3f(x) + 5f(y) + 7f(z)$$

$$15\sqrt{\frac{3x + 5y + 7z}{15} - \left(\frac{3x + 5y + 7z}{15}\right)^2} \geq 3\sqrt{x - x^2} + 5\sqrt{y - y^2} + 7\sqrt{z - z^2}$$

$$\begin{aligned}
 3\sqrt{x-x^2} + 5\sqrt{y-y^2} + 7\sqrt{z-z^2} &\leq 15 \sqrt{\frac{3x+5y+7z}{15} - \left(\frac{3x+5y+7z}{15}\right)^2} \\
 \sqrt{9x-9x^2} + \sqrt{25y-25y^2} + \sqrt{49z-49z^2} &\leq \\
 &\leq \sqrt{225 \left(\frac{3x+5y+7z}{15} - \left(\frac{3x+5y+7z}{15}\right)^2 \right)} \\
 \sqrt{3 \cdot 3x - (3x)^2} + \sqrt{5 \cdot 5y - (5y)^2} + \sqrt{7 \cdot 7z - (7z)^2} &\leq \\
 &\leq \sqrt{15(3x+5y+7z) - (3x+5y+7z)^2} \\
 \sqrt{3a-a^2} + \sqrt{5b-b^2} + \sqrt{7c-c^2} &\leq \\
 &\leq \sqrt{15(a+b+c) - (a+b+c)^2}
 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

$$\sqrt{3a-a^2} + \sqrt{5b-b^2} + \sqrt{7c-c^2} \leq \sqrt{15(a+b+c) - (a+b+c)^2}$$

Using CBS inequality, we obtain:

$$\begin{aligned}
 LHS &= \sqrt{3a-a^2} + \sqrt{5b-b^2} + \sqrt{7c-c^2} = \sqrt{a}\sqrt{3-a} + \sqrt{b}\sqrt{5-b} + \sqrt{c}\sqrt{7-c} \stackrel{CBS}{\leq} \\
 &\stackrel{CBS}{\leq} \sqrt{(a+b+c)(3-a+5-b+7-c)} = \sqrt{(a+b+c)(15-a-b-c)} = \\
 &= \sqrt{15(a+b+c) - (a+b+c)^2} = RHS
 \end{aligned}$$

Observation: $a, b, c \in (0, 1) \Rightarrow \sqrt{a}, \sqrt{b}, \sqrt{c} > 0$ and $\sqrt{3-a}, \sqrt{5-b}, \sqrt{7-c} > 0$

Equality holds if and only if $\frac{a}{3-a} = \frac{b}{5-b} = \frac{c}{7-c} \Leftrightarrow \frac{a}{3} = \frac{b}{5} = \frac{c}{7}$

Remark: The problem can be developed.

If $a, b, c \in (0, 1)$ and $x, y, z \in (1, \infty)$ then:

$$\sqrt{xa-a^2} + \sqrt{yb-b^2} + \sqrt{zc-c^2} \leq \sqrt{(x+y+z)(a+b+c) - (a+b+c)^2}$$

Marin Chirciu

Solution: Using CBS inequality, we obtain:

$$\begin{aligned}
 LHS &= \sqrt{xa-a^2} + \sqrt{yb-b^2} + \sqrt{zc-c^2} = \sqrt{a}\sqrt{x-a} + \sqrt{b}\sqrt{y-b} + \sqrt{c}\sqrt{z-c} \stackrel{CBS}{\leq} \\
 &\stackrel{CBS}{\leq} \sqrt{(a+b+c)(x-a+y-b+z-c)} = \sqrt{(a+b+c)(x+y+z-a-b-c)} = \\
 &= \sqrt{(x+y+z)(a+b+c) - (a+b+c)^2} = RHS
 \end{aligned}$$

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Observation: $a, b, c \in (0, 1) \Rightarrow \sqrt{a}, \sqrt{b}, \sqrt{c} > 0$ and $\sqrt{x-a}, \sqrt{y-b}, \sqrt{z-c} > 0$

Equality holds if and only if $\frac{a}{x-a} = \frac{b}{y-b} = \frac{c}{z-c} \Leftrightarrow \frac{a}{x} = \frac{b}{y} = \frac{c}{z}$.

Note: For $x = 3, y = 5, z = 7$ we obtain Problem SP.546 from RMM Nr. 37 – Summer 2025

SP.547 Prove that if $x, y > 1$ then:

$$\ln x \cdot \ln y \left(\sqrt[3]{\log_x y} + \sqrt[3]{\log_y x} \right)^3 \leq 2 \ln^2(xy)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\left(\sqrt[3]{\log_x y} + \sqrt[3]{\log_y x} \right)^3 \leq \frac{2(\ln x + \ln y)^2}{\ln x \cdot \ln y}$$

$$\left(\sqrt[3]{\frac{\ln y}{\ln x}} + \sqrt[3]{\frac{\ln x}{\ln y}} \right)^3 \leq \frac{2(\ln x + \ln y)^2}{\ln x \cdot \ln y}$$

Denote: $\ln x = u^3; \ln y = v^3$

$$\left(\sqrt[3]{\frac{v^3}{u^3}} + \sqrt[3]{\frac{u^3}{v^3}} \right)^3 \leq \frac{2(u^3 + v^3)^2}{u^3 \cdot v^3}$$

$$\left(\frac{v}{u} + \frac{u}{v} \right)^3 \leq \frac{2(u^3 + v^3)^2}{(uv)^3}$$

$$(u^2 + v^2)^3 \leq 2(u^3 + v^3)^2 \quad (\text{to prove})$$

$$(u^2 + v^2)^3 = (u^2 + v^2)(u^2 + v^2)^2 = (u^2 + v^2) \left(\sqrt{u} \cdot \sqrt{u^3} + \sqrt{v} \cdot \sqrt{v^3} \right)^2 \leq$$

$$\stackrel{CBS}{\leq} (u^2 + v^2)(u + v)(u^3 + v^3) \leq$$

$$\stackrel{CEBYSHEV}{\leq} 2(u^2 \cdot u + v^2 \cdot v)(u^3 + v^3) = 2(u^3 + v^3)^2$$

Equality holds for $x = y$.

Solution 2 by Marin Chirciu-Romania

With the substitution $(a, b) = (\ln x, \ln y)$ the inequality can be written:

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$$ab \left(\sqrt[3]{\frac{a}{b}} + \sqrt[3]{\frac{b}{a}} \right)^3 \leq 2(a+b)^2$$

Denoting $(a, b) = (u^3, v^3)$ the above inequality becomes:

$$u^3 v^3 \left(\frac{u}{v} + \frac{v}{u} \right)^3 \leq 2(u^3 + v^3)^2 \Leftrightarrow u^3 v^3 \left(\frac{u^3}{v^3} + 3 \frac{u}{v} + 3 \frac{v}{u} + \frac{v^3}{u^3} \right) \leq 2(u^6 + 2u^3 v^3 + v^6)$$

$\Leftrightarrow$

$$\Leftrightarrow u^6 + 3u^4 v^2 + 3u^2 v^4 + v^6 \leq 2u^6 + 4u^3 v^3 + 2v^6 \Leftrightarrow$$

$$\Leftrightarrow u^6 - 3u^4 v^2 + 4u^3 v^3 - 3u^2 v^4 + v^6 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (u - v)^2 (u^4 + 2u^3 v + 2uv^3 + v^4) \geq 0, \text{ with equality for } u = v \Leftrightarrow a = b \Leftrightarrow$$

$$\Leftrightarrow \ln x = \ln y \Leftrightarrow x = y$$

Equality holds if and only if $x = y$.

Remark.

The problem can be developed.

If $x, y > 1$ then:

$$\ln x \ln y \left(\sqrt[4]{\log_x y} + \sqrt[4]{\log_y x} \right)^4 \leq 4 \ln^2(xy).$$

Marin Chirciu

Solution

With the substitution $(a, b) = (\ln x, \ln y)$ the inequality can be written:

$$ab \left(\sqrt[4]{\frac{a}{b}} + \sqrt[4]{\frac{b}{a}} \right)^4 \leq 4(a+b)^2.$$

Denoting $(a, b) = (u^4, v^4)$ the above inequality becomes:

$$u^4 v^4 \left(\frac{u}{v} + \frac{v}{u} \right)^4 \leq 4(u^4 + v^4)^2 \Leftrightarrow u^4 v^4 \left(\frac{u^4}{v^4} + 4 \frac{u^2}{v^2} + 6 + 4 \frac{v^2}{u^2} + \frac{v^4}{u^4} \right) \leq$$

$$\leq 4(u^8 + 2u^4 v^4 + v^8) \Leftrightarrow$$

$$\Leftrightarrow u^8 + 4u^6 v^2 + 6u^4 v^4 + 4u^2 v^6 + v^8 \leq 4u^8 + 8u^4 v^4 + 4v^8 \Leftrightarrow$$

$$\Leftrightarrow 3u^8 - 4u^6 v^2 + 2u^4 v^4 - 4u^2 v^6 + 3v^8 \geq 0 \Leftrightarrow (u^2 - v^2)^2 (3u^4 + 2u^2 v^2 + 3v^4) \geq 0$$

with equality for $u = v \Leftrightarrow a = b \Leftrightarrow \ln x = \ln y \Leftrightarrow x = y$.

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Equality holds if and only if $x = y$.

Solution 3 by Angel Plaza-Spain

First, we change variables by doing $\ln x = a$ and $\ln y = b$, so the inequality reads as

Prove that if $a, b > 0$ then:

$$a \cdot b \left(\sqrt[3]{\frac{b}{a}} + \sqrt[3]{\frac{a}{b}} \right)^3 \leq 2(a+b)^2$$

and since the last expression is homogenous, we may assume additionally that $a + b = 1$,

so

$$a \cdot (1-a) \left(\sqrt[3]{\frac{1-a}{a}} + \sqrt[3]{\frac{a}{1-a}} \right)^3 \leq 2, \quad \sqrt[3]{a \cdot (1-a)} \left(\sqrt[3]{\frac{1-a}{a}} + \sqrt[3]{\frac{a}{1-a}} \right) \leq \sqrt[3]{2}$$

$$(1-a)^{\frac{2}{3}} + a^{\frac{2}{3}} \leq 2^{\frac{1}{3}}, \quad \frac{(1-a)^{\frac{2}{3}} + a^{\frac{2}{3}}}{2} \leq 2^{-\frac{2}{3}}$$

$$\left(\frac{(1-a)^{\frac{2}{3}} + a^{\frac{2}{3}}}{2} \right)^{\frac{3}{2}} \leq \frac{1}{2}$$

$$\text{which it is true since } \left(\frac{(1-a)^{\frac{2}{3}} + a^{\frac{2}{3}}}{2} \right)^{\frac{3}{2}} \leq \frac{1-a+a}{2} = \frac{1}{2}$$

SP.548 If $a, b, c > 0$ then:

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2 \left(\frac{ab}{a+b} + \frac{bc}{c+a} + \frac{ca}{a+b} \right) \geq 4(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\text{For } a, b > 0; (a+b-2\sqrt{ab})^2 \geq 0$$

$$(a+b)^2 - 2(a+b) \cdot 2\sqrt{ab} + 4ab \geq 0$$

$$(a+b)^2 + 4ab \geq 4(a+b)\sqrt{ab}, \quad \frac{(a+b)^2}{2(a+b)} + \frac{4ab}{2(a+b)} \geq 2\sqrt{ab}$$

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$$\frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{ab}$$

$$\sum_{cyc} \frac{a+b}{2} + 2 \sum_{cyc} \frac{ab}{a+b} \geq 2 \sum_{cyc} \sqrt{ab}$$

$$a+b+c + 2 \sum_{cyc} \frac{ab}{a+b} \geq 2 \sum_{cyc} \sqrt{ab}$$

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 - 2 \sum_{cyc} \sqrt{ab} + 2 \sum_{cyc} \frac{ab}{a+b} \geq 2 \sum_{cyc} \sqrt{ab}$$

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2 \sum_{cyc} \frac{ab}{a+b} \geq 4 \sum_{cyc} \sqrt{ab}$$

Equality holds for $a = b = c$.

Solution 2 by Marin Chirciu-Romania

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2 \sum_{cyc} \frac{ab}{a+b} \geq 4(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \Leftrightarrow$$

$$\Leftrightarrow a+b+c + 2 \sum_{cyc} \sqrt{ab} + 2 \sum_{cyc} \frac{ab}{a+b} \geq 4 \sum_{cyc} \sqrt{ab} \Leftrightarrow a+b+c + 2 \sum_{cyc} \frac{ab}{a+b} \geq$$

$$\geq 2 \sum_{cyc} \sqrt{ab} \Leftrightarrow \sum_{cyc} \frac{a+b}{2} + 2 \sum_{cyc} \frac{ab}{a+b} \geq 2 \sum_{cyc} \sqrt{ab}$$

$$\text{which follows from } \frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{ab}$$

Indeed:

$$\frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{ab} \Leftrightarrow (a+b)^2 + 4ab \geq 4\sqrt{ab}(a+b) \Leftrightarrow$$

$$\Leftrightarrow a^2 + 6ab + b^2 \geq 4\sqrt{ab}(a+b) \Leftrightarrow$$

$$\Leftrightarrow (a^2 + 6ab + b^2)^2 \geq 16ab(a+b)^2 \Leftrightarrow a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4 \geq 0 \Leftrightarrow$$

$$\Leftrightarrow (a-b)^4 \geq 0 \text{ with equality for } a = b.$$

$$\text{Summing the inequalities } \frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{ab} \text{ we obtain } \sum_{cyc} \frac{a+b}{2} + 2 \sum_{cyc} \frac{ab}{a+b} \geq 2 \sum_{cyc} \sqrt{ab}$$

Equality holds if and only if $a = b = c$.

Remark.

The problem can be developed.

If $a, b, c > 0$ then:

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$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + \lambda \sum \frac{ab}{a+b} \geq \left(3 + \frac{\lambda}{2}\right) (\sqrt{ab} + \sqrt{bc} + \sqrt{ca})$$

Marin Chirciu

Solution

$$\begin{aligned} & (\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + \lambda \sum \frac{ab}{a+b} \geq \left(3 + \frac{\lambda}{2}\right) (\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \Leftrightarrow \\ & \Leftrightarrow a + b + c + 2 \sum \sqrt{ab} + \lambda \sum \frac{ab}{a+b} \geq \left(3 + \frac{\lambda}{2}\right) \sum \sqrt{ab} \Leftrightarrow \\ & a + b + c + \lambda \sum \frac{ab}{a+b} \geq \left(1 + \frac{\lambda}{2}\right) \sum \sqrt{ab} \Leftrightarrow \\ & \sum \frac{a+b}{2} + \lambda \sum \frac{ab}{a+b} \geq \left(1 + \frac{\lambda}{2}\right) \sum \sqrt{ab}, \text{ which follows from } \frac{a+b}{2} + \frac{\lambda}{a+b} \geq \left(1 + \frac{\lambda}{2}\right) \sqrt{ab} \end{aligned}$$

Indeed:

$$\begin{aligned} & \frac{a+b}{2} + \frac{\lambda ab}{a+b} \geq \left(1 + \frac{\lambda}{2}\right) \sqrt{ab} \Leftrightarrow (a+b)^2 + 2\lambda ab \geq (\lambda+2)(a+b)\sqrt{ab} \Leftrightarrow \\ & a^2 + 2(\lambda+1)ab + b^2 \geq (\lambda+2)(a+b)\sqrt{ab} \Leftrightarrow \\ & \Leftrightarrow (a^2 + 2(\lambda+1)ab + b^2)^2 \geq (\lambda+2)^2(a+b)^2 ab \Leftrightarrow \\ & a^4 - \lambda^2 a^3 b + 2(\lambda^2 - 1)a^2 b^2 - \lambda^2 ab^3 + b^4 \geq 0 \Leftrightarrow \\ & (a-b)^2(a^2 + (2-\lambda^2)ab + b^2) \geq 0 \end{aligned}$$

with equality for $a = b$.

Summing the inequalities $\frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{ab}$ we obtain $\sum \frac{a+b}{2} + 2 \sum \frac{ab}{a+b} \geq 2 \sum \sqrt{ab}$

Equality holds if and only if $a = b = c$.

Solution 3 by Tapas Das – India

$$\begin{aligned} & (\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2 \left(\frac{ab}{a+b} + \frac{bc}{c+a} + \frac{ca}{a+b} \right) \stackrel{\text{Bergstrom}}{\geq} \\ & \geq (a+b+c) + 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) + 2 \frac{(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})^2}{2(a+b+c)} \\ & = 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) + \frac{(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})^2}{(a+b+c)} + (a+b+c) \\ & \stackrel{AM-GM}{\geq} 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) + 2 \sqrt{\frac{(a+b+c)(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})^2}{a+b+c}} \geq \end{aligned}$$

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$$\geq 4(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})$$

Equality holds if and only if $a = b = c$.

Solutions 4, 5 by Kunihiko Chikaya-Tokyo-Japan

Solution 4:

Using the AM-GM inequality, $\frac{a+b}{2} + \frac{2ab}{a+b} \geq 2\sqrt{\frac{a+b}{2} \cdot \frac{2ab}{a+b}} = 2\sqrt{ab}$.

Equality $\frac{a+b}{2} = \frac{2ab}{a+b}$ and $a > 0, b > 0 \Leftrightarrow (a-b)^2 = 0$ ($a, b > 0$) $\Leftrightarrow a = b$.

Analogously, we have $\frac{b+c}{2} + \frac{2bc}{b+c} \geq 2\sqrt{bc}$ and $\frac{c+a}{2} + \frac{2ca}{c+a} \geq 2\sqrt{ca}$.

Adding these three inequalities gives

$$\begin{aligned} a + b + c + 2\left(\frac{ab}{a+b} + \frac{bc}{b+c} + \frac{ca}{c+a}\right) &\geq 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \Leftrightarrow \\ (\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2\left(\frac{ab}{a+b} + \frac{bc}{b+c} + \frac{ca}{c+a}\right) &\geq 4(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}). \end{aligned}$$

Equality attains if only if $a = b = c$.

Solution 5:

$$\begin{aligned} &(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 + 2\left(\frac{ab}{a+b} + \frac{bc}{b+c} + \frac{ca}{c+a}\right) - 4(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \\ &= a + b + c + 2\left(\frac{ab}{a+b} + \frac{bc}{b+c} + \frac{ca}{c+a}\right) - 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \\ &= \left(\sqrt{\frac{a+b}{2}} - \sqrt{\frac{2ab}{a+b}}\right)^2 + \left(\sqrt{\frac{b+c}{2}} - \sqrt{\frac{2bc}{b+c}}\right)^2 + \left(\sqrt{\frac{c+a}{2}} - \sqrt{\frac{2ca}{c+a}}\right)^2 \geq 0 \end{aligned}$$

Equality attains if only if

$$\frac{a+b}{2} = \frac{2ab}{a+b}, \frac{b+c}{2} = \frac{2bc}{b+c}, \frac{c+a}{2} = \frac{2ca}{c+a} \text{ and}$$

$$a > 0, b > 0, c > 0 \Leftrightarrow a = b = c.$$

SP.549 Let be $\triangle ABC$ with sides a, b, c . Let be $x, y, z \in \mathbb{R}$ such that:

$$\cos x = \frac{a}{b+c}; \cos y = \frac{b}{c+a}; \cos z = \frac{c}{a+b}$$

Prove that:

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$$\tan \frac{x}{2} \tan \frac{y}{2} \tan \frac{z}{2} \leq \frac{\sqrt{3}}{9}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned} \tan \frac{x}{2} &= \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \sqrt{\frac{1 - \frac{a}{b+c}}{1 + \frac{a}{b+c}}} = \sqrt{\frac{b+c-a}{a+b+c}} = \\ &= \sqrt{\frac{a+b+c-2a}{a+b+c}} = \sqrt{\frac{2s-2a}{2s}} = \sqrt{\frac{s-a}{s}} \\ s &= \frac{a+b+c}{2} - \text{semiperimeter; } F - \text{area} \\ \tan \frac{x}{2} \tan \frac{y}{2} \tan \frac{z}{2} &= \sqrt{\frac{s-a}{s}} \cdot \sqrt{\frac{s-b}{s}} \cdot \sqrt{\frac{s-c}{s}} = \\ &= \sqrt{\frac{(s-a)(s-b)(s-c)}{s^3}} = \sqrt{\frac{s(s-a)(s-b)(s-c)}{s^4}} = \\ &\stackrel{\text{HERON}}{=} \frac{F}{s^2} = \frac{rs}{s^2} = \frac{r}{s} \stackrel{\text{MITRINOVIC}}{\leq} \frac{r}{3\sqrt{3}r} = \frac{1}{3\sqrt{3}} = \frac{\sqrt{3}}{9} \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

Using $\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x} = \frac{1 - \frac{a}{b+c}}{1 + \frac{a}{b+c}} = \frac{b+c-a}{b+c+a} = \frac{2s-2a}{2s} = \frac{s-a}{s}$ we obtain:

$$LHS = \prod \sqrt{\frac{s-a}{s}} = \sqrt{\frac{r^2 s}{s^3}} \stackrel{\text{Mitrinovic}}{\leq} \frac{r}{3\sqrt{3}r} = \frac{1}{3\sqrt{3}} = \frac{\sqrt{3}}{9} = RHS$$

Equality holds if and only if the triangle is equilateral.

Remark.

In the same way.

Let be ΔABC with a, b, c sides and $x, y, z \in \mathbb{R}$ such that

$$x = \frac{a}{b+c}, \cos y = \frac{b}{c+a}, \cos z = \frac{c}{a+b}$$

Prove that:

$$\cot \frac{x}{2} + \cot \frac{y}{2} + \cot \frac{z}{2} \geq 3\sqrt{3}$$

Marin Chirciu

Solution:

Using $\cot^2 \frac{x}{2} = \frac{1+\cos x}{1-\cos x} = \frac{1+\frac{a}{b+c}}{1-\frac{a}{b+c}} = \frac{b+c+a}{b+c-a} = \frac{2s}{2s-2a} = \frac{s}{s-a}$ we obtain:

$$\begin{aligned} LHS &= \sum \sqrt{\frac{s}{s-a}} \stackrel{AM-GM}{\geq} 3 \sqrt[3]{\prod \sqrt{\frac{s}{s-a}}} = 3 \sqrt[3]{\sqrt{\frac{s^3}{r^2 s}}} = 3 \sqrt[3]{\frac{s}{r}} \stackrel{\text{Mitrinovic}}{\geq} \\ &\geq 3 \sqrt[3]{\frac{3\sqrt{3}r}{r}} = 3 \sqrt[3]{3\sqrt{3}} = 3 \sqrt[3]{(\sqrt{3})^3} = 3\sqrt{3} = RHS \end{aligned}$$

Equality holds if and only if the triangle is equilateral.

SP.550 Prove that $(\forall)x \in (0; 1)$ and $n \in \mathbb{N}^*$, we have the inequality:

$$x(1+x)(1-x^n) < \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n+1}$$

Proposed by Gheorghe Molea – Romania

Solution 1 by proposer

$$\begin{aligned} (1-x^2)(x^1+x^3+x^5+\dots+x^{2n+1}) &= x - x^{2n+3} = x(1-x^{2n+2}) < x \Rightarrow \\ \frac{x}{1-x^2} &> x^1+x^3+x^5+\dots+x^{2n+1} > (n+1) \sqrt[n+1]{x^1 \cdot x^3 \cdot x^5 \cdot \dots \cdot x^{2n+1}} = \\ &= (n+1) \sqrt[n+1]{x^{(n+1)^2}} = (n+1)x^{n+1} \Rightarrow \frac{x}{1-x^2} > (n+1)x^{n+1} \Rightarrow \\ \frac{1}{1-x^2} &> (n+1)x^n \Rightarrow x^n - x^{n+2} < \frac{1}{n+1}, (\forall)x \in (0; 1) \text{ and } n \in \mathbb{N}^* \end{aligned}$$

$$x^1 - x^3 < \frac{1}{2}; x^2 - x^4 < \frac{1}{3}; x^3 - x^5 < \frac{1}{4}; \dots$$

$$x^{n-2} - x^n < \frac{1}{n-1}; x^{n-1} - x^{n+1} < \frac{1}{n};$$

$$x^n - x^{n+2} < \frac{1}{n+1}$$

By adding we obtain:

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$$x + x^2 - x^{n+1} - x^{n+2} < \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n+1}$$

$$\Rightarrow x(1+x)(1-x^n) < \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n+1}$$

Solution 2 by Marin Chirciu-Romania

We prove through mathematical induction after $n \in \mathbb{N}^*$, the inequality from enunciation.

For $n = 1$ we write the inequality $x(1+x)(1-x) < \frac{1}{2} \Leftrightarrow 2x(1-x^2) < 1 \Leftrightarrow$
 $\Leftrightarrow 2x^3 - 2x + 1 > 0$.

We consider the function $f: (0, 1) \rightarrow \mathbb{R}, f(x) = 2x^3 - 2x + 1$, we have:

$$f'(x) = 6x^2 - 2; f'(x) = 0 \Leftrightarrow x = \frac{1}{\sqrt{3}}.$$

Preparing the variation table of the function we have:

$$f'(x) < 0 \text{ for } x \in \left(0, \frac{1}{\sqrt{3}}\right) \Rightarrow f \downarrow \text{ and } f'(x) > 0 \text{ for } x \in \left(\frac{1}{\sqrt{3}}, 1\right) \Rightarrow f \uparrow.$$

We have $f(0) = f(1) = 0$ and $f\left(\frac{1}{\sqrt{3}}\right) = 1 - \frac{4}{3\sqrt{3}} > 0 \Rightarrow$ point $A\left(\frac{1}{\sqrt{3}}, 1 - \frac{4}{3\sqrt{3}}\right)$ is a point of minimum $\Rightarrow \text{Im } f = \left(1 - \frac{4}{3\sqrt{3}}, 1\right) \subset (0, \infty) \Rightarrow f(x) > 0, x \in (0, 1)$.

Next, we assume that the inequality holds for $k \in \mathbb{N}^*$ and we show that it is true for $k + 1$.

We prove: $x(1+x)(1-x^k) < \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k+1} \Rightarrow$

$$\Rightarrow x(1+x)(1-x^{k+1}) < \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k+1} + \frac{1}{k+2}.$$

It remains to prove that:

$$x(1+x)(1-x^{k+1}) < x(1+x)(1-x^k) + \frac{1}{k+2} \Leftrightarrow$$

$$\Leftrightarrow x(1+x)(x^k - x^{k+1}) < \frac{1}{k+2} \Leftrightarrow$$

$$\Leftrightarrow x^{k+1}(1-x^2) < \frac{1}{k+2}.$$

We consider the function $f: (0, 1) \rightarrow \mathbb{R}, f(x) = x^{k+1}(1-x^2) - \frac{1}{k+2}$;

we have $f'(x) = x^k(k+1 - (k+3)x^2); f'(x) = 0 \Leftrightarrow \sqrt{\frac{k+1}{k+3}}$.

Preparing the variation table of the function we have:

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$$f'(x) > 0 \text{ for } x \in \left(0, \sqrt{\frac{k+1}{k+3}}\right) \Rightarrow f \uparrow \text{ and } f'(x) < 0 \text{ for } x \in \left(\sqrt{\frac{k+1}{k+3}}, 1\right) \Rightarrow f \downarrow$$

We have $f(0) = f(1) = \frac{-1}{k+2}$ and $f\left(\sqrt{\frac{k+1}{k+3}}\right) < 0 \Rightarrow$ point $A\left(\sqrt{\frac{k+1}{k+3}}, f\sqrt{\frac{k+1}{k+3}}\right)$ is a point of

maximum with $f\sqrt{\frac{k+1}{k+3}} < 0 \Rightarrow f(x) < 0, x \in (0, 1)$

This issue is closed.

SP.551 Find:

$$\Omega = \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^3(1 + x^{4n-8})}{(1 + x^4)^n} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Denote:

$$\Omega(n) = \int_0^1 \frac{x^3(1 + x^{4n-8})}{(1 + x^4)^n} dx$$

$$\text{For: } y = x^4 \Rightarrow dy = 4x^3 dx \Rightarrow x^3 dx = \frac{1}{4} dy$$

$$x = 0 \Rightarrow y = 0$$

$$x = 1 \Rightarrow y = 1$$

$$\begin{aligned} \Omega(n) &= \int_0^1 \frac{(1 + y^{n-2}) \cdot \frac{1}{4} dy}{(1 + dy)^n} = \\ &= \frac{1}{4} \int_0^1 \frac{dy}{(1 + y)^n} + \frac{1}{4} \int_0^1 \frac{y^{n-2}}{(1 + y)^n} dy \end{aligned}$$

Let's observe that:

$$\begin{aligned} \left(\frac{y}{1+y}\right)' &= \frac{y'(y+1) - y(y+1)'}{(y+1)^2} = \frac{y+1-y}{(y+1)^2} = \frac{1}{(y+1)^2} \\ \Omega(n) &= \frac{1}{4} \int_0^1 (1+y)' (1+y)^{-n} dy + \frac{1}{4} \int_0^1 \left(\frac{y}{1+y}\right)^{n-2} \cdot \frac{1}{(1+y)^2} dy \\ \Omega(n) &= \frac{1}{4} \cdot \frac{(1+y)^{-n+1}}{-n+1} \Big|_0^1 + \frac{1}{4} \int_0^1 \left(\frac{y}{1+y}\right)^{n-2} \cdot \left(\frac{y}{1+y}\right)' dy \end{aligned}$$

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$$\Omega(n) = \frac{1}{4} \left(\frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} \right) + \frac{1}{4(n-1)} \cdot \left(\frac{y}{1+y} \right)^{n-1} \Big|_0^1$$

$$\Omega(n) = \frac{1}{4(1-n)2^{n-1}} + \frac{1}{4(n-1)} + \frac{1}{4(n-1)} \left(\frac{1}{2^{n-1}} - 0 \right)$$

$$n\Omega(n) = \frac{n}{4(1-n)} \cdot \frac{1}{2^{n-1}} + \frac{n}{4(n-1)} + \frac{n}{4(n-1)} \cdot \frac{1}{2^{n-1}}$$

$$\Omega = \lim_{n \rightarrow \infty} n\Omega(n) = \frac{1}{4} \cdot 0 + \frac{1}{4} + \frac{1}{4} \cdot 0$$

$$\Omega = \frac{1}{4}$$

Solution 2 by Marin Chirciu-Romania

We find $\int_0^1 \frac{x^3(1+x^{4n-8})}{(1+x^4)^n} dx$. We make the substitution $x^4 = t \Rightarrow 4x^3 dx = dt$.

We obtain:

$$\begin{aligned} \int_0^1 \frac{x^3(1+x^{4n-8})}{(1+x^4)^n} dx &= \frac{1}{4} \int_0^1 \frac{1+t^{n-2}}{(1+t)^n} dt = \frac{1}{4} \left(\int_0^1 \frac{1}{(1+t)^n} dt + \int_0^1 \frac{t^{n-2}}{(1+t)^n} dt \right) = \\ &= \frac{1}{4} \left(\int_0^1 (t+1)^{-n} dt + \int_0^1 \left(\frac{t}{t+1} \right)^{n-2} \frac{1}{t^2} dt \right) = \\ &= \frac{1}{4} \left(\frac{(t+1)^{-n+1}}{-n+1} \Big|_0^1 + \int_0^1 \left(\frac{t}{t+1} \right)^{n-2} \left(\frac{t}{t+1} \right)' dt \right) = \\ &= \frac{1}{4} \left(\frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} + \frac{\left(\frac{t}{t+1} \right)^{n-2+1}}{n-2+1} \Big|_0^1 \right) = \\ &= \frac{1}{4} \left(\frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} + \frac{\left(\frac{1}{2} \right)^{n-1}}{n-1} - 0 \right) = \frac{1}{4} \left(\frac{2^{-n+1}}{-n+1} + \frac{1}{n-1} + \frac{2^{-n+1}}{n-1} \right) = \\ &= \frac{1}{4} \cdot \frac{1}{n-1} = \frac{1}{4(n-1)} \end{aligned}$$

$$\text{It follows } \int_0^1 \frac{x^3(1+x^{4n-8})}{(1+x^4)^{6n}} dx = \frac{1}{4(n-1)}.$$

$$\text{Finally, } \Omega = \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^3(1+x^{4n-8})}{(1+x^4)^n} dx = \lim_{n \rightarrow \infty} n \frac{1}{4(n-1)} = \frac{1}{4}$$

Remark.

Let $k > 1$ fixed. Find:

$$\Omega = \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^{k-1}(1+x^{kn-2k})}{(1+x^k)^n} dx$$

Marin Chirciu

Solution

We find $\int_0^1 \frac{x^{k-1}(1+x^{kn-2k})}{(1+x^k)^n}$. We make the substitution $x^k = t \Rightarrow kx^{k-1}dx = dt$.

We obtain

$$\begin{aligned} \int_0^1 \frac{x^{k-1}(1+x^{kn-2k})}{(1+x^k)^n} &= \frac{1}{k} \int_0^1 \frac{1+t^{n-2}}{(1+t)^n} dt = \\ &= \frac{1}{k} \left(\int_0^1 \frac{1}{(1+t)^n} dt + \int_0^1 \frac{t^{n-2}}{(1+t)^n} dt \right) = \frac{1}{k} \left(\int_0^1 (t+1)^{-n} dt + \int_0^1 \left(\frac{t}{t+1} \right)^{n-2} \frac{1}{t^2} dt \right) = \\ &= \frac{1}{k} \left(\frac{(t+1)^{-n+1}}{-n+1} \Big|_0^1 + \int_0^1 \left(\frac{t}{t+1} \right)^{n-2} \left(\frac{t}{t+1} \right)' dt \right) = \\ &= \frac{1}{k} \left(\frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} + \frac{\left(\frac{t}{t+1} \right)^{n-2+1}}{n-2+1} \Big|_0^1 \right) = \frac{1}{k} \left(\frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} + \frac{\left(\frac{1}{2} \right)^{n-1}}{n-1} - 0 \right) = \\ &= \frac{1}{k} \left(\frac{2^{-n+1}}{-n+1} + \frac{1}{n-1} + \frac{2^{-n+1}}{n-1} \right) = \frac{1}{k} \cdot \frac{1}{n-1} = \frac{1}{k(n-1)} \end{aligned}$$

It follows $\int_0^1 \frac{x^{k-1}(1+x^{kn-2k})}{(1+x^k)^n} = \frac{1}{k(n-1)}$. Finally, $\Omega = \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^{k-1}(1+x^{kn-2k})}{(1+x^k)^n} dx =$

$$\lim_{n \rightarrow \infty} \frac{1}{k(n-1)} = \frac{1}{k}$$

Note: For $k = 4$ we obtain Problem SP.551 from RMM Nr. 37 – Summer 2025

SP.551. Find:

$$\Omega = \lim_{n \rightarrow \infty} \frac{x^3(1+x^{4n-8})}{(1+x^4)^n} dx$$

Daniel Sitaru – Romania

Solution 3 by Angel Plaza-Spain

After changing variable $x^4 = t$ we have:

$$\begin{aligned}\Omega &= \lim_{n \rightarrow \infty} \frac{n}{4} \int_0^1 \frac{1 + t^{n-2}}{(1+t)^n} dt \\ \int_0^1 \frac{1 + t^{n-2}}{(1+t)^n} dt &= \int_0^1 \frac{t^2 + t^n}{t^2(1+t)^n} dt = \int_0^1 \frac{1}{(1+t)^n} dt + \int_0^1 \frac{t^n}{t^2(1+t)^n} dt \\ &= \left. \frac{(1+x)^{-n+1}}{-n+1} \right|_0^1 + \frac{1}{n-1} \cdot \left. \left(\frac{x}{1+x} \right)^{n-1} \right|_0^1 = \frac{2^{-n+1}}{-n+1} - \frac{1}{-n+1} + \frac{1}{(n-1)2^{n-1}} \\ &= \frac{1 - 2^{-n+1}}{n-1} - \frac{1}{(n-1)2^{n-1}} = \frac{1}{n-1} \\ \text{Therefore, } \Omega &= \lim_{n \rightarrow \infty} \frac{n}{4} \frac{1}{n-1} = \frac{1}{4}.\end{aligned}$$

SP.552 If $a > 0$ then find:

$$\Omega = \int_{-a}^a \log_a \left(\sqrt{a^2 x^2 + 1} - ax \right) dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\Omega = \int_{-a}^a \log_a \left(\sqrt{a^2 x^2 + 1} - ax \right) dx$$

For $y = -x \Rightarrow$

$$x = -a \Rightarrow y = a$$

$$x = a \Rightarrow y = -a$$

$$dx = -dy$$

$$\Omega = \int_a^{-a} \log_a \left(\sqrt{a^2 y^2 + 1} + ay \right) (-dy)$$

$$\Omega = \int_{-a}^a \log_a \left(\sqrt{a^2 y^2 + 1} + ay \right) dy = \int_{-a}^a \log_a \left(\sqrt{a^2 x^2 + 1} + ax \right) dx$$

$$2\Omega = \int_{-a}^a \log_a \left(\sqrt{a^2 x^2 + 1} - ax \right) dx + \int_{-a}^a \log_a \left(\sqrt{a^2 x^2 + 1} + ax \right) dx$$

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$$2\Omega = \int_{-a}^a \log_a \left(\left(\sqrt{a^2 x^2 + 1} \right)^2 - (ax)^2 \right) dx$$

$$2\Omega = \int_{-a}^a \log_a (a^2 x^2 + 1 - a^2 x^2) dx$$

$$2\Omega = \int_{-a}^a \log_a 1 dx \Rightarrow 2\Omega = 0 \Rightarrow \Omega = 0$$

Solution 2 by Marin Chirciu-Romania

The function $f: [-a, a] \rightarrow \mathbb{R}, f(x) = \log_a(\sqrt{a^2 x^2 + 1} - ax)$ is odd, see:

$$\begin{aligned} f(x) + f(-x) &= \log_a(\sqrt{a^2 x^2 + 1} - ax) + \log_a(\sqrt{a^2 x^2 + 1} + ax) = \\ &= \log_a(\sqrt{a^2 x^2 + 1} - ax)(\sqrt{a^2 x^2 + 1} + ax) = \\ &= \log_a(a^2 x^2 + 1 - a^2 x^2) = \log_a 1 = 0 \end{aligned}$$

Using the fact that the integral over a symmetric interval of an odd function is zero, we

deduce that:

$$\Omega = \int_{-a}^a \log_a(\sqrt{a^2 x^2 + 1} - ax) dx = 0$$

Remark.

If $a > 0, a \neq 1$ then find:

$$\Omega = \int_a^a \log_a(\sqrt{a^2 x^2 + 1} + ax) dx$$

Solution

Analogous

$$\Omega = \int_{-a}^a \log_a(\sqrt{a^2 x^2 + 1} + ax) dx = 0$$

SP.553 If $0 \leq a \leq b < 1$ then:

$$6 \int_a^b \log \left(\frac{1+x}{1-x} \right) dx \geq (b^2 - a^2)(b^2 + a^2 + 6)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let be $f: [a, b] \rightarrow \mathbb{R}; f(x) = \log\left(\frac{1+x}{1-x}\right) - 2x - \frac{2x^3}{3}$

$$f(x) = \log(1+x) - \log(1-x) - 2x - \frac{2x^3}{3}$$

$$f'(x) = \frac{1}{1+x} + \frac{1}{1-x} - 2 - 2x^2$$

$$f'(x) = \frac{2}{1-x^2} - 2(1+x^2) = 2\left(\frac{1}{1-x^2} - (1+x^2)\right)$$

$$f'(x) = 2 \cdot \frac{1-1+x^4}{1-x^2} = \frac{2x^4}{1-x^2} \geq 0$$

f increasing on $[a, b] \subset [0, 1] \Rightarrow$

$$\min_{x \in [a, b]} f(x) = f(0) = 0 \Rightarrow f(x) \geq 0; (\forall) x \in [a, b]$$

$$\log\left(\frac{1+x}{1-x}\right) \geq 2x + \frac{2x^3}{3}$$

$$\begin{aligned} \int_a^b \log\left(\frac{1+x}{1-x}\right) dx &\geq \int_a^b 2x dx + \frac{2}{3} \int_a^b x^3 dx \\ &= b^2 - a^2 + \frac{2}{3} \cdot \frac{b^4 - a^4}{4} = b^2 - a^2 + \frac{1}{6} (b^2 - a^2)(b^2 + a^2) = \\ &= \frac{1}{6} (b^2 - a^2)(6 + b^2 + a^2) \end{aligned}$$

$$6 \int_a^b \log\left(\frac{1+x}{1-x}\right) dx \geq (b^2 - a^2)(6 + b^2 + a^2)$$

Equality holds for $a = b$.

Solution 2 by Marin Chirciu-Romania

We prove that $\log\left(\frac{1+x}{1-x}\right) \geq 2x + \frac{2}{3}x^3, 0 \leq x < 1$.

Indeed: We consider the function $f(x) = \log\left(\frac{1+x}{1-x}\right) - 2x - \frac{2}{3}x^3, 0 \leq x < 1$

We have $f'(x) = \frac{2x^4}{1-x^2} \geq 0 \Rightarrow f$ is increasing on $[0, 1) \Rightarrow f(x) \geq f(0) = 0 \Rightarrow$

$$\Rightarrow \log\left(\frac{1+x}{1-x}\right) - 2x - \frac{2}{3}x^3 \geq 0 \Rightarrow \log\left(\frac{1+x}{1-x}\right) \geq 2x + \frac{2}{3}x^3.$$

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Using $\log\left(\frac{1+x}{1-x}\right) \geq 2x + \frac{2}{3}x^3$ we obtain:

$$\begin{aligned} 6 \int_a^b \log\left(\frac{1+x}{1-x}\right) dx &\geq 6 \int_a^b \left(2x + \frac{2}{3}x^3\right) dx = 6 \int_a^b \left(x^2 + \frac{x^4}{6}\right) dx = \\ &= 6 \left(b^2 - a^2 + \frac{b^4 - a^4}{6}\right) = (b^2 - a^2)(b^2 + a^2 + 6). \end{aligned}$$

Remark: In the same way.

If $0 \leq a \leq b$ then:

$$6 \int_a^b \log(1+x) dx \geq (a-b)(a^2 + ab + b^2 - 3a - 3b)$$

Marin Chirciu

Solution

We prove that $\log(1+x) \geq x - \frac{1}{2}x^2, x \geq 0$

Indeed: We consider the function $f(x) = \log(1+x) - x + \frac{1}{2}x^2, x \geq 0$.

We have $f'(x) = \frac{x^2}{1+x} \geq 0 \Rightarrow f$ is increasing on $[0, \infty) \Rightarrow f(x) \geq f(0) = 0 \Rightarrow$

$$\Rightarrow \log(1+x) - x + \frac{1}{2}x^2 \geq 0 \Rightarrow \log(1+x) \geq x - \frac{1}{2}x^2$$

Using $\log(1+x) \geq x - \frac{1}{2}x^2$ we obtain:

$$\begin{aligned} 6 \int_a^b \log(1+x) dx &\geq 6 \int_a^b \left(x - \frac{1}{2}x^2\right) dx = 6 \left(\frac{x^2}{2} - \frac{x^3}{6}\right) \Big|_a^b = 6 \left(\frac{b^2 - a^2}{2} - \frac{b^3 - a^3}{6}\right) = \\ &= 3(b^2 - a^2) - (b^3 - a^3) = (a-b)(a^2 + ab + b^2 - 3a - 3b) \end{aligned}$$

SP.554 Find:

$$\Omega = \int \frac{8x-1}{e^{8x}+7x} dx; x \in (0, \infty)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\Omega = \int \frac{8x-1}{e^{8x}+7x} dx = \frac{1}{7} \int \frac{7(8x-1)}{e^{8x}+7x} dx =$$

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$$\begin{aligned}
 &= \frac{1}{7} \int \frac{56x - 7}{e^{8x} + 7x} dx = \frac{1}{7} \int \frac{8e^{8x} + 56x - 8e^{8x} - 7}{e^{8x} + 7x} dx = \\
 &= \frac{1}{7} \int \frac{8(e^{8x} + 7x)}{e^{8x} + 7x} dx - \frac{1}{7} \int \frac{8e^{8x} + 7}{e^{8x} + 7x} dx = \\
 &= \frac{1}{7} \int 8 dx - \frac{1}{7} \int \frac{(e^{8x} + 7x)'}{e^{8x} + 7x} dx = \frac{8x}{7} - \frac{1}{7} \ln(e^{8x} + 7x) + C
 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

$$\Omega = \int \frac{8x - 1}{e^{8x} + 7x} dx = \int \left(\frac{8}{7} - \frac{1}{7} \cdot \frac{8e^{8x} + 7}{e^{8x} + 7x} \right) dx = \frac{8}{7}x - \frac{1}{7} \ln(e^{8x} + 7x) + C$$

Remark:

If $m \in \mathbb{R}, n \in (0, \infty)$ then find:

$$\Omega = \int \frac{mx - 1}{e^{mx} + nx} dx.$$

Solution.

$$\Omega = \int \frac{mx - 1}{e^{mx} + nx} dx = \int \left(\frac{m}{n} - \frac{1}{n} \cdot \frac{me^{mx} + n}{e^{mx} + nx} \right) dx = \frac{m}{n}x - \frac{1}{n} \ln(e^{mx} + nx) + C$$

SP.555 Find:

$$\Omega = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x \frac{dt}{t + e^t} - 1 \right)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let be $f, F: (0, \infty) \rightarrow \mathbb{R}$

$$f(x) = \frac{1}{x + e^x}; F'(x) = f(x)$$

$$f'(x) = \frac{-1 - e^x}{(x + e^x)^2}; f'(0) = \frac{-1 - 1}{(0 + 1)^2} = -2$$

$$\begin{aligned}
 \Omega &= \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x f(t) dt - 1 \right) = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} (F(x) - F(0) - 1) \right) = \\
 &= \lim_{x \rightarrow 0} \frac{F(x) - F(0) - x}{x^2} = \lim_{x \rightarrow 0} \frac{F'(x) - (F(0))' - x'}{2x} =
 \end{aligned}$$

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$$= \lim_{x \rightarrow 0} \frac{f(x) - 0 - 1}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \frac{1}{2} f'(0) = \frac{1}{2} \cdot (-2) = -1$$

Solution 2 by Marin Chirciu-Romania

$$\begin{aligned} \Omega &= \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x \frac{dt}{t + e^t} - 1 \right) \\ \Omega &= \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x \frac{dt}{t + e^t} - 1 \right) = \lim_{x \rightarrow 0} \frac{\int_0^x \frac{dt}{t + e^t} - x}{x^2} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{x + e^x} - 1}{2x} = \\ &= \lim_{x \rightarrow 0} \frac{1 - x - e^x}{2x(x + e^x)} = \lim_{x \rightarrow 0} \frac{1}{2(x + e^x)} \lim_{x \rightarrow 0} \frac{1 - x - e^x}{x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{1 - x - e^x}{x} \stackrel{\left(\frac{0}{0}\right)}{=} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{-1 - e^x}{1} = \frac{1}{2} (-2) = -1 \end{aligned}$$

Remark: In the same way.

Find:

$$\Omega = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x \frac{dt}{\sin t + e^t} - 1 \right)$$

Marin Chirciu

Solution

$$\begin{aligned} \Omega &= \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \int_0^x \frac{dt}{\sin t + e^t} - 1 \right) = \lim_{x \rightarrow 0} \frac{\int_0^x \frac{dt}{\sin t + e^t} - x}{x^2} \stackrel{\left(\frac{0}{0}\right)}{=} \\ &= \lim_{x \rightarrow 0} \frac{\frac{1}{\sin x + e^x} - 1}{2x} = \lim_{x \rightarrow 0} \frac{1 - \sin x - e^x}{2x(\sin x + e^x)} = \\ &= \lim_{x \rightarrow 0} \frac{1}{2(\sin x + e^x)} \lim_{x \rightarrow 0} \frac{1 - \sin x - e^x}{x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{1 - \sin x - e^x}{x} \stackrel{\left(\frac{0}{0}\right)}{=} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{-\cos x - e^x}{1} = \frac{1}{2} (-2) = -1 \end{aligned}$$

Solution 3 by Angel Plaza-Spain

The limit follows by the L'Hospital rule and the fundamental theorem of the integral calculus

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$$\Omega = \lim_{n \rightarrow 0} \frac{\int_0^x \frac{dt}{t+e^t} - x}{x^2} = \lim_{n \rightarrow 0} \frac{\frac{1}{x+e^x} - 1}{2x} = \lim_{n \rightarrow 0} \frac{\frac{-(1+e^x)}{(x+e^x)^2}}{2} = 1.$$

Alternatively, the result may be obtained by using the Taylor expansion at $x = 0$ of function $f(x) = \int_0^x \frac{dt}{t+e^t}$ of the second order with Lagrange remainder.

UNDERGRADUATE PROBLEMS

UP.541 Find:

$$\Omega = \int_1^{\sqrt{3}} \frac{x - \tan^{-1} x}{(1+x^2)^2 (\tan^{-1} x)^3} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let by $y = \tan^{-1} x \Rightarrow dy = \frac{1}{\cos^2 y} dy$; $x = \tan y$

For $x = 1 \Rightarrow y = \frac{\pi}{4}$. For $x = \sqrt{3} \Rightarrow y = \frac{\pi}{3}$.

$$\begin{aligned} \Omega &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan y - y}{(1 + \tan^2 y)^2 \cdot y^3} \cdot \frac{1}{\cos^2 y} dy = \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\frac{\sin y}{\cos y} - y}{\frac{1}{\cos^4 y} \cdot y^3} \cdot \frac{1}{\cos^2 y} dy = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{(\sin y - y \cos y) \cos y}{y^3} dy = \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin y \cos y - y \cos^2 y}{y^3} dy = -\frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sin y \cos y - y \cos^2 y) \cdot \left(\frac{-2}{y^3}\right) dy = \\ &= -\frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sin y \cos y - y \cos^2 y) \cdot \left(\frac{1}{y^2}\right)' dy \stackrel{IBP}{=} -\frac{1}{2} (\sin y \cos y - y \cos^2 y) \cdot \frac{1}{y^2} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \\ &\quad + \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\cos^2 y - \sin^2 y - \cos^2 y + 2y \sin y \cos y) \cdot \frac{1}{y^2} dy = \end{aligned}$$

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$$\begin{aligned}
 &= -\frac{1}{2y^2}(\sin y \cos y - y \cos^2 y) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \\
 &+ \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (2y \sin y \cos y - \sin^2 y) \cdot \frac{1}{y^2} dy = -\frac{1}{2 \cdot \frac{\pi^2}{9}} \left(\sin \frac{\pi}{3} \cos \frac{\pi}{3} - \frac{\pi}{3} \cos^2 \frac{\pi}{3} \right) + \\
 &+ \frac{1}{2 \cdot \frac{\pi^2}{16}} \left(\sin \frac{\pi}{4} \cos \frac{\pi}{4} - \frac{\pi}{4} \cos^2 \frac{\pi}{4} \right) + \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{\sin^2 y}{y} \right)' dy = \\
 &= -\frac{9}{2\pi^2} \left(\frac{\sqrt{3}}{4} - \frac{\pi}{12} \right) + \frac{8}{\pi^2} \left(\frac{1}{2} - \frac{\pi}{8} \right) + \frac{1}{2} \left(\frac{\sin^2 \frac{\pi}{3}}{\frac{\pi}{3}} - \frac{\sin^2 \frac{\pi}{4}}{\frac{\pi}{4}} \right) = \\
 &= \frac{1}{\pi^2} \left(-\frac{9\sqrt{3}}{8} + \frac{9\pi}{24} + 4 - \pi \right) + \frac{1}{\pi} \left(3 \cdot \frac{3}{4} - 4 \cdot \frac{1}{2} \right) = \frac{1}{\pi^2} \left(4 - \frac{9\sqrt{3}}{8} - \frac{5\pi}{8} \right) + \frac{1}{2\pi} \cdot \frac{1}{4} = \\
 &= \frac{1}{8\pi^2} (32 - 9\sqrt{3} - 5\pi) + \frac{1}{8\pi}
 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

Using the substitution $t = \arctan x \Rightarrow x = \tan t \Rightarrow dx = (1 + \tan^2 t) dt \Rightarrow$

$$\begin{aligned}
 \Omega &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan t - t}{(1 + \tan^2 t)^2 t^3} (1 + \tan^2 t) dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan t - t}{(1 + \tan^2 t) t^3} dt = \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan t}{(1 + \tan^2 t) t^3} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{t}{(1 + \tan^2 t) t^3} dt = \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\frac{\sin t}{\cos t}}{\frac{1}{\cos^2 t} t^3} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{t}{\frac{1}{\cos^2 t} t^3} dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin t \cos t}{t^3} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^2 t}{t^2} dt = \\
 &= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin 2t}{t^3} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1 + \cos 2t}{2t^2} dt = \\
 &= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{-1}{2t^2} \right)' \sin 2t dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2t^2} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} dt = \\
 &= \frac{1}{2} \left[\left(\frac{-1}{2t^2} \right) \sin 2t \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{-1}{2t^2} \right) \cdot 2 \cos 2t dt \right] - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2t^2} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} dt =
 \end{aligned}$$

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$$\begin{aligned}
 &= \frac{1}{2} \left[\left(\frac{-1}{2 \left(\frac{\pi}{3} \right)^2} \right) \sin \frac{2\pi}{3} + \left(\frac{1}{2 \left(\frac{\pi}{4} \right)^2} \right) \sin \frac{2\pi}{4} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{t^2} \cos 2t \, dt \right] + \frac{1}{2t} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} \, dt = \\
 &= \frac{1}{2} \left[\left(\frac{-1}{2 \left(\frac{\pi}{3} \right)^2} \right) \sin \frac{2\pi}{3} + \left(\frac{1}{2 \left(\frac{\pi}{4} \right)^2} \right) \sin \frac{2\pi}{4} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{t^2} \cos 2t \, dt \right] + \frac{1}{2 \cdot \frac{\pi}{3}} - \frac{1}{2 \cdot \frac{\pi}{4}} \\
 &\quad - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} \, dt \\
 &= \frac{1}{2} \left(\frac{-1}{2 \left(\frac{\pi}{3} \right)^2} \right) \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \left(\frac{1}{2 \left(\frac{\pi}{4} \right)^2} \right) \cdot 1 + \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{t^2} \cos 2t \, dt + \frac{1}{\frac{2\pi}{3}} - \frac{1}{\frac{2\pi}{4}} - \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{t^2} \, dt = \\
 &= \frac{-\sqrt{3}}{8 \left(\frac{\pi}{3} \right)^2} + \frac{1}{4 \left(\frac{\pi}{4} \right)^2} + \frac{3}{2\pi} - \frac{4}{2\pi} = \frac{-\sqrt{3}}{8 \frac{\pi^2}{9}} + \frac{1}{4 \frac{\pi^2}{16}} - \frac{1}{2\pi} = \frac{-9\sqrt{3}}{8\pi^2} + \frac{4}{\pi^2} - \frac{1}{2\pi}
 \end{aligned}$$

We deduce that $\Omega = \int_1^{\sqrt{3}} \frac{x - \arctan x}{(1+x^2)(\arctan x)^3} \, dx = \frac{4\pi}{\pi^2} - \frac{9\sqrt{3}}{8\pi^2} - \frac{1}{2\pi}$

Remark: The problem can be developed.

If $k \in \mathbb{N}$ find:

$$\Omega = \int_1^{\sqrt{3}} \frac{kx - \arctan x}{(1+x^2)^2 (\arctan x)^{2k+1}} \, dx$$

Marin Chirciu

Solution

For $k = 0$ we have $\Omega = \int_1^{\sqrt{3}} \frac{-\arctan x}{(1+x^2)(\arctan x)} \, dx = - \int_1^{\sqrt{3}} \frac{1}{(1+x^2)} \, dx$

Integrating by parts we have

$$\begin{aligned}
 \int \frac{1}{x^2+1} \, dx &= \int x' \frac{1}{x^2+1} \, dx = x \cdot \frac{1}{x^2+1} - \int x \left(\frac{1}{x^2+1} \right)' \, dx = \\
 &= \frac{x}{x^2+1} - \int x \cdot \frac{-2x}{(x^2+1)^2} \, dx = \\
 &= \frac{x}{x^2+1} + 2 \frac{x^2}{(x^2+1)^2} \, dx = \frac{x}{x^2+1} + 2 \int \frac{x^2+1-1}{(x^2+1)^2} \, dx = \frac{x}{x^2+1} + 2 \int \frac{1}{x^2+1} -
 \end{aligned}$$

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$$-2 \int \frac{1}{(x^2+1)^2} dx$$

It follows $\int \frac{1}{x^2+1} dx = \frac{x}{x^2+1} + 2 \int \frac{1}{x^2+1} - 2 \int \frac{1}{(x^2+1)} dx \Leftrightarrow 2 \int \frac{1}{(x^2+1)^2} dx = \frac{x}{x^2+1} + \int \frac{1}{x^2+1}$

wherefrom $\int \frac{1}{(x^2+1)^2} dx = \frac{1}{2} \cdot \frac{x}{x^2+1} + \frac{1}{2} \arctan x$

We obtain

$$\begin{aligned} \int_1^{\sqrt{3}} \frac{1}{(1+x^2)^2} dx &= \left(\frac{1}{2} \cdot \frac{x}{x^2+1} + \frac{1}{2} \arctan x \right) \Big|_1^{\sqrt{3}} = \\ &= \frac{1}{2} \cdot \frac{\sqrt{3}}{3+1} + \frac{1}{2} \arctan \sqrt{3} - \frac{1}{2} \cdot \frac{1}{1+1} - \frac{1}{2} \arctan 1 = \\ &= \frac{\sqrt{3}}{8} + \frac{1}{2} \cdot \frac{\pi}{3} - \frac{1}{4} - \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\sqrt{3}}{8} - \frac{1}{4} + \frac{1}{2} \cdot \frac{\pi}{12} = \frac{\sqrt{3}}{8} - \frac{1}{4} + \frac{\pi}{24} \\ \Omega &= - \int_1^{\sqrt{3}} \frac{1}{(1+x^2)^2} dx = - \left(\frac{\sqrt{3}}{8} - \frac{1}{4} + \frac{\pi}{24} \right) = \frac{1}{4} - \frac{\sqrt{3}}{8} - \frac{\pi}{24}. \end{aligned}$$

Next, let be $k \geq 1$.

Using the substitution $t = \arctan x \Rightarrow x = \tan t \Rightarrow dx = (1 + \tan^2 t) dt \Rightarrow$

$$\begin{aligned} \Omega &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{k \tan t - 1}{(1 + \tan^2 t)^2} (1 + \tan^2 t) dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{k \tan t - t}{(1 + \tan^2 t) t^{2k+1}} dt = \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{k \tan t}{(1 + \tan^2 t) t^{2k+1}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{t}{(1 + \tan^2 t) t^{2k+1}} dt = \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{k \frac{\sin t}{\cos t}}{\frac{1}{\cos^2 t} t^{2k+1}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{t}{\frac{1}{\cos^2 t} t^{2k+1}} dt = \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{k \sin t \cos t}{t^{2k+1}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^2 t}{t^{2k+1}} dt = \frac{k}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin 2t}{t^{2k+1}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1 + \cos 2t}{2 t^{2k+1}} dt = \\ &= \frac{k}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{-1}{2kt^{2k}} \right)' \sin 2t dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2t^{2k}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^{2k}} dt = \\ &= \frac{k}{2} \left[\left(-\frac{1}{2kt^{2k}} \right) \sin 2t \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{-1}{2kt^{2k}} \right) \cdot 2 \cos 2t dt \right] - \end{aligned}$$

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$$\begin{aligned}
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2t^{2k}} dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^{2k}} dt = \frac{k}{2} \left[\left(-\frac{1}{2k \left(\frac{\pi}{3}\right)^{2k}} \right) \sin \frac{2\pi}{3} + \left(\frac{1}{2k \left(\frac{\pi}{4}\right)^{2k}} \right) \sin \frac{2\pi}{4} + \right. \\
 &\quad \left. + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{2kt^{2k-1}} \right) \cdot 2 \cos 2t dt \right] + \\
 &+ \frac{1}{2(2k-1)t^{2k-1}} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^{2k}} dt = \frac{k}{2} \left[\left(\frac{-1}{2k \left(\frac{\pi}{3}\right)^{2k}} \right) \cdot \frac{\sqrt{3}}{2} + \left(\frac{1}{2 \left(\frac{\pi}{4}\right)^2} \right) \cdot 1 + \right. \\
 &\quad \left. + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{t^2} \cos 2t dt \right] + \\
 &+ \frac{1}{2(2k-1) \cdot \left(\frac{\pi}{3}\right)^{2k-1}} - \frac{1}{2 \cdot \left(\frac{\pi}{4}\right)^{2k-1}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} dt = \\
 &= \frac{-\frac{k}{2}}{2k \left(\frac{\pi}{3}\right)^{2k}} \cdot \frac{\sqrt{3}}{2} + \frac{\frac{k}{2}}{2k \left(\frac{\pi}{4}\right)^{2k}} \cdot 1 + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{2t^2} \cos 2t dt + \\
 &+ \frac{1}{2(2k-1) \left(\frac{\pi}{3}\right)^{2k-1}} - \frac{1}{2(2k-1) \left(\frac{\pi}{4}\right)^{2k-1}} - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos 2t}{2t^2} dt = \\
 &= \frac{-\sqrt{3}}{8 \left(\frac{\pi}{3}\right)^{2k}} + \frac{1}{4 \left(\frac{\pi}{4}\right)^{2k}} + \frac{1}{2(2k-1) \left(\frac{\pi}{3}\right)^{2k-1}} - \frac{1}{2(2k-1) \left(\frac{\pi}{4}\right)^{2k-1}} = \\
 &= \frac{-\sqrt{3} \cdot 3^{2k}}{8\pi^{2k}} + \frac{4^{2k-1}}{\pi^{2k}} + \frac{3^{2k-1} - 4^{2k-1}}{2(2k-1)\pi^{2k-1}} =
 \end{aligned}$$

We deduce that $\Omega = \int_1^{\sqrt{3}} \frac{kx - \arctan x}{(1+x^2)(\arctan x)^{2k+1}} dx = \frac{4^{2k-1}}{\pi^{2k}} - \frac{\sqrt{3} \cdot 3^{2k}}{8\pi^{2k}} + \frac{3^{2k-1} - 4^{2k-1}}{2(2k-1)\pi^{2k-1}}$

Note: For $k = 1$ we obtain Problem UP.541 from RMM Nr.37 – Summer 2025.

Solution 3 by Angel Plaza-Spain

First, we change variable by doing $t = \tan^{-1} x$, so $dt = \frac{1}{1+x^2} dx$. Then

$$\Omega = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan t - t}{(1 + \tan^2 t)t^3} dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin t \cos t - t \cos^2 t}{t^3} dt$$

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Now, by parts, with $u = \sin t \cos t - t \cos^2 t$, and $dv = \frac{1}{t^3} dt$, so $v = -\frac{1}{2t^2}$ and

$$\begin{aligned} du &= (\cos^2 t - \sin^2 t - \cos^2 t + 2t \cos t \sin t) dt = \\ &= (-\sin^2 t + 2t \cos t \sin t) dt \end{aligned}$$

Then

$$\Omega = -\frac{\sin t \cos t - t \cos^2 t}{2t^2} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{-\sin^2 t + 2t \cos t \sin t}{2t^2} dt$$

The new integral may be solved also by parts with $u = -\sin^2 t + t \cos t \sin t$,

$dv = \frac{1}{2t^2} dt$, so $du = 2t(\cos^2 t - \sin^2 t) dt$, $v = \frac{-1}{2t}$. Therefore

$$\begin{aligned} &\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{-\sin^2 t + 2t \cos t \sin t}{2t^2} dt = \\ &= -\frac{\sin^2 t - 2t \sin t \cos t}{2t} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos 2t dt = -\frac{\sin^2 t}{2t} - \frac{\sin 2t}{2} + \frac{\sin 2t}{2} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} \end{aligned}$$

$$\text{Therefore, } \Omega = \frac{t - \sin t \cos t}{2t^2} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \frac{\frac{\pi}{3} - \frac{\sqrt{3}}{4}}{2 \cdot \frac{\pi^2}{9}} - \frac{\frac{\pi}{4} - \frac{1}{2}}{2 \cdot \frac{\pi^2}{16}} = \frac{32 - 9\sqrt{3} - 4\pi}{8\pi^2} \sim 0.04869 \dots$$

UP.542 Find:

$$\Omega = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{49}{49 + (7n+1)(7n+8)} \right)$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned} \Omega &= \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{7((7n+8) - (7n+1))}{49 + (7n+1)(7n+8)} \right) = \\ &= \sum_{n=1}^{\infty} \left(\frac{(7n+1)(7n+8) \cdot 7 \cdot \left(\frac{1}{7n+1} - \frac{1}{7n+8} \right)}{(7n+1)(7n+8) \left(\frac{49}{(7n+1)(7n+8)} + 1 \right)} \right) = \end{aligned}$$

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$$\begin{aligned}
 &= \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{\frac{7}{7n+1} - \frac{7}{7n+8}}{1 + \frac{7}{7n+1} \cdot \frac{7}{7n+8}} \right) = \sum_{n=1}^{\infty} \left(\tan^{-1} \left(\frac{7}{7n+1} \right) - \tan^{-1} \left(\frac{7}{7n+8} \right) \right) = \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\tan^{-1} \left(\frac{7}{7k+1} \right) - \tan^{-1} \left(\frac{7}{7k+8} \right) \right) = \lim_{n \rightarrow \infty} \left(\tan^{-1} \left(\frac{7}{8} \right) - \tan^{-1} \left(\frac{7}{7n+8} \right) \right) = \\
 &= \tan^{-1} \left(\frac{7}{8} \right) - \tan^{-1} 0 = \tan^{-1} \left(\frac{7}{8} \right)
 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

Using $\arctan \frac{1}{x} - \arctan \frac{1}{x+1} = \arctan \frac{1}{x^2+x+1}$, $x > 0$, for $x = n + \frac{1}{7}$ we obtain:

$$\begin{aligned}
 \arctan \frac{1}{n + \frac{1}{7}} - \arctan \frac{1}{n + \frac{1}{7} + 1} &= \arctan \frac{1}{\left(n + \frac{1}{7}\right)^2 + n + \frac{1}{7} + 1} \\
 &= \frac{49}{49 + (7n+1)(7n+8)}
 \end{aligned}$$

It follows:

$$\begin{aligned}
 \Omega &= \sum_{n=1}^{\infty} \arctan \frac{49}{49 + (7n+1)(7n+8)} = \sum_{n=1}^{\infty} \left(\arctan \frac{1}{n + \frac{1}{7}} - \arctan \frac{1}{n + \frac{1}{7} + 1} \right) = \\
 &= \arctan \frac{1}{1 + \frac{1}{7}} - \arctan \frac{1}{2 + \frac{1}{7}} + \arctan \frac{1}{2 + \frac{1}{7}} - \arctan \frac{1}{3 + \frac{1}{7}} + \dots + \\
 &\quad + \arctan \frac{1}{n + \frac{1}{7}} - \arctan \frac{1}{n + 1 + \frac{1}{7}} + \dots = \\
 &= \lim_{n \rightarrow \infty} \left(\arctan \frac{1}{1 + \frac{1}{7}} - \arctan \frac{1}{n + 1 + \frac{1}{7}} \right) = \arctan \frac{7}{8}
 \end{aligned}$$

Remark: The problem can be developed.

Let $\lambda > 0$. Find:

$$\Omega = \sum_{n=1}^{\infty} \arctan \frac{\lambda^2}{\lambda^2 + (\lambda n + 1)(\lambda n + \lambda + 1)}$$

Marin Chirciu

Solution

Using $\arctan \frac{1}{x} - \arctan \frac{1}{x+1} = \arctan \frac{1}{x^2+x+1}$, $x > 0$, for $x = n + \frac{1}{\lambda}$ we obtain:

$$\begin{aligned} \arctan \frac{1}{n + \frac{1}{\lambda}} - \arctan \frac{1}{n + \frac{1}{\lambda} + 1} &= \arctan \frac{1}{\left(n + \frac{1}{\lambda}\right)^2 + n + \frac{1}{\lambda} + 1} = \\ &= \frac{\lambda^2}{\lambda^2 + (\lambda n + 1)(\lambda n + \lambda + 1)} \end{aligned}$$

It follows:

$$\begin{aligned} \Omega &= \sum_{n=1}^{\infty} \arctan \frac{\lambda^2}{\lambda^2 + (\lambda n + 1)(\lambda n + \lambda + 1)} = \sum_{n=1}^{\infty} \left(\arctan \frac{1}{n + \frac{1}{\lambda}} - \arctan \frac{1}{n + \frac{1}{\lambda} + 1} \right) \\ &= \\ &= \arctan \frac{1}{1 + \frac{1}{\lambda}} - \arctan \frac{1}{2 + \frac{1}{\lambda}} + \arctan \frac{1}{2 + \frac{1}{\lambda}} - \arctan \frac{1}{3 + \frac{1}{\lambda}} + \dots + \\ &\quad + \arctan \frac{1}{n + \frac{1}{\lambda}} - \arctan \frac{1}{n + 1 + \frac{1}{\lambda}} + \dots = \\ &= \lim_{n \rightarrow \infty} \left(\arctan \frac{1}{1 + \frac{1}{\lambda}} - \arctan \frac{1}{n + 1 + \frac{1}{\lambda}} \right) = \arctan \frac{\lambda}{\lambda + 1} \end{aligned}$$

Note: For $\lambda = 7$ we obtain Problem SP.542 from RMM Nr. 37 – Summer 2025.

UP.543 Prove:

$$\int_0^1 \left(\frac{x^2 \log(x)}{1 + x^2} \right)^2 dx = G + 2 - \frac{3\pi^3}{32}$$

G represents the Catalan's constant.

Proposed by Said Attaoui – Oran – Algeria

Solution by proposer

Recall that $G = \beta(2)$ and $\frac{\pi^3}{32} = \beta(3)$, with $\beta(p)$ is the Dirichlet's beta function given by

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$$\beta(p) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^p}$$

We can write

$$\begin{aligned} \int_0^1 \left(\frac{\log(x)}{1+x^2} \right)^2 dx &= \int_0^1 \left(1 - \frac{x^2}{1+x^2} \right)^2 \log^2(x) dx \\ &= \int_0^1 \log^2(x) dx + \int_0^1 \left(\frac{x^2 \log(x)}{1+x^2} \right)^2 dx - 2 \int_0^1 \frac{x^2 \log^2(x)}{1+x^2} dx \\ &= \int_0^1 \log^2(x) dx + \int_0^1 \left(\frac{x^2 \log(x)}{1+x^2} \right)^2 dx + 2 \int_0^1 \frac{\log^2(x)}{1+x^2} dx - 2 \int_0^1 \log^2(x) dx \\ &= \int_0^1 \left(\frac{x^2 \log(x)}{1+x^2} \right)^2 dx + 2 \underbrace{\int_0^1 \frac{\log^2(x)}{1+x^2} dx}_{=2\beta(3)} - \underbrace{\int_0^1 \log^2(x) dx}_2 \end{aligned}$$

Hence

$$\int_0^1 \left(\frac{x^2 \log(x)}{1+x^2} \right)^2 dx = \int_0^1 \left(\frac{\log(x)}{1+x^2} \right)^2 dx - 4\beta(3) + 2$$

Now, it is sufficient to demonstrate that

$$\int_0^1 \left(\frac{\log(x)}{1+x^2} \right)^2 dx = \beta(2) + \beta(3)$$

Indeed, by the substitution $x = e^{-t}$, $dx = -e^{-t} dt$, we get

$$\begin{aligned} \int_0^1 \left(\frac{\log(x)}{1+x^2} \right)^2 dx &= \int_0^{\infty} \left(\frac{-t}{1+e^{-2t}} \right)^2 e^{-t} dt \\ &= \int_0^{\infty} \frac{t^2 e^{-t}}{(1+e^{-2t})^2} dt. \end{aligned}$$

Applying the expansion series $\sum_{n=0}^{\infty} n(-y)^{n-1} = \frac{1}{(1+y)^2}$, $|y| < 1$ for $y = e^{-t}$, we obtain

$$\begin{aligned} \int_0^1 \left(\frac{\log(x)}{1+x^2} \right)^2 dx &= \int_0^{\infty} t^2 e^{-t} \left(\sum_{n=0}^{\infty} (-1)^{n-1} n e^{-2(n-1)t} \right) dt \\ &= \sum_{n=0}^{\infty} (-1)^{n-1} n \left(\int_0^{\infty} t^2 e^{-(2n-1)t} dt \right) \end{aligned}$$

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$$\begin{aligned}
 &= \sum_{n=1}^{\infty} (-1)^{n-1} n \frac{2!}{(2n-1)^3} \\
 &= \sum_{n=0}^{\infty} (-1)^n (n+1) \frac{2!}{(2n+1)^3} \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{2n+2}{(2n+1)^3} \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^2} + \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^3} \\
 &= \beta(2) + \beta(3).
 \end{aligned}$$

Substituting this result into equation (1), we ultimately reach:

$$\int_0^1 \left(\frac{x^2 \log(x)}{1+x^2} \right)^2 dx = G + 2 - \frac{3\pi^3}{32}.$$

UP.544 For $x, y, z > 0$ let us denote:

$$F(x, y, z) = \frac{xyz(xy + yz + zx)}{x^3y^3 + y^3z^3 + z^3x^3} [x^2(y-z)^2 + y^2(z-x)^2 + z^2(x-y)^2]$$

If $u, v, w \geq 1$, prove that:

$$F(u, v, w) \geq F\left(\frac{1}{u}, \frac{1}{v}, \frac{1}{w}\right)$$

Proposed by Vasile Mircea Popa – Romania

Solution 1 by proposer

Let us denote $A = F(u, v, w)$. Performing the calculations, we obtain:

$$A = 2 \left(uvw - \frac{3u^3v^3w^3}{u^3v^3 + v^3w^3 + w^3u^3} \right)$$

We make the substitutions: $u = \sqrt[3]{a}, v = \sqrt[3]{b}, w = \sqrt[3]{c}$, where $a, b, c \geq 1$.

We obtain:

$$A = 2 \left(\sqrt[3]{abc} - \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \right); A = 2[G(a, b, c) - H(a, b, c)]$$

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where $G(a, b, c)$ is the geometric mean of numbers a, b, c and $H(a, b, c)$ is the harmonic mean of numbers a, b, c . Let us denote $B = F\left(\frac{1}{u}, \frac{1}{v}, \frac{1}{w}\right)$.

Performing the calculations, we obtain:

$$B = 2\left(\frac{1}{uvw} - \frac{3}{u^3 + v^3 + w^3}\right)$$

We make the same substitutions: $u = \sqrt[3]{a}, v = \sqrt[3]{b}, w = \sqrt[3]{c}$, where $a, b, c \geq 1$.

$$B = 2\left(\frac{1}{\sqrt[3]{abc}} - \frac{3}{a+b+c}\right); B = 2\left[G\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right) - H\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)\right]$$

where $G\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$ is the geometric mean of numbers $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ and $H\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$ is the harmonic mean of numbers $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$.

So the inequality required in the problem statement is equivalent to the inequality:

$$G(a, b, c) - H(a, b, c) \geq G\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right) - H\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$$

where $a, b, c \geq 1$.

This inequality is true. See the American Mathematical Monthly, Volume 127, 2020, Issue 7 problem 12196, case $n = 3$. Thus, proposed inequality is proved.

Solution 2 by Soumava Chakraborty-Kolkata-India

$$\begin{aligned} F(u, v, w) &\geq F\left(\frac{1}{u}, \frac{1}{v}, \frac{1}{w}\right) \Leftrightarrow \frac{uvw(\sum_{cyc} uv)}{\sum_{cyc} u^3 v^3} \left(\sum_{cyc} u^2 (v-w)^2 \right) \geq \\ &\frac{\frac{1}{uvw}(\sum_{cyc} \frac{1}{uv})}{\sum_{cyc} \frac{1}{u^3 v^3}} \left(\sum_{cyc} \frac{1}{u^2} \left(\frac{1}{v} - \frac{1}{w} \right)^2 \right) \Leftrightarrow \frac{uvw(\sum_{cyc} uv)}{\sum_{cyc} u^3 v^3} \left(\sum_{cyc} u^2 (v-w)^2 \right) \\ &\stackrel{(*)}{\geq} \frac{(\sum_{cyc} u)(\sum_{cyc} (v-w)^2)}{uvw(\sum_{cyc} u^3)} \\ &\Leftrightarrow u^2 v^2 w^2 \left(\sum_{cyc} uv \right) \left(\sum_{cyc} u^3 \right) \left(\sum_{cyc} u^2 v^2 - uvw \left(\sum_{cyc} u \right) \right) \\ &\stackrel{(**)}{\geq} \left(\sum_{cyc} u \right) \left(\sum_{cyc} u^3 v^3 \right) \left(\sum_{cyc} u^2 - \sum_{cyc} uv \right) \end{aligned}$$

Assigning $v + w = a, w + u = b, u + v = c \Rightarrow a + b - c = 2w > 0, b + c - a = 2u > 0$ and $c + a - b = 2v > 0 \Rightarrow a + b > c, b + c > a, c + a > b \Rightarrow a, b, c$ form

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sides of a triangle with semiperimeter, circumradius and inradius = s, R, r (say)

$$\text{yielding } 2 \sum_{\text{cyc}} u = \sum_{\text{cyc}} a = 2s \Rightarrow \sum_{\text{cyc}} u \stackrel{(*)}{=} s \Rightarrow u = s - a, v = s - b, w = s - c$$

$$\therefore uvw \stackrel{(**)}{=} r^2 s \text{ and } \sum_{\text{cyc}} uv = \sum_{\text{cyc}} (s - a)(s - b) = 4Rr + r^2 \Rightarrow \sum_{\text{cyc}} uv \stackrel{(***)}{=} 4Rr + r^2$$

$$\text{and also, } \sum_{\text{cyc}} u^2 = \left(\sum_{\text{cyc}} u \right)^2 - 2 \sum_{\text{cyc}} uv \stackrel{\text{via } (*) \text{ and } (***)}{=} s^2 - 2(4Rr + r^2)$$

$$\Rightarrow \sum_{\text{cyc}} u^2 \stackrel{(***)}{=} s^2 - 8Rr - 2r^2 \text{ and also, } \sum_{\text{cyc}} u^2 v^2 = \left(\sum_{\text{cyc}} uv \right)^2 - 2uvw \left(\sum_{\text{cyc}} u \right) \\ \stackrel{\text{via } (*), (**), \text{ and } (***)}{=} (4Rr + r^2)^2 - 2r^2 s^2 \Rightarrow \sum_{\text{cyc}} u^2 v^2 \stackrel{(***)}{=} r^2 ((4R + r)^2 - 2s^2)$$

$$\text{and, } \sum_{\text{cyc}} u^3 = \left(\sum_{\text{cyc}} u \right)^3 - 3(u + v)(v + w)(w + u) \stackrel{\text{via } (*)}{=} s^3 - 3 \cdot 4Rrs$$

$$\Rightarrow \sum_{\text{cyc}} u^3 \stackrel{(***)}{=} s(s^2 - 12Rr) \text{ and finally, } \sum_{\text{cyc}} u^3 v^3 =$$

$$\left(\sum_{\text{cyc}} uv \right)^3 - 3uvw(u + v)(v + w)(w + u) \stackrel{\text{via } (**), \text{ and } (***)}{=} (4Rr + r^2)^3 - 3r^2 s \cdot 4Rrs$$

$$\Rightarrow \sum_{\text{cyc}} u^3 v^3 \stackrel{(***)}{=} r^3 ((4R + r)^3 - 12Rs^2)$$

$$\boxed{\text{Case 1}} \quad \left[\left(\sum_{\text{cyc}} uv \right)^3 \geq uvw \left(\sum_{\text{cyc}} u \right)^3 \right] \text{ and } \therefore u^2 v^2 w^2 \geq 1 \therefore$$

via $(*)$, $(**)$, $(***)$, $(****)$, $(*****)$, $(*****)$ and $(*****)$, in order to prove $(**)$, it suffices to prove :

$$\geq sr^3 ((4R + r)^3 - 12Rs^2) (s^2 - 8Rr - 2r^2 - 4Rr - r^2) \stackrel{\text{simplifying}}{\Leftrightarrow}$$

$$3r(r(4R + r)^3 - s^4) \geq 0 \rightarrow \text{true} \therefore \left(\sum_{\text{cyc}} uv \right)^3 \geq uvw \left(\sum_{\text{cyc}} u \right)^3$$

$$\Rightarrow (4Rr + r^2)^3 \geq r^2 s^4 \Rightarrow r(4R + r)^3 \geq s^4 \therefore (**) \Rightarrow (*) \text{ is true}$$

$$\boxed{\text{Case 2}} \quad \left[\left(\sum_{\text{cyc}} uv \right)^3 \leq uvw \left(\sum_{\text{cyc}} u \right)^3 \right] \text{ and } \therefore u, v, w \geq 1 \therefore$$

$$\sum_{\text{cyc}} u^2 (v - w)^2 \geq \sum_{\text{cyc}} (v - w)^2 \therefore \text{in order to prove } (*), \text{ it suffices to prove :}$$

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$$\begin{aligned}
 & u^2 v^2 w^2 \left(\sum_{\text{cyc}} uv \right) \left(\sum_{\text{cyc}} u^3 \right) \geq \left(\sum_{\text{cyc}} u \right) \left(\sum_{\text{cyc}} u^3 v^3 \right) \text{ and } \because uvw \geq 1 \Rightarrow \\
 & \quad u^2 v^2 w^2 \geq \sqrt[3]{u^2 v^2 w^2} \therefore \text{it suffices to prove :} \\
 & \sqrt[3]{u^2 v^2 w^2} \cdot \left(\sum_{\text{cyc}} uv \right) \left(\sum_{\text{cyc}} u^3 \right) \geq \left(\sum_{\text{cyc}} u \right) \left(\sum_{\text{cyc}} u^3 v^3 \right) \text{ via } (\bullet), (\bullet\bullet), (\bullet\bullet\bullet), (\bullet\bullet\bullet\bullet) \text{ and } (\bullet\bullet\bullet\bullet\bullet) \Leftrightarrow \\
 & \quad r^4 s^2 \cdot r^3 (4R + r)^3 \cdot s^3 (s^2 - 12Rr)^3 \geq s^3 r^9 ((4R + r)^3 - 12Rs^2)^3 \\
 & \Leftrightarrow \boxed{s^8 (4R + r)^3 (s^2 - 12Rr)^3 \stackrel{(***)}{\geq} r^2 (s^2 (4R + r)^3 - 12Rs^4)^3} \\
 & \text{Now, } \left(\sum_{\text{cyc}} uv \right)^3 \leq uvw \left(\sum_{\text{cyc}} u \right)^3 \Rightarrow r^2 s^4 \geq (4Rr + r^2)^3 \Rightarrow \mathbf{s^4 \geq r(4R + r)^3} \\
 & \Rightarrow r^2 (s^2 (4R + r)^3 - 12Rs^4)^3 \leq r^2 (s^2 (4R + r)^3 - 12Rr(4R + r)^3)^3 \\
 & = r^2 (4R + r)^9 (s^2 - 12Rr)^3 \stackrel{?}{\leq} s^8 (4R + r)^3 (s^2 - 12Rr)^3 \Leftrightarrow s^8 \stackrel{?}{\geq} r^2 (4R + r)^6 \\
 & \Leftrightarrow s^4 \stackrel{?}{\geq} r(4R + r)^3 \rightarrow \text{true under case 2} \Rightarrow (***) \Rightarrow (*) \text{ is true} \\
 & \therefore (*) \text{ is true considering both cases} \Rightarrow F(u, v, w) \geq F\left(\frac{1}{u}, \frac{1}{v}, \frac{1}{w}\right) \forall u, v, w \geq 1 \\
 & \text{where } F(x, y, z) = \frac{xyz(xy + yz + zx)}{x^3 y^3 + y^3 z^3 + z^3 x^3} (x^2(y - z)^2 + y^2(z - x)^2 + z^2(x - y)^2), \\
 & \quad \quad \quad " = " \text{ iff } u = v = w = 1 \text{ (QED)}
 \end{aligned}$$

Solution 3 by Mohamed Amine Ben Ajiba-Tanger-Morocco

We have

$$F\left(\frac{1}{x}, \frac{1}{y}, \frac{1}{z}\right) = \frac{x + y + z}{xyz(x^3 + y^3 + z^3)} [(y - z)^2 + (z - x)^2 + (x - y)^2],$$

so to prove that $F(u, v, w) \geq F\left(\frac{1}{u}, \frac{1}{v}, \frac{1}{w}\right)$, it suffices to prove that

$$\frac{uvw(uv + vw + wu)}{u^3 v^3 + v^3 w^3 + w^3 u^3} \cdot u^2 \geq \frac{u + v + w}{uvw(u^3 + v^3 + w^3)} \quad (\text{and analogs}) \quad (1)$$

The inequality (1) is equivalent to

$$u^4 v^2 w^2 (uv + vw + wu)(u^3 + v^3 + w^3) \geq (u + v + w)(u^3 v^3 + v^3 w^3 + w^3 u^3).$$

Since $u, v, w \geq 1$, then we have $u^3 + v^3 + w^3 \geq u + v + w$, and

$$\begin{aligned}
 u^4 v^2 w^2 (uv + vw + wu) &= u^3 v^3 \cdot u^2 w^2 + v^3 w^3 \cdot u^4 + w^3 u^3 \cdot u^2 v^2 \\
 &\geq u^3 v^3 + v^3 w^3 + w^3 u^3,
 \end{aligned}$$

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then (1) is true and the proof is complete. Equality holds iff $u = v = w = 1$.

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$$\Omega = \int_0^{\frac{1}{2}} \frac{x^5 - 3x^3}{3x^6 - x^4 - 3x^2 + 1} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$x \in \left(0, \frac{1}{2}\right) \Rightarrow (\exists) y \in \left(0, \arctan \frac{1}{2}\right); x = \tan y$$

$$dx = \frac{1}{\cos^2 y} dy; dx = (1 + \tan^2 y) dy$$

$$x = 0 \Rightarrow y = 0; x = \frac{1}{2} \Rightarrow y = \arctan \frac{1}{2}$$

$$\Omega = \int_0^{\arctan \frac{1}{2}} \frac{(\tan^5 y - 3 \tan^3 y)(1 + \tan^2 y)}{3 \tan^6 y - \tan^4 y - 3 \tan^2 y + 1} dy$$

$$\Omega = - \int_0^{\arctan \frac{1}{2}} \frac{\tan^3 y (3 - \tan^2 y)(1 + \tan^2 y)}{3 \tan^2 y (\tan^4 y - 1) - (\tan^4 y - 1)} dy$$

$$\Omega = - \int_0^{\arctan \frac{1}{2}} \frac{\tan^3 y (3 - \tan^2 y)(1 + \tan^2 y)}{(\tan^2 y - 1)(\tan^2 y + 1)(3 \tan^2 y - 1)} dy$$

$$\Omega = \int_0^{\arctan \frac{1}{2}} \frac{\tan y (3 - \tan^2 y)}{3 \tan^2 y - 1} \cdot \frac{\tan^2 y}{\tan^2 y - 1} dy$$

$$\Omega = \frac{1}{2} \int_0^{\arctan \frac{1}{2}} \tan(3y) \cdot \frac{2 \tan y}{\tan^2 y - 1} \cdot \tan y dy$$

$$\Omega = \frac{1}{2} \int_0^{\arctan \frac{1}{2}} \tan(3y) \cdot \tan(2y) \cdot \tan y dy$$

$$\Omega = \frac{1}{2} \int_0^{\arctan \frac{1}{2}} (\tan 3y - \tan 2y - \tan y) dy$$

$$\Omega = \frac{1}{2} \int_0^{\arctan \frac{1}{2}} \tan 3y dy - \frac{1}{2} \int_0^{\arctan \frac{1}{2}} \tan 2y dy - \frac{1}{2} \int_0^{\arctan \frac{1}{2}} \tan y dy$$

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$$\Omega = \frac{1}{2} \ln \left(\cos \left(\arctan \frac{1}{2} \right) \right) + \frac{1}{4} \ln \left(\cos \left(2 \arctan \frac{1}{2} \right) \right) - \frac{1}{6} \ln \left(\cos \left(3 \arctan \frac{1}{2} \right) \right)$$

Solution 2 by Marin Chirciu-Romania

With the substitution $x^2 = t$ we obtain:

$$\begin{aligned} \Omega &= \frac{1}{2} \int_0^{\frac{1}{4}} \frac{t^2 - 3t}{3t^3 - t^2 - 3t + 1} dt = \frac{1}{2} \int_1^{\frac{1}{4}} \frac{t^2 - 3t}{(t-1)(t+1)(3t-1)} dt = \\ &= \frac{1}{2} \int_1^{\frac{1}{4}} \left(\frac{-\frac{1}{2}}{t-1} + \frac{-1}{3t-1} + \frac{-1}{t+1} \right) dt = \\ &= \frac{1}{2} \left(-\frac{1}{2} \ln|t-1| - \frac{1}{3} \ln|3t-1| - \ln|t+1| \right) \Big|_0^{\frac{1}{4}} = \\ &= -\frac{1}{2} \left(\frac{1}{2} \ln|t-1| + \frac{1}{3} \ln|3t-1| + \ln|t+1| \right) \Big|_0^{\frac{1}{4}} = \\ &= -\frac{1}{2} \left(\frac{1}{2} \ln \left| \frac{1}{4} - 1 \right| + \frac{1}{3} \ln \left| \frac{3}{4} - 1 \right| + \ln \left| \frac{1}{4} + 1 \right| \right) = -\frac{1}{2} \left(\frac{1}{2} \ln \frac{3}{4} + \frac{1}{3} \ln \frac{1}{4} + \ln \frac{5}{4} \right) = \\ &= -\frac{1}{2} \left(\frac{1}{2} \ln 3 - \ln 2 - \frac{2}{3} \ln 2 + \ln 5 + \ln 2 - 2 \ln 2 \right) = -\frac{1}{2} \left(\frac{1}{2} \ln 3 - \frac{11}{3} \ln 2 + \ln 5 \right) = \\ &= \frac{11}{6} \ln 2 - \frac{1}{4} \ln 3 - \frac{1}{2} \ln 5 \end{aligned}$$

I broke it down into simple fractions:

$$\frac{t^2-3t}{(t-1)(3t-1)(t+1)} = \frac{A}{t-1} + \frac{B}{3t-1} + \frac{C}{t+1}, \text{ with } A = -\frac{1}{2}, B = -1, C = -1$$

Solution 3 by Angel Plaza-Spain

Since $\frac{x^5-3x^3}{3x^6-x^4-3x^2+1} = \frac{x}{2(x^2+1)} + \frac{x}{3x^2+1} - \frac{1}{4(x-1)} - \frac{1}{4(x+1)}$, then

$$\begin{aligned} \Omega &= \int_0^{\frac{1}{2}} \left(\frac{x}{2(x^2+1)} + \frac{x}{3x^2+1} - \frac{1}{4(x-1)} - \frac{1}{4(x+1)} \right) dx \\ &= \frac{1}{4} \ln|x^2+1| + \frac{1}{6} \ln|3x^2-1| - \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| \Big|_0^{\frac{1}{2}} = \frac{-1}{12} \ln \left(\frac{423}{125} \right) \\ &\sim -0.10334. \end{aligned}$$

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UP.546 Prove without any software:

$$\int_0^{2/3} \sqrt{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx + \int_0^{2/3} \sqrt{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx > \frac{5}{4}$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

Let be $f: [0, 1] \rightarrow [0, 2]; f(x) = x^3 + x$

f continuous; f – increasing because $f'(x) = 3x^2 + 1 > 0$

$$f \text{ bijectif; } f^{-1}(x) = \sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} + \sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}}$$

$$f(0) = 0; f(1) = 2$$

By Young's inequality – integral form:

$$\int_0^1 (x^3 + x) dx + \int_0^2 \left(\sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} + \sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} \right) dx \geq 1 \cdot 2$$

$$\frac{1}{4} + \frac{1}{2} + \int_0^2 \left(\sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} \right) dx + \int_0^2 \left(\sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} \right) dx > 2$$

$$\int_0^{2/3} \sqrt{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx + \int_0^{2/3} \sqrt{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx > \frac{5}{4}$$

Solution 2 by Marin Chirciu-Romania

We use Cardano's formulas for the cubic equation $x^3 + px + q = 0$.

The discriminator is $\Delta = -(4p^3 + 27q^2)$

If $\Delta = -(4p^3 + 27q^2) < 0$, the equation has a single real root.

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$$x_1 = \sqrt[3]{\frac{-q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{\frac{-q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \text{ and } x_{2,3} \in \mathbb{C} - \mathbb{R}.$$

For $p = 1, q = -1$ the third-grade equation $x^3 + x - 1 = 0$ has $\Delta = -(4 + 27) < 0 \Rightarrow$

$$x_1 = \sqrt[3]{\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{27}}} + \sqrt[3]{\frac{1}{2} - \sqrt{\frac{1}{4} + \frac{1}{27}}} \text{ and } x_{2,3} \in \mathbb{C} - \mathbb{R}$$

We have:

$$\begin{aligned} f(x) &= \sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} + \sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} \geq \\ &\geq f(1) = \sqrt[3]{\frac{1}{2} + \sqrt{\frac{1}{27} + \frac{1}{4}}} + \sqrt[3]{\frac{1}{2} - \sqrt{\frac{1}{27} + \frac{1}{4}}} = x_1 \end{aligned}$$

We prove that $x_1 > \frac{5}{8}$

We consider the function $g: [0, 6] \rightarrow \mathbb{R}, g(x) = x^3 + x - 1$; we have

$$g'(x) = 3x^2 + 1 > 0 \Rightarrow g \uparrow$$

Because $g\left(\frac{5}{8}\right) = \left(\frac{5}{8}\right)^3 + \frac{5}{8} - 1 = \frac{-323}{512} < 0$ and $g(1) = 1 > 0 \Rightarrow \frac{5}{8} < x_1 < 1$.

We obtain:

$$\begin{aligned} &\int_0^2 \sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx + \int_0^2 \sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} dx = \\ &= \int_0^2 \left(\sqrt[3]{\frac{x}{2} + \sqrt{\frac{1}{27} + \frac{x^2}{4}}} + \sqrt[3]{\frac{x}{2} - \sqrt{\frac{1}{27} + \frac{x^2}{4}}} \right) dx > \\ &> \int_0^2 x_1 dx > \int_0^2 \frac{5}{8} dx = \frac{5}{8} x \Big|_0^2 = \frac{5}{8} \cdot 2 = \frac{5}{4} \end{aligned}$$

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UP.547 Find without any software:

$$\Omega = \int_1^2 \frac{4x^4 - 6x - 9}{x^4} e^{2x + \frac{3}{x}} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned} \frac{4x^4 - 6x - 9}{x^4} e^{2x + \frac{3}{x}} &= \left(4 - \frac{6}{x^3} - \frac{9}{x^4}\right) e^{2x + \frac{3}{x}} = \left(\left(2 - \frac{3}{x^2}\right)\left(2 + \frac{3}{x^2}\right) - \frac{6}{x^3}\right) e^{2x + \frac{3}{x}} = \\ &= -\frac{6}{x^3} e^{2x + \frac{3}{x}} + \left(2 + \frac{3}{x^2}\right)\left(2 - \frac{3}{x^2}\right) e^{2x + \frac{3}{x}} = \left(2 + \frac{3}{x^2}\right)' e^{2x + \frac{3}{x}} + \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right)' = \\ &= \left(\left(2 + \frac{3}{x^2}\right) e^{2x + \frac{3}{x}}\right)' \\ \Omega &= \int_1^2 \frac{4x^4 - 6x - 9}{x^4} e^{2x + \frac{3}{x}} dx = \int_1^2 \left(\left(2 + \frac{3}{x^2}\right) e^{2x + \frac{3}{x}}\right)' dx = \\ &= \left(2 + \frac{3}{x^2}\right) e^{2x + \frac{3}{x}} \Big|_1^2 = \frac{11}{4} \sqrt{e^{11}} - 5e^5 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

$$\text{We have } \frac{4x^4 - 6x - 9}{x^4} = 4 - \frac{6}{x^3} - \frac{9}{x^4} = \left(2 - \frac{3}{x^2}\right)\left(2 + \frac{3}{x^2}\right) - \frac{6}{x^3}.$$

We obtain:

$$\begin{aligned} \Omega &= \int_1^2 \frac{4x^4 - 6x - 9}{x^4} e^{2x + \frac{3}{x}} dx = \int_1^2 \left(\left(2 - \frac{3}{x^2}\right)\left(2 + \frac{3}{x^2}\right) - \frac{6}{x^3}\right) e^{2x + \frac{3}{x}} dx = \\ &= \int_1^2 \left(2 - \frac{3}{x^2}\right)\left(2 + \frac{3}{x^2}\right) e^{2x + \frac{3}{x}} dx - \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx = \int_1^2 \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right)' dx - \\ &\quad - \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx \stackrel{\text{Parts}}{=} \int_1^2 \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right)' dx - \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx = \\ &= \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right) \Big|_1^2 - \int_1^2 \left(2 + \frac{3}{x^2}\right)' e^{2x + \frac{3}{x}} dx - \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx = \\ &= \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right) \Big|_1^2 + \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx - \int_1^2 \frac{6}{x^3} e^{2x + \frac{3}{x}} dx = \left(2 + \frac{3}{x^2}\right) \left(e^{2x + \frac{3}{x}}\right) \Big|_1^2 = \end{aligned}$$

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$$= \left(2 + \frac{3}{4}\right) \left(e^{4+\frac{3}{2}}\right) - \left(2 + \frac{3}{1}\right) \left(e^{2+\frac{3}{1}}\right) = \frac{11}{4} e^{\frac{11}{2}} - 5e^5$$

Remark.

The problem can be developed.

Let $a, b \in \mathbb{R}^*$. Find:

$$\Omega = \int_1^2 \frac{a^2 x^4 - 2bx - b^2}{x^4} e^{ax + \frac{b}{x}} dx$$

Marin Chirciu

Solution

We have $\frac{a^2 x^4 - 2bx - b^2}{x^4} = a^2 - \frac{2b}{x^3} - \frac{b^2}{x^4} = \left(a - \frac{b}{x^2}\right) \left(a + \frac{b}{x^2}\right) - \frac{2b}{x^3}$.

We obtain:

$$\begin{aligned} \Omega &= \int_1^2 \frac{a^2 x^4 - 2bx - b^2}{x^4} e^{ax + \frac{b}{x}} dx = \int_1^2 \left(\left(a - \frac{b}{x^2}\right) \left(a + \frac{b}{x^2}\right) - \frac{2b}{x^3} \right) e^{ax + \frac{b}{x}} dx = \\ &= \int_1^2 \left(a - \frac{b}{x^2}\right) \left(a + \frac{b}{x^2}\right) e^{ax + \frac{b}{x}} dx - \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx = \int_1^2 \left(a + \frac{b}{x^2}\right) \left(e^{ax + \frac{b}{x}}\right)' dx - \\ &\quad - \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx \stackrel{\text{Parts}}{=} \int_1^2 \left(a + \frac{b}{x^2}\right) \left(e^{ax + \frac{b}{x}}\right)' dx - \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx = \\ &= \left(a + \frac{b}{x^2}\right) \left(e^{ax + \frac{b}{x}}\right) \Big|_1^2 - \int_1^2 \left(a + \frac{b}{x^2}\right)' e^{ax + \frac{b}{x}} dx - \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx = \\ &= \left(a + \frac{b}{x^2}\right) \left(e^{ax + \frac{b}{x}}\right) \Big|_1^2 + \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx - \int_1^2 \frac{2b}{x^3} e^{ax + \frac{b}{x}} dx = \left(a + \frac{b}{x^2}\right) \left(e^{ax + \frac{b}{x}}\right) \Big|_1^2 = \\ &= \left(a + \frac{b}{4}\right) \left(e^{2a + \frac{b}{2}}\right) - \left(a + \frac{b}{1}\right) \left(e^{a + \frac{b}{1}}\right) = \frac{4a + b}{4} e^{\frac{4a+b}{2}} - (a + b) e^{a+b} \end{aligned}$$

Note: For $a = 2, b = 3$ we obtain the problem UP.547 from RMM – 37 – Summer 2025

Solution 3 by Angel Plaza-Spain

Since $\left(2x + \frac{3}{x}\right)' = 2 - \frac{3}{x^2} = \frac{2x^2 - 3}{x^2}$, let us try a solution of the form $\frac{ax^2 + bx + c}{x^2} e^{2x + \frac{3}{x}}$ for the

indefinite integral, where a, b, c are coefficients to be determined. Then, by differentiating

it is obtained

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$$\frac{4x^4 - 2x - 9}{x^4} e^{2x + \frac{3}{x}} =$$

$$= \left(\frac{(2ax + b)x^2 - 2x(ax^2 + bx + c)}{x^4} + \frac{ax^2 + bx + c}{x^2} \cdot \frac{2x^2 - 3}{x^2} \right) e^{2x + \frac{3}{x}}$$

Hence

$$4x^4 - 2x - 9 = (2ax + b)x^2 - 2x(ax^2 + bx + c) + (ax^2 + bx + c)(2x^2 - 3)$$

$$4x^4 - 2x - 9 = 2ax^4 + 2bx^3 + (-3a - b + 2x)x^2 + (-3b - 2x)x - 3c$$

from where $a = 2, b = 0, c = 3$. Then

$$\Omega = \int_1^2 \frac{4x^4 - 2x - 9}{x^4} e^{2x + \frac{3}{x}} dx = \frac{2x^2 + 3}{x^2} e^{2x + \frac{3}{x}} \Big|_1^2 = \frac{11e^{\frac{11}{2}}}{4} - 5e^5$$

UP.548 Find:

$$\Omega = \int_1^e \frac{1 - \ln x}{x^2 + \ln^2 x} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\Omega = \int_1^e \frac{1 - \ln x}{x^2 + \ln^2 x} dx = \int_1^e \frac{1 - \ln x + 1 - \ln x}{(x - \ln x)^2 + (x + \ln x)^2} dx =$$

$$= \int_1^e \frac{x - \ln x + 1 - \frac{\ln x}{x} - x + 1 - \ln x + \frac{\ln x}{x}}{(x - \ln x)^2 \cdot \left(1 + \left(\frac{x + \ln x}{x - \ln x}\right)^2\right)} dx =$$

$$= \int_1^e \frac{\left(1 + \frac{1}{x}\right)(x - \ln x) - (x + \ln x)\left(1 - \frac{1}{x}\right)}{(x - \ln x)^2} \cdot \frac{1}{1 + \left(\frac{x + \ln x}{x - \ln x}\right)^2} dx =$$

$$= \int_1^e \frac{\left(\frac{x + \ln x}{x - \ln x}\right)'}{1 + \left(\frac{x + \ln x}{x - \ln x}\right)^2} dx = \arctan \left(\frac{x + \ln x}{x - \ln x} \right) \Big|_1^e =$$

$$= \arctan \left(\frac{e + 1}{e - 1} \right) - \arctan \left(\frac{1 + \ln 1}{1 - \ln 1} \right) = \arctan \left(\frac{e + 1}{e - 1} \right) - \frac{\pi}{4}$$

Solution 2 by Marin Chirciu-Romania

$$\begin{aligned}\Omega &= \int_1^e \frac{1 - \ln x}{x^2 + \ln^2 x} dx = \int_1^e \frac{\frac{1 - \ln x}{\ln^2 x}}{\left(\frac{x}{\ln x}\right)^2 + 1} dx = \int_1^e \frac{-\left(\frac{x}{\ln x}\right)'}{\left(\frac{x}{\ln x}\right)^2 + 1} dx = -\arctan \frac{x}{\ln x} \Big|_1^e = \\ &= -\left(\arctan \frac{e}{\ln e} - \arctan \frac{1}{\ln 1}\right) = -\left(\arctan e - \frac{\pi}{2}\right) = \frac{\pi}{2} - \arctan e.\end{aligned}$$

Remark.

In the same way.

Find:

$$\Omega = \int_1^e \frac{1 - \ln x}{x^2 - \ln^2 x} dx$$

Marin Chirciu

Solution

$$\begin{aligned}\Omega &= \int_1^e \frac{1 - \ln x}{x^2 - \ln^2 x} dx = \int_1^e \frac{\frac{1 - \ln x}{\ln^2 x}}{\left(\frac{x}{\ln x}\right)^2 - 1} dx = \int_1^e \frac{-\left(\frac{x}{\ln x}\right)'}{\left(\frac{x}{\ln x}\right)^2 - 1} dx = -\frac{1}{2} \ln \left| \frac{\frac{x}{\ln x} - 1}{\frac{x}{\ln x} + 1} \right| \Big|_1^e = \\ &= -\frac{1}{2} \ln \left| \frac{x - \ln x}{x + \ln x} \right| \Big|_1^e = -\frac{1}{2} \ln \left| \frac{x - \ln x}{x + \ln x} \right| \Big|_1^e = -\frac{1}{2} \left(\ln \frac{e - \ln e}{e + \ln e} - \ln \frac{1 - \ln 1}{1 + \ln 1} \right) = \\ &= -\frac{1}{2} \left(\ln \frac{e - 1}{e + 1} - \ln 1 \right) = -\frac{1}{2} \ln \frac{e - 1}{e + 1} = \frac{1}{2} \ln \frac{e + 1}{e - 1}\end{aligned}$$

UP.549 If $0 < a \leq b$ then find:

$$\Omega(a, b) = \int_a^b \frac{\ln x}{x^2 + (a + b)x + ab} dx$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\text{Let be } y = \frac{ab}{x} \Rightarrow x = \frac{ab}{y} \Rightarrow dx = -\frac{ab}{y^2} dy$$

$$\text{If } x = a \text{ then } y = b.$$

$$\text{If } x = b \text{ then } y = a.$$

$$\Omega(a, b) = \int_a^b \frac{\ln\left(\frac{ab}{y}\right)}{\frac{a^2 b^2}{y^2} + \frac{(a+b)ab}{y} + ab} \cdot \frac{-ab}{y^2} dy$$

$$\Omega(a, b) = - \int_b^a \frac{\ln(ab) - \ln y}{\left(\frac{ab}{y^2} + \frac{a+b}{y} + 1\right) y^2} dy$$

$$\Omega(a, b) = \int_a^b \frac{\ln(ab) - \ln y}{ab + (a+b)y + y^2} dy$$

$$\Omega(a, b) = \int_a^b \frac{\ln(ab)}{y^2 + (a+b)y + ab} dy - \int_a^b \frac{\ln y}{y^2 + (a+b)y + ab} dy$$

$$\Omega(a, b) = \ln(ab) \int_a^b \frac{dy}{\left(y + \frac{a+b}{2}\right)^2 - \frac{(a+b)^2 - 4ab}{4}} - \Omega(a, b)$$

$$2\Omega(a, b) = \ln(ab) \cdot \int_a^b \frac{dy}{\left(y + \frac{a+b}{2}\right)^2 - \frac{(a-b)^2}{4}}$$

$$\Omega(a, b) = \frac{1}{2} \ln(ab) \cdot \frac{1}{2 \cdot \frac{b-a}{2}} \ln \left| \frac{y + \frac{a+b}{2} - \frac{a-b}{2}}{y + \frac{a+b}{2} + \frac{a-b}{2}} \right| \Bigg|_a^b$$

$$\Omega(a, b) = \frac{\ln(ab)}{2(b-a)} \left(\ln \left| \frac{a+b}{2b} \right| - \ln \left| \frac{2a}{a+b} \right| \right)$$

$$\Omega(a, b) = \frac{\ln(ab)}{2(b-a)} \ln \frac{(a+b)^2}{4ab}$$

Solution 2 by Marin Chirciu-Romania

Using the substitution $x = \frac{ab}{t} \Rightarrow dx = \frac{-ab}{t^2} dt \Rightarrow$

$$\begin{aligned} \Rightarrow \Omega(a, b) &= \int_a^b \frac{\ln x}{x^2 + (a+b)x + ab} dx = \int_a^b \frac{\ln \frac{ab}{t}}{\left(\frac{ab}{t}\right)^2 + (a+b)\frac{ab}{t} + ab} \left(-\frac{ab}{t^2} dt\right) = \\ &= \int_a^b \frac{\ln(ab) - \ln t}{t^2 + (a+b)t + ab} dt = \int_a^b \frac{\ln(ab)}{t^2 + (a+b)t + ab} dt - \int_a^b \frac{\ln t}{t^2 + (a+b)t + ab} dt = \end{aligned}$$

$$= \int_a^b \frac{\ln(ab)}{t^2 + (a+b)t + ab} dt - \Omega(a, b).$$

It follows that:

$$\begin{aligned} \Omega(a, b) &= \int_a^b \frac{\ln(ab)}{t^2 + (a+b)t + ab} dt - \Omega(a, b) \Leftrightarrow \\ \Leftrightarrow 2\Omega(a, b) &= \int_a^b \frac{\ln(ab)}{t^2 + (a+b)t + ab} dt \Leftrightarrow \Omega(a, b) = \frac{\ln(ab)}{2} \int_a^b \frac{1}{t^2 + (a+b)t + ab} dt \\ \int_a^b \frac{1}{t^2 + (a+b)t + ab} dt &= \int_a^b \frac{1}{(t+a)(t+b)} dt = \frac{1}{b-a} \int_a^b \left(\frac{1}{t+a} - \frac{1}{t+b} \right) dt = \\ &= \frac{1}{b-a} \int_a^b \left(\frac{1}{t+a} - \frac{1}{t+b} \right) dt = \frac{1}{b-a} (\ln(t+a) - \ln(t+b)) = \\ &= \frac{1}{b-a} \ln \left(\frac{t+a}{t+b} \right) \Big|_a^b = \frac{1}{b-a} \left(\ln \left(\frac{b+a}{b+b} \right) - \ln \left(\frac{a+a}{a+b} \right) \right) = \frac{1}{b-a} \ln \frac{(a+b)^2}{4ab}, \text{ for } a < b \end{aligned}$$

For $a = b$ we have $\int_a^b \frac{1}{t^2 + (a+b)t + ab} dt = 0$.

We deduce that:

$$\begin{aligned} \Omega(a, b) &= \frac{\ln(ab)}{2} \int_a^b \frac{1}{t^2 + (a+b)t + ab} dt = \frac{\ln(ab)}{2} \cdot \frac{1}{b-a} \ln \frac{(a+b)^2}{4ab} = \\ &= \frac{\ln(ab)}{2(b-a)} \ln \frac{(a+b)^2}{4ab} \end{aligned}$$

for $a < b$ and $\Omega(a, b) = 0$ for $a = b$.

Remark: In the same way.

If $0 < a \leq b$ then find:

$$\Omega(a, b) = \int_a^b \frac{\ln x}{(x+a)^2} dx$$

Marin Chirciu

Solution

Using the substitution $x = \frac{a^2}{t} \Rightarrow dx = \frac{-a^2}{t^2} dt \Rightarrow$

$$\Omega(a, b) = \int_a^b \frac{\ln x}{(x+a)^2} dx = \int_a^b \frac{\ln \frac{a^2}{t}}{\left(\frac{a^2}{t} + a \right)^2} \left(-\frac{a^2}{t^2} dt \right) = \int_a^b \frac{\ln(a^2) - \ln t}{(t+a)^2} dt =$$

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$$= \int_a^b \frac{\ln(a^2)}{(t+a)^2} dt - \int_a^b \frac{\ln t}{(t+a)^2} dt = \int_a^b \frac{2 \ln a}{(t+a)^2} dt - \Omega(a, b)$$

It follows that: $\Omega(a, b) = \int_a^b \frac{2 \ln a}{(t+a)^2} dt - \Omega(a, b) \Leftrightarrow 2\Omega(a, b) = 2 \ln a \int_a^b \frac{1}{(t+a)^2} dt \Leftrightarrow$

$$\Leftrightarrow \Omega(a, b) = \ln a \cdot \int_a^b \frac{1}{(t+a)^2} dt$$

$$\int_a^b \frac{1}{(t+a)^2} dt = \left. \frac{-1}{t+a} \right|_a^b = \frac{-1}{b+a} + \frac{1}{2a} = \frac{-2a+b+a}{2a(a+b)} = \frac{b-a}{2a(a+b)}$$

We deduce that: $\Omega(a, b) = \ln a \cdot \frac{b-a}{2a(a+b)} = \frac{(b-a) \ln a}{2a(a+b)}$

UP.550 If $f: [0, 1] \rightarrow \mathbb{R}$; f continuous and

$$\int_0^1 xf(x) dx = a; \int_0^1 f(x) dx = b; a, b \in \mathbb{R}$$

then:

$$\int_0^1 f^2(x) dx \geq 3(a-b)^2$$

Proposed by Daniel Sitaru – Romania

Solution 1 by proposer

$$\begin{aligned} \int_0^1 f^2(x) dx \cdot \int_0^1 (x-1)^2 dx &\stackrel{CBS}{\geq} \left(\int_0^1 f(x) \cdot (x-1) dx \right)^2 \\ \int_0^1 f^2(x) dx \cdot \int_0^1 (x^2 - 2x + 1) dx &\geq \left(\int_0^1 xf(x) dx - \int_0^1 f(x) dx \right)^2 \\ \int_0^1 f^2(x) dx \cdot \left(\frac{1^3}{3} - \frac{0^3}{3} - 2 \left(\frac{1^2}{2} - \frac{0^2}{2} \right) + (1-0) \right) &\geq (b-a)^2 \\ \int_0^1 f^2(x) dx \cdot \left(\frac{1}{3} - 1 + 1 \right) &\geq (b-a)^2 \\ \int_0^1 f^2(x) dx &\geq 3(b-a)^2 \end{aligned}$$

Solution 2 by Marin Chirciu-Romania

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Using CBS inequality under integral form:

If $f, g: [a, b] \rightarrow \mathbb{R}$, f, g integrable then

$$\int_a^b f^2(x) dx \int_a^b g^2(x) dx \geq \left(\int_a^b f(x) g(x) dx \right)^2$$

Putting $[a, b] = [0, 1]$ and $g(x) = x - 1$ we obtain:

$$\begin{aligned} \int_0^1 f^2(x) dx \int_0^1 (x-1)^2 dx &\geq \left(\int_0^1 f(x) (x-1) dx \right)^2 \Rightarrow \\ \Rightarrow \int_0^1 f^2(x) dx \cdot \frac{1}{3} &\geq \left(\int_0^1 x f(x) dx - \int_0^1 f(x) dx \right)^2 \Rightarrow \\ \Rightarrow \int_0^1 f^2(x) dx \cdot \frac{1}{3} &\geq (a-b)^2 \Rightarrow \int_0^1 f^2(x) dx \geq 3(a-b)^2. \end{aligned}$$

Remark: In the same way.

If $f: [0, 1] \rightarrow \mathbb{R}$, f continuous and $\int_0^1 x f(x) dx = a$, $\int_0^1 f(x) dx = b$, $a, b \in \mathbb{R}$ then:

$$\int_0^1 f^2(x) dx \geq (3a - 2b)^2$$

Marin Chirciu

Solution

Using CBS inequality under integral form:

If $f, g: [a, b] \rightarrow \mathbb{R}$, f, g integrable, then $\int_a^b f^2(x) dx \int_a^b g^2(x) dx \geq \left(\int_a^b f(x) g(x) dx \right)^2$.

Putting $[a, b] = [0, 1]$ and $g(x) = 3x - 2$ we obtain:

$$\begin{aligned} \int_0^1 f^2(x) dx \int_0^1 (3x-2)^2 dx &\geq \left(\int_0^1 f(x) (3x-2) dx \right)^2 \Rightarrow \\ \Rightarrow \int_0^1 f^2(x) dx \cdot 1 &\geq \left(3 \int_0^1 x f(x) dx - 2 \int_0^1 f(x) dx \right)^2 \Rightarrow \int_0^1 f^2(x) dx \geq (3a - 2b)^2. \end{aligned}$$

UP.551 Calculate the integral:

$$\int_0^\infty \frac{\arctan(x)}{\sqrt{3x^4 + x^2 + 3}} dx$$

Solution 1 by proposer

We make the notation: $A = \int_0^{\infty} \frac{\arctan(x)}{\sqrt{3x^4+x^2+3}} dx$.

We also consider the integral: $B = \int_0^{\infty} \frac{\operatorname{arccot}(x)}{\sqrt{3x^4+x^2+3}} dx$.

We have:

$$A + B = \int_0^{\infty} \frac{\arctan(x) + \operatorname{arccot}(x)}{\sqrt{3x^4+x^2+3}} dx = \frac{\pi}{2} \int_0^{\infty} \frac{1}{\sqrt{3x^4+x^2+3}} dx$$

We are going to calculate the integral:

$$C = \int_0^{\infty} \frac{1}{\sqrt{3x^4+x^2+3}} dx.$$

We can write:

$$C = \int_0^{\infty} \frac{1}{\sqrt{3x^4+x^2+3}} dx = \int_0^1 \frac{1}{\sqrt{3x^4+x^2+3}} dx + \int_1^{\infty} \frac{1}{\sqrt{3x^4+x^2+3}} dx$$

In the second integral we make the variable change $x = \frac{1}{t}$.

$$\int_1^{\infty} \frac{1}{\sqrt{3x^4+x^2+3}} dx = \int_1^0 \frac{1}{\sqrt{\frac{3}{t^4} + \frac{1}{t^2} + 3}} \left(-\frac{1}{t^2}\right) dt = \int_0^1 \frac{1}{\sqrt{3t^4+t^2+3}} dt$$

So we have:

$$C = 2 \int_0^1 \frac{1}{\sqrt{3t^4+t^2+3}} dt.$$

We will show that the C integral can be expressed using the complete elliptic integral the first kind.

The complete elliptic integral of the first kind is defined by relationship:

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2 \sin^2 \theta}}, \text{ with } -1 < k < 1.$$

Substitute:

$$t = \tan \frac{\theta}{2}, \text{ so } \sin \theta = \frac{2t}{1+t^2} \text{ and } d\theta = \frac{2}{1+t^2} dt.$$

We have:

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$$K(k) = \int_0^1 \frac{1}{\sqrt{1 - k^2 \frac{4t^2}{(1+t^2)^2}}} 2 \frac{1}{1+t^2} dt = 2 \int_0^1 \frac{1}{\sqrt{t^4 + (2-4k^2)t^2 + 1}} dt$$

We can write:

$$C = 2 \int_0^1 \frac{1}{\sqrt{3t^4 + t^2 + 3}} dt = 2 \frac{1}{\sqrt{3}} \int_0^1 \frac{1}{\sqrt{t^4 + \frac{1}{3}t^2 + 1}} dt$$

We put the condition:

$$2 - 4k^2 = \frac{1}{3}, \text{ so } k = \frac{1}{2}\sqrt{\frac{5}{3}} \text{ (we choose } k > 0 \text{)}.$$

We obtain:

$$C = \frac{1}{\sqrt{3}} K\left(\frac{1}{2}\sqrt{\frac{5}{3}}\right)$$

In the B integral we make the variable change $x = \frac{1}{t}$ and we immediately obtain $B = A$.

$$\text{So we have: } 2A = \frac{\pi}{2\sqrt{3}} K\left(\frac{1}{2}\sqrt{\frac{5}{3}}\right).$$

We obtained the value of the integral required in the problem statement:

$$A = \frac{\pi}{4\sqrt{3}} K\left(\frac{1}{2}\sqrt{\frac{5}{3}}\right).$$

Solution 2 by Pham Duc Nam-Vietnam

$$\begin{aligned} I &= \int_0^\infty \frac{\arctan(x)}{\sqrt{3x^4 + x^2 + 3}} dx = \int_0^\infty \frac{\arctan\left(\frac{1}{x}\right)}{\sqrt{3x^4 + x^2 + 3}} dx \Rightarrow \\ &\quad \frac{1}{x} \rightarrow x \\ I &= \frac{1}{2} \int_0^\infty \frac{\arctan(x) + \arctan\left(\frac{1}{x}\right)}{\sqrt{3x^4 + x^2 + 3}} dx \\ &= \frac{\pi}{4} \int_0^\infty \frac{1}{\sqrt{3x^4 + x^2 + 3}} dx = \frac{\pi}{4\sqrt{3}} \int_0^\infty \frac{1}{\sqrt{x^4 + \frac{1}{3}x^2 + 1}} dx = \end{aligned}$$

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$$= \frac{\pi}{4\sqrt{3}} \int_0^{\infty} \frac{1}{\sqrt{(x^2 + \omega_+^2)(x^2 + \omega_-^2)}} dx$$

$$\text{Where: } \omega_{\pm}^2 = \frac{1}{6}(1 \pm i\sqrt{35}), p = \sqrt{\omega_+ \omega_-} = 1, q = \frac{1}{2}(\omega_+ + \omega_-)$$

$$\Rightarrow p^2 = 1 \text{ and } q^2 = \frac{7}{12} \text{ now letting: } x = t + \sqrt{t^2 + 1} \Rightarrow dx = \frac{tdt}{\sqrt{t^2 + 1}}$$

$$\text{And: } \sqrt{(x^2 + \omega_+^2)(x^2 + \omega_-^2)} = 2x\sqrt{t^2 + q^2} = 2x\sqrt{t^2 + \frac{7}{12}}$$

$$\Rightarrow I = \frac{\pi}{8\sqrt{3}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{t^2 + 1}\sqrt{t^2 + \frac{7}{12}}} dt = \frac{\pi}{4\sqrt{3}} \int_0^{\infty} \frac{1}{\sqrt{t^2 + 1}\sqrt{t^2 + \frac{7}{12}}} dt$$

$$\text{Now letting: } t = \sqrt{\frac{7}{12}} \tan(u)$$

$$\begin{aligned} \Rightarrow I &= \frac{\pi}{4\sqrt{3}} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\cos^2(u) + \frac{7}{12} \sin^2(u)}} du = \frac{\pi}{4\sqrt{3}} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - \frac{5}{12} \sin^2(u)}} du = \\ &= \frac{\pi}{4\sqrt{3}} K\left(\sqrt{\frac{5}{12}}\right) \end{aligned}$$

Where: $K(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - k^2 \sin^2(x)}} dx$ is the complete elliptic integral of the first kind.

UP.552 Calculate the integral:

$$\int_1^{\infty} \frac{\sqrt{x} \ln x}{(x+1)(x^2+1)} dx$$

Proposed by Vasile Mircea Popa – Romania

Solution 1 by proposer

$$\text{Let us denote: } A = \int_1^{\infty} \frac{\sqrt{x} \ln x}{(x+1)(x^2+1)} dx$$

In the integral A we make the variable change: $x = \frac{1}{y}$.

$$\text{We obtain: } A = - \int_0^1 \frac{\sqrt{y} \ln y}{(y+1)(y^2+1)} dy$$

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Let us denote: $B = \int_0^1 \frac{\sqrt{y} \ln y}{(y+1)(y^2+1)} dx$

We have, successively:

$$B = \int_0^1 \frac{(1-y)y^{\frac{1}{2}} \ln y}{1-y^4} dy = \int_0^1 \frac{y^{\frac{1}{2}} \ln y}{1-y^4} dy - \int_0^1 \frac{y^{\frac{3}{2}} \ln y}{1-y^4} dy$$

$$\begin{aligned} B &= \int_0^1 \sum_{n=0}^{\infty} y^{4n+\frac{1}{2}} \ln y dy - \int_0^1 \sum_{n=0}^{\infty} y^{4n+\frac{3}{2}} \ln y dy = \\ &= \sum_{n=0}^{\infty} \left(\int_0^1 y^{4n+\frac{1}{2}} \ln y dy - \int_0^1 y^{4n+\frac{3}{2}} \ln y dy \right) \end{aligned}$$

We will use the following relationship:

$$\int_0^1 x^a \ln x dx = -\frac{1}{(a+1)^2}, \text{ where } a \in \mathbb{R}, a \geq 0.$$

We obtain:

$$B = \sum_{n=0}^{\infty} \left[\frac{1}{\left(4n + \frac{5}{2}\right)^2} - \frac{1}{\left(4n + \frac{3}{2}\right)^2} \right] = \sum_{n=0}^{\infty} \left[\frac{\frac{1}{16}}{\left(n + \frac{5}{8}\right)^2} - \frac{\frac{1}{16}}{\left(n + \frac{3}{8}\right)^2} \right]$$

We now use the following relationship:

$$\psi(x) = \sum_{n=0}^{\infty} \frac{1}{(x+n)^2}, \text{ where } \psi_1(x) \text{ is the trigamma function.}$$

$$\text{We obtained the value of the integral B: } B = \frac{1}{16} \left[\psi_1\left(\frac{5}{8}\right) - \psi_1\left(\frac{3}{8}\right) \right]$$

So:

$$A = \frac{1}{16} \left[\psi\left(\frac{3}{8}\right) - \psi\left(\frac{5}{8}\right) \right]$$

Thus, the problem is solved.

Solution 2 by Said Attaoui-Algeria

Let $J(a) = \int_1^{\infty} \frac{x^a}{(x+1)(x^2+1)} dx$, $a > 0$, so by replacing x with $\frac{1}{x}$, we have

$$\begin{aligned} J(a) &= \int_1^{\infty} \frac{x^{-a}}{(x^{-1}+1)(x^{-2}+1)} \frac{dx}{x^2} = \int_0^1 \frac{x^{1-a}}{(x+1)(x^2+1)} dx \\ &= \frac{1}{2} \int_0^1 \frac{x^{1-a}}{x+1} dx - \frac{1}{2} \int_0^1 \frac{x^{1-a}(x-1)}{x^2+1} dx \end{aligned}$$

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$$\begin{aligned}
 &= \frac{1}{2} \int_0^1 \frac{x^{1-a}}{x+1} dx - \frac{1}{2} \int_0^1 \frac{x^{2-a}}{x^2+1} dx + \frac{1}{2} \int_0^1 \frac{x^{1-a}}{x^2+1} dx \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \int_0^1 x^{n+1-a} dx - \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \int_0^1 x^{2n+2-a} dx + \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \int_0^1 x^{2n+1-a} dx \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2-a)} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+3-a)} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+2-a)}
 \end{aligned}$$

Differentiating with respect to a , we obtain

$$\begin{aligned}
 J'(a) &= \int_1^{\infty} \frac{x^a \log x}{(x+1)(x^2+1)} dx = \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2-a)^2} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+3-a)^2} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+2-a)^2}
 \end{aligned}$$

Fixing $a = \frac{1}{2}$, we get

$$\begin{aligned}
 &\int_1^{\infty} \frac{\sqrt{x} \log x}{(x+1)(x^2+1)} dx = J'\left(\frac{1}{2}\right) = \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{\left(n+2-\frac{1}{2}\right)^2} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{\left(2n+3-\frac{1}{2}\right)^2} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{\left(2n+2-\frac{1}{2}\right)^2} = \\
 &= 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+3)^2} - 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(4n+5)^2} + 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(4n+3)^2} = \\
 &= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n+1)^2} - 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(4n+5)^2} + 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(4n+3)^2}
 \end{aligned}$$

Thereby

$$\begin{aligned}
 &\int_1^{\infty} \frac{\sqrt{x} \log x}{(x+1)(x^2+1)} dx = \\
 &= -2(G-1) - 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+5)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+9)^2} \right) + 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+3)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+7)^2} \right) = \\
 &= -2(G-1) - 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+5)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+1)^2} + 1 \right) + 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+3)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+7)^2} \right) =
 \end{aligned}$$

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$$\begin{aligned}
 &= -2G - 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+5)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+1)^2} \right) + 2 \left(\sum_{n=0}^{\infty} \frac{1}{(8n+3)^2} - \sum_{n=0}^{\infty} \frac{1}{(8n+7)^2} \right) = \\
 &= -2G - \frac{1}{32} \left(\sum_{n=0}^{\infty} \frac{1}{\left(n+\frac{5}{8}\right)^2} - \sum_{n=0}^{\infty} \frac{1}{\left(n+\frac{1}{8}\right)^2} \right) + \frac{1}{32} \left(\sum_{n=0}^{\infty} \frac{1}{\left(n+\frac{3}{8}\right)^2} - \sum_{n=0}^{\infty} \frac{1}{\left(n+\frac{7}{8}\right)^2} \right) = \\
 &= -2G + \frac{1}{32} \left(\psi^{(1)}\left(\frac{1}{8}\right) + \psi^{(1)}\left(\frac{3}{8}\right) - \psi^{(1)}\left(\frac{5}{8}\right) - \psi^{(1)}\left(\frac{7}{8}\right) \right)
 \end{aligned}$$

Here $G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$ denotes the Catalan's constant, and $\psi^{(1)}(\cdot)$ is the trigamma function defined as

$$\psi^{(1)}(z) = \frac{\partial^2}{\partial z^2} \log(\Gamma(z)) = \sum_{n=0}^{\infty} \frac{1}{(n+z)^2}, \text{ where } \Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt \text{ is the Gamma function.}$$

UP.553 Prove the equality:

$$\int_0^{\infty} \frac{|\cos(x)|}{1+x^2} dx = 1 - 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{(4n^2-1)e^{2n}}$$

Proposed by Vasile Mircea Popa – Romania

Solution by proposer

Let us denote:

$$I = \int_0^{\infty} \frac{|\cos(x)|}{1+x^2} dx$$

The function $f(x) = |\cos(x)|$ is periodic with period π and satisfies Dirichlet's conditions.

Also, the function is even. We expand the function in the Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2nx), \text{ where:}$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} |\cos(x)| dx; a_n = \frac{2}{\pi} \int_0^{\pi} |\cos(x)| \cos(2nx) dx$$

Calculating these integrals, we obtain:

$$a_0 = \frac{2}{\pi}; a_n = -\frac{4}{\pi} \cdot \frac{(-1)^n}{4n^2-1}$$

We have:

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$$f(x) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1} \cos(2nx)$$

Substituting $f(x)$ is the expression of I , results:

$$I = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1+x^2} dx - \frac{4}{\pi} \int_0^{\infty} \sum_{n=1}^{\infty} \frac{(-1)^n \cos(2nx)}{(4n^2 - 1)(1+x^2)} dx$$

So:

$$I = 1 - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1} \int_0^{\infty} \frac{\cos(2nx)}{1+x^2} dx$$

We now use the following relationship:

$$\int_0^{\infty} \frac{\cos(mx)}{1+x^2} dx = \frac{\pi}{2} e^{-m}, \text{ where } m > 0.$$

This relation is Laplace's integral and is well known.

It is easily proved for example using the properties of the Laplace transform.

We obtained the value of the integral I :

$$I = 1 - 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{(4n^2 - 1)e^{2n}}$$

Thus, the problem is solved.

UP.554 We consider the equation: $(1 + iz)^{2n} = i \cdot (1 + z^2)^n$, where $n \geq 1$ natural number and $i^2 = -1$.

a. Prove that the complex number i is a solution of the equation for any $n \geq 1$.

b. Solve the equation in the case $n = 1$ and in one of the cases $n = 2$ or $n = 3$.

c. Find the solution of the equation in the general case $n \in \mathbb{N}^*$

Proposed by Adalbert Kovacs – Romania

Solution 1 by proposer

1. a. By replacing we can check that $z = i$ is a solution of the equation.

1.b. Case $n = 1$: We have the equation: $(1 + iz)^2 = i \cdot (1 + z^2)$, with the roots i and 1 .

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Case $n = 2$. We use the decomposition: $(1 + z^2) = (1 + iz)(1 - iz)$ the given equation becomes: $(1 + iz)^4 = i(1 + iz)^2 \cdot (1 - iz)^2 \Leftrightarrow (1 + iz)^2 \cdot [(1 + iz)^2 - i \cdot (1 - iz)^2] = 0$
 $\Leftrightarrow (1 + iz)^2 = 0$ and $(1 + iz)^2 = i \cdot (1 - iz)^2 = 0$

The first equation has the double root: i

The second equation: $1 + 2iz - z^2 - i - 2z + iz^2 = 0 \Leftrightarrow$

$$\Leftrightarrow (1 - i)z^2 + 2(1 - i)z - (1 - i) = 0 \Leftrightarrow$$

$$\Leftrightarrow z^2 + 2z - 1 = 0, \text{ with the roots: } -1 + \sqrt{2} \text{ and } -1 - \sqrt{2}.$$

Case $n = 3$: We have the equation: $(1 + i \cdot z)^6 = i \cdot (1 + z^2)^3$.

Using the decomposition: $(1 + z^2) = (1 + iz)(1 - iz)$ the equation becomes:

$$(1 + iz)^6 = i(1 + iz)^3 \cdot (1 - iz)^3$$

$$\Leftrightarrow (1 + iz)^3 \cdot [(1 + iz)^3 - i \cdot (1 - iz)^3] = 0 \Leftrightarrow (1 + iz)^3 = 0 \text{ or}$$

$$(1 + iz)^3 - i \cdot (1 - iz)^3 = 0$$

The first equation has a triples solution: $z = 1$.

The second equation: $(1 + iz)^3 + i^3 \cdot (1 - iz)^3 = 0 \Leftrightarrow$

$$(1 + iz + i(1 - iz))(1 + iz)^2 - i(1 + iz)(1 - iz) + i^2 \cdot (1 - iz)^2 = 0 \Leftrightarrow$$

$$1 + i + z(1 + i) = 0 \text{ or } 1 + 2iz - z^2 - i - iz^2 - 1 + 2iz + z^2 = 0 \Leftrightarrow$$

$$z + 1 = 0 \text{ or } z^2 - 4z + 1 = 0,$$

with the roots: -1 respectively: $2 + \sqrt{3}$ and $2 - \sqrt{3}$

So the solutions of the equations are real numbers: $-1, 2 + \sqrt{3}, 2 - \sqrt{3}$ and the complex number: i , imaginary unit

1.c. General case: We have the equation: $(1 + iz)^{2n} = i \cdot (1 + z^2)^n \Leftrightarrow$

$$\Leftrightarrow (1 + iz)^{2n} = i(1 + iz)^n \cdot (1 - iz)^n \Leftrightarrow (1 + iz)^n [(1 + iz)^n - i(1 - iz)^n] = 0 \Leftrightarrow$$

$$(1 + iz)^n = 0 \text{ or } (1 + iz)^n - i(1 - iz)^n = 0$$

From the first equation it follows the solution $z = i$, which is a multiple solution of order n

The second equation can be written: $\left(\frac{1+iz}{1-iz}\right)^n = i$, it is a binomial equation, it goes to the

trigonometric form: $\left(\frac{1+iz}{1-iz}\right)^n = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right)$ and with the extraction of the radical of order n :

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$$\begin{aligned}\frac{1+iz}{1-iz} &= \cos\left(\frac{\frac{\pi}{2} + 2k\pi}{n}\right) + i \sin\left(\frac{\frac{\pi}{2} + 2k\pi}{n}\right) \Leftrightarrow \frac{1+iz}{1-iz} \\ &= \cos\left(\frac{\pi + 4k\pi}{2n}\right) + i \sin\left(\frac{\pi + 4k\pi}{2n}\right)\end{aligned}$$

Solving for the z variable, the result is obtained in the form:

$$z = -\frac{1 - \cos t - i \cdot \sin t}{i \cdot (1 + \cos t + i \cdot \sin t)} = \frac{i \cdot (1 - \cos t - i \cdot \sin t)}{1 + \cos t + i \cdot \sin t}$$

where $t = \frac{\pi + 4k\pi}{2n}$. Using trigonometric formulas and performing the calculations we arrive

at the final result: $z = \tan \frac{t}{2}$. We obtain the solutions: $z_k = \tan\left(\frac{\pi + 4k\pi}{4n}\right)$,

with $k = 0, 1, 2, \dots, n-1$ (the number of solutions in this form is n). Particular cases:

For $n = 1$ we have the solution: $\tan \frac{\pi}{4} = 1$

For $n = 2$ we have the solutions: $\tan \frac{\pi}{8} = \sqrt{2} + 1$ and $\tan \frac{5\pi}{8} = \sqrt{2} - 1$

For $n = 3$ we have the solutions: $\tan \frac{\pi}{12} = 2 - \sqrt{3}$, $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$ and $\tan \frac{3\pi}{4} = -1$

Solution 2 by Marin Chirciu-Romania

a. It checks $(1 + i \cdot i)^{2n} = i \cdot (1 + i^2)^n \Leftrightarrow (1 - 1)^{2n} = i \cdot (1 - 1)^n \Leftrightarrow 0^{2n} = i \cdot 0^n \Leftrightarrow$
 $\Leftrightarrow 0 = 0$

b. For $n = 1$ we have the equation $(1 + iz)^2 = i \cdot (1 + z^2) \Leftrightarrow (1 + i)z^2 - 2iz + i - 1 =$
 0 , with $\Delta = 4$. We obtain $z_1 = 2, z + 2 = 2i$.

For $n = 2$ we have the equation $(1 + iz)^4 = i \cdot (1 + z^2)^2 \Leftrightarrow$
 $\Leftrightarrow (1 - i)z^4 - 4iz^3 - (6 + 2i)z^2 + 4iz + 1 - i = 0 \Leftrightarrow (z - i)^2(z^2 + 2z - 1) = 0$

We obtain $z_1 = i, z_{2,3} = -1 \pm \sqrt{2}$

For $n = 3$ we have the equation $(1 + iz)^6 = i \cdot (1 + z^2)^3 \Leftrightarrow$
 $\Leftrightarrow (1 + i)z^6 - 6iz^5 + (3i - 15)z^4 + 20iz^3 + (3i + 15)z^2 - 6iz + i - 1 = 0$
 $\Leftrightarrow (z - i)^3(z^3 - 3z^2 - 3z + 1) = 0 \Leftrightarrow (z - i)^3(z + 1)(z^2 - 4z + 1) = 0$

We obtain $z_1 = i, z_2 = -1, z_{3,4} = 2 \pm \sqrt{3}$

c. For solving the equation $(1 + iz)^{2n} = i \cdot (1 + z^2)^n$ we distinguish the cases:

i. Case 1. $1 + z^2 \neq 0$

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$$(1 + iz)^{2n} = i \cdot (1 + z^2)^n \Leftrightarrow \left(\frac{1 + 2iz - z^2}{1 + z^2} \right)^n = i, 1 + z^2 \neq 0$$

Denoting $w = \frac{1+2iz-z^2}{1+z^2}$ we solve the equation $w^n = i \Leftrightarrow w^n = 1 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$, with the solutions

$$w_k = \cos \frac{\frac{\pi}{2} + 2k\pi}{n} + i \sin \frac{\frac{\pi}{2} + 2k\pi}{n}, k = \overline{0, n-1}$$

Going back to the notation we have:

$$w = \frac{1+2iz-z^2}{1+z^2} \Leftrightarrow (1+w)z^2 - iz + w - 1 = 0, \text{ with } \Delta = 3 - 4w^2 \Rightarrow z = i \pm \sqrt{3 - 4w^2},$$

where

$$w = \cos \frac{\frac{\pi}{2} + 2k\pi}{n} + i \sin \frac{\frac{\pi}{2} + 2k\pi}{n}, k = \overline{0, n-1}$$

ii. Case 2. $1 + z^2 = 0$

$$1 + z^2 = 0 \Leftrightarrow z^2 = -1 \Leftrightarrow z = \pm i$$

The problem is completely solved.

UP.555 Find the solutions of the system:

$$\begin{aligned} \sqrt{8x+5} + \sqrt{9y+6} &= \sqrt{8x+9y+29} \\ \sqrt{12x+19} - \sqrt{3y+15} &= \sqrt{12x+3y-6} \end{aligned}$$

Proposed by Bela Kovacs – Romania

Solution 1 by proposer

We set the conditions of existence: The expressions under the radicals must be positive, both members of the equations must also be positive. Through consecutive squaring we

obtain the following system:

$$\begin{aligned} (8x+5)(9y+6) &= 81 \text{ and } (12x+19)(3y+15) = 400 \\ \Leftrightarrow (8x+5)(3y+2) &= 27 \text{ and } (12x+19)(3y+15) = 400 \end{aligned}$$

We do the calculus: $24xy + 16x + 15y = 17$ and $36xy + 180x + 57y = 115$

We multiply with -3 , respectively with 2 :

$$-72xy - 48x - 45y = -51 \text{ and } 72xy + 360x + 114y = 230$$

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By adding we obtain: $312x + 69y = 179$

We express the value of y that we substitute in the first equation: $y = \frac{179-312x}{69}$

$$(8x + 5) \left(3 \cdot \frac{179 - 312x}{69} + 2 \right) = 27 \Leftrightarrow (8x + 5) \left(\frac{179 - 312x}{23} + 2 \right) = 27 \Leftrightarrow$$

$$(8x + 5)(225 - 312x) = 621 \Leftrightarrow (8x + 5)(75 - 104x) = 207 \Leftrightarrow$$

$$-832x^2 + 600x - 520x + 375 - 207 = 0 \Leftrightarrow 832x^2 - 80x - 168 = 0 \Leftrightarrow$$

$$208x^2 - 20x - 42 = 0 \Leftrightarrow 104x^2 - 10x - 21 = 0$$

We solve the quadratic equation obtained: $\Delta_1 = 25 + 2184 = 2209 = 47^2$

$$x_1 = \frac{5+47}{104} = \frac{52}{104} = \frac{1}{2} \text{ and } x_2 = \frac{5-47}{104} = \frac{-42}{104} = \frac{-21}{52}$$

We return for the corresponding y values.

$$y_1 = \frac{179-156}{69} = \frac{23}{69} = \frac{1}{3} \text{ and } y_2 = \frac{179-312\left(\frac{-21}{52}\right)}{69} = \frac{179+126}{69} = \frac{305}{69}$$

By substitution, we find that the first result verifies the system, so it is a solution, and the second result does not verify the system, i.e. in the second equation, a negative number is obtained in the left member, so it is not a solution. So the only system solution is: $\left(\frac{1}{2}; \frac{1}{3}\right)$

Solution 2 by Marin Chirciu-Romania

For admissible (x, y) , the square is successively raised.

$$\begin{cases} \sqrt{8x+5} + \sqrt{9y+6} = \sqrt{8x+9y+29} \\ \sqrt{12x+19} - \sqrt{3y+15} = \sqrt{12x+3y-6} \end{cases} \Leftrightarrow \begin{cases} \sqrt{8x+5} \cdot \sqrt{9y+6} = 9 \\ \sqrt{12x+19} \cdot \sqrt{3y+15} = 20 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} (8x+5) \cdot (9y+6) = 81 \\ (12x+19) \cdot (3y+15) = 400 \end{cases} \Leftrightarrow \begin{cases} (8x+5) \cdot (3y+2) = 27 \\ (12x+19) \cdot (3y+15) = 400 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 24xy + 16x + 15y = 17 \\ 36xy + 180x + 57y = 115 \end{cases} \Leftrightarrow \begin{cases} 24xy + 16x + 15y = 17 \\ 312x + 69y = 179 \end{cases}$$

Replacing $y = \frac{179-312x}{69}$ in $24xy + 16x + 15y = 17$ we obtain the equation:

$$936x^2 - 90x - 189 = 0 \text{ with the solutions } x_{12} = \frac{45 \pm 423}{936} \Rightarrow x_1 = \frac{1}{2} \text{ and } x_2 = \frac{-21}{52}$$

We obtain $(x, y) = \left(\frac{1}{2}, \frac{1}{3}\right)$ also checks $(x, y) = \left(\frac{-21}{52}, \frac{305}{69}\right)$ doesn't check

$$\sqrt{12x+19} - \sqrt{3y+15} \geq 0$$

We deduce that $(x, y) = \left(\frac{1}{2}, \frac{1}{3}\right)$ is the unique solution of the system.